CONSUMER INDUSTRY ANALYSIS · EVIDENCE RECORD · 18 OCT 2026

Early warning signals for consumer credit: what was tested, what was kept, and why the rest was set aside

Over 16 and 17 October 2026 the analyst asked whether FICO should be a rated dimension of the consumer book, and the question widened to every public signal that might come before card and mortgage losses. This is the record: each signal drawn against what later happened, in its own units and rescaled for timing, with the numbers behind it. Public data only. The outcome is at the top; the evidence follows in the order the decision was made.

The decision, measure by measure

Measure	Outcome	How	Why	Section
30+ to charge-off (card, other consumer): industry, peer group, KeyBank	**Kept**	Rated by level	Smooth, moves with every source, and rates on the book's lines. The inherent-risk reading: how much delinquency turns into loss.	A1
Gap to peers and gap to industry on 30+ to charge-off (card, other consumer)	**Kept**	Rated by level, no floor at zero	The relative reading: when KeyBank turns more of its delinquency into loss than peers do. Common forces cancel out.	A1
Share of card accounts paying only the minimum (large banks)	**Kept**	Rated by level	Moves with 30–89 and about a year ahead of charge-offs; it carries what stretched utilization carries (0.92).	A3
Mortgage: rate on nonaccrual rather than charge-offs	**Kept**	Lines on the existing measure	Mortgage charge-offs are near zero since 2014 and the industry's lines are refused; nonaccrual still separates quarters and leaves out government-guaranteed 90+.	A4
Card origination score; new accounts under 660; banks easing standards	**Context**	Shown, not rated	They come 2 to 3 years before card losses, but they are lending policy, closer to an economic indicator; Book 2's territory.	A5
Other consumer 30–89 as a mortgage signal	**Context**	Shown, not rated	Comes about 3 quarters before mortgage 90+, but mortgage's own 30–89 turns at the same time: it confirms, it does not lead.	A6
Refreshed FICO (the weakest tenth's score; KeyCorp's share under 660)	**Set aside**	Set aside	Moves in the same quarter as 30–89: its year's lead on charge-offs is only the time delinquency takes to become a loss. Adds nothing 30–89 does not show.	B1
Card balances under 660 as a rating	**Set aside**	Kept as a peer comparison only	Orders banks only roughly against losses (0.55 across five banks); mix and charge-off practice move the loss rate as much.	B2
Purchase volume by accounts under 660	**Set aside**	Set aside	Falls in the same quarter delinquency rises (−0.7 to −0.85): a reaction, and purchases, not balances.	B3
Mortgage origination terms (DTI, LTV, score) and current LTV	**Set aside**	Set aside	DTI moves with rates and refinancing; LTV and score only match through 2005–2010; a large equity cushion is why high DTI makes no foreclosures now. Economic data, Book 2.	B4
Roll steps 1 and 2 on their own (30–89 to 90+; 90+ to charge-off)	**Set aside**	Watch only	Step 2 is mechanical (180 and 120 days) and matches across sources; step 1 is where cures differ, which the bank's own cohort data measures far better.	A2

Every “why” is drawn in the section named, from public data, so a reader can disagree with the chart rather than the sentence. The settings used throughout are the book’s agreed ones (width 0.5, Moderate-Low 1.0, z 2.5) unless a row says the Check refused them.

A. What was kept, and the evidence for it

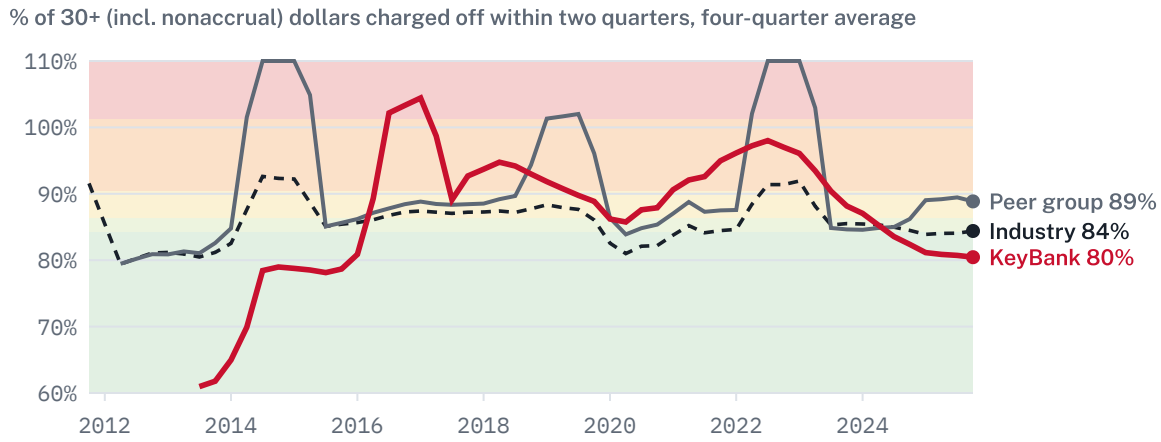
Each measure here is in the book, or shown beside its ratings. The charts are the test.

A1 · 30+ TO CHARGE-OFF, AND THE GAPS TO PEERS AND INDUSTRY

30+ to charge-off as a rating: the level, and the gap to peers

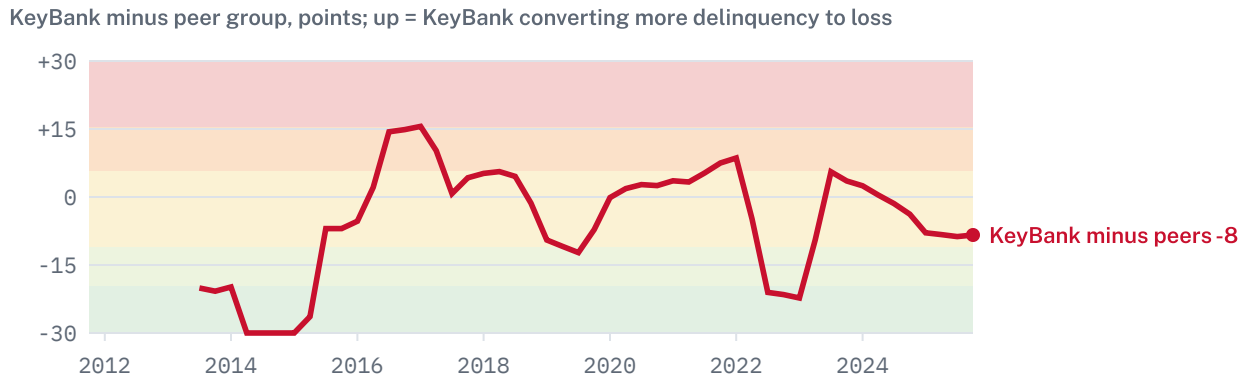
Two ratings from the same measure, both by level, as the book rates everything else. The level: KeyBank's share of 30+ (including nonaccrual) dollars charged off within two quarters, on rating zones drawn from the peer group's history at the agreed settings. The gap: KeyBank minus the peer group in the same quarter, on zones drawn from the gap's own history; this is "performing worse than peers".

Card: the level



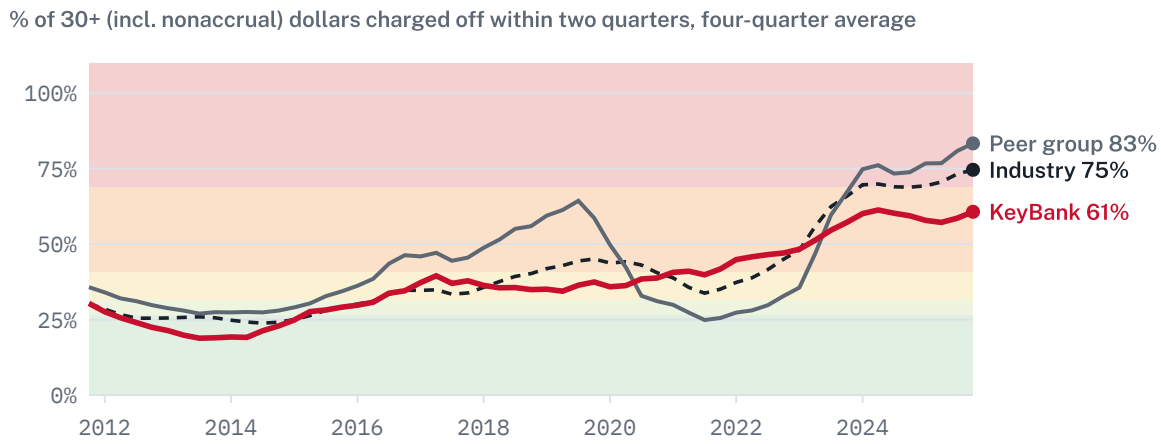
Zones from the peer group: 2 begins 84.3%, 3 86.3%, 4 90.4%, High 101.4%. KeyBank 2025Q4: rates 1. KeyBank's 30+ includes its nonaccrual card loans, which most banks do not use, so its share reads lower than it would on the same definition.

Card: KeyBank minus peers



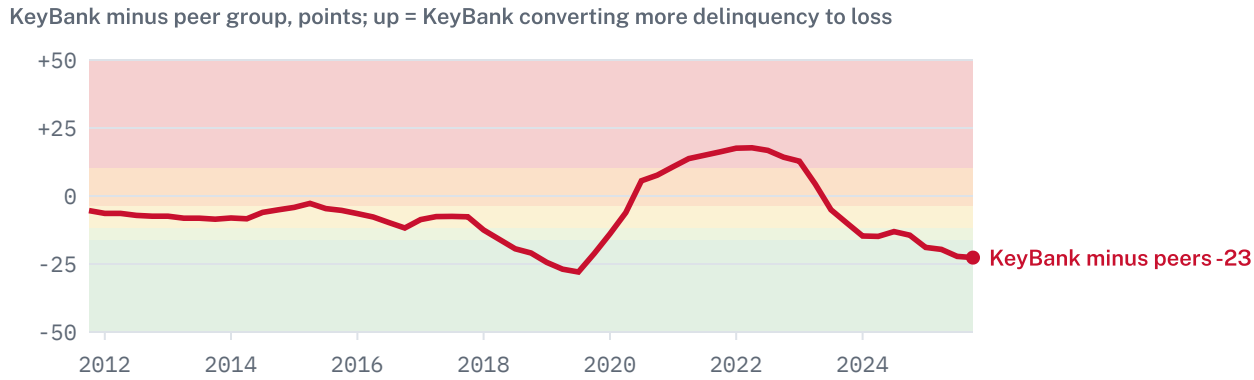
Now -8.4 points: rates 3. At z 2.5 the method refuses the gap's own history; zones shown at z 1.79, the highest that fits.

Other consumer: the level



Zones from the peer group: 2 begins 26.7%, 3 31.5%, 4 41.0%, High 69.2%. KeyBank 2025Q4: rates 4. Every source stepped up from about 35% to 60–85% from 2023, which looks like reporting and product mix rather than a collapse in cures, so lines drawn across the whole history flag everything since.

Other consumer: KeyBank minus peers



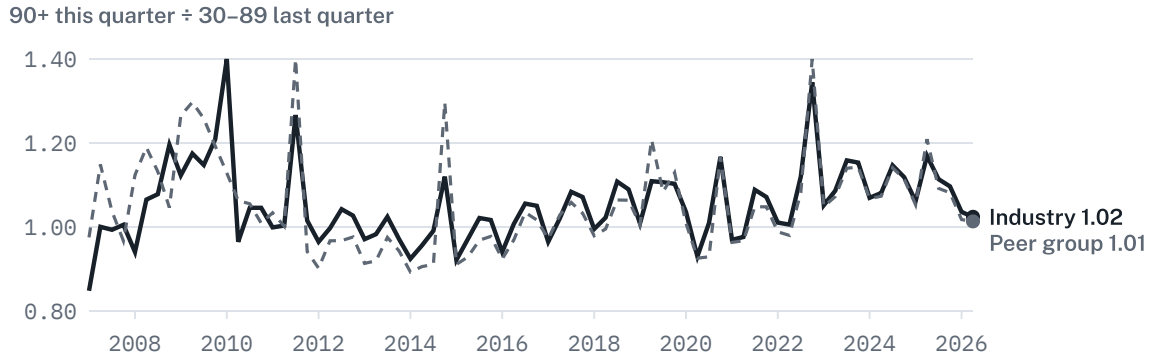
Now -22.7 points: rates 1. Zones at the agreed settings. The gap is measured in the same quarter, so the industry-wide step in 2023 largely cancels out.

Estimated roll and cure, card and other consumer

Built from every bank's Call Report: 30–89, 90+ and nonaccrual dollars at each quarter-end, and gross charge-offs each quarter. Not tracked cohorts, so these are estimates of the waterfall, not true roll rates. Industry solid black, peer group dashed, KeyBank red.

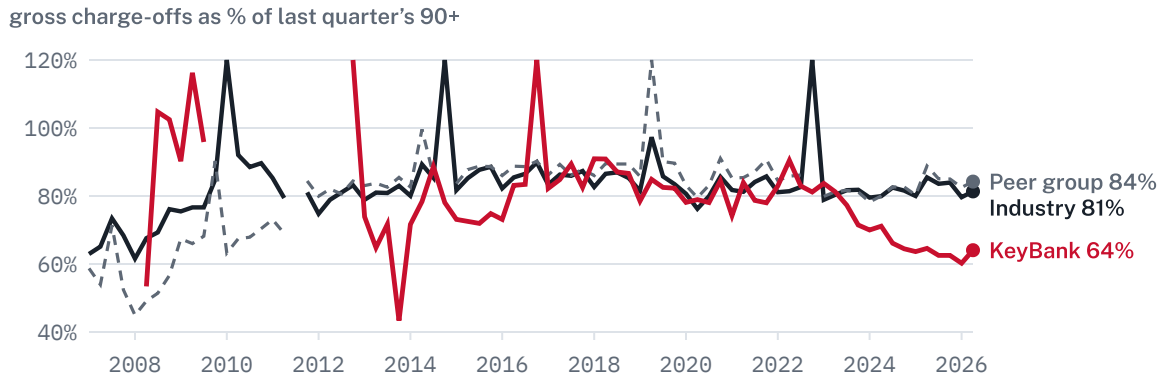
All charge-offs here are gross (before recoveries): recoveries arrive later and mostly come from loans charged off in earlier quarters. The waterfall in three steps: 30–89 to 90+, 90+ to charge-off, and the two together.

Step 1, card: the roll ratio, 90+ this quarter over 30–89 last quarter



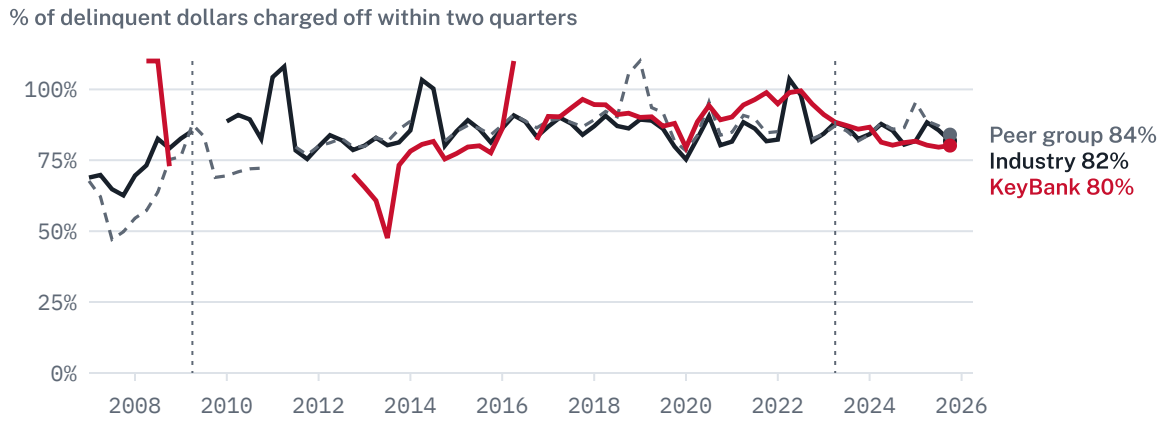
Above 1 because the 90+ bucket holds about three months of loans and 30–89 two. Its rise came before rising card charge-offs by 2–3 quarters, loosely (0.40 to 0.43 industry, 0.25 peers).

Step 2, card: gross charge-offs this quarter over 90+ last quarter



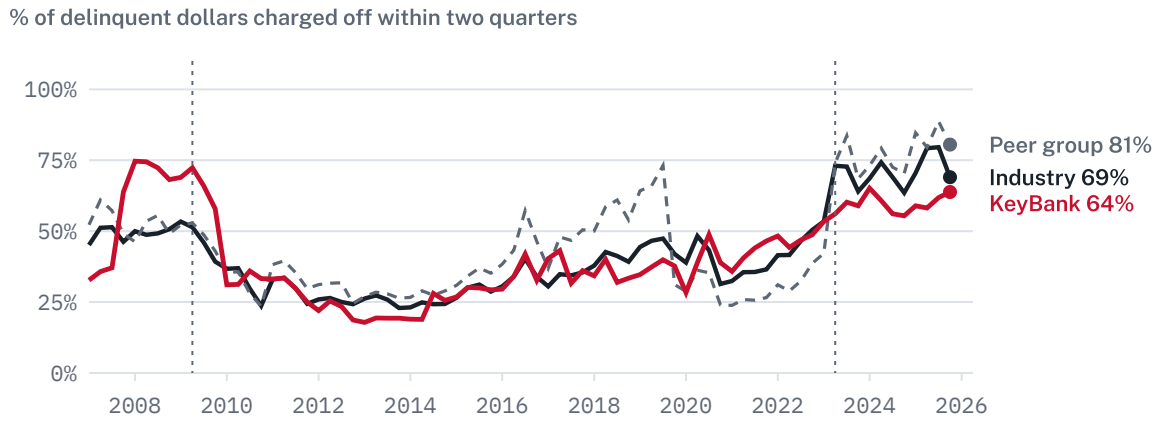
Industry about 80–90%: most of the 90+ bucket charges off in the next quarter, at 180 days. KeyBank has fallen to about 64% because its 90+ includes nonaccrual card loans, which most banks do not use and which are not all on the way to charge-off. Peers ran low in 2008–2010 as delinquency piled up faster than it was charged off.

30+ to charge-off, card: share of delinquent dollars that end up charged off (steps 1 and 2 together)



Charge-offs over the next two quarters ÷ (30–89 + 90+) at the quarter-end. Cards charge off at 180 days, so every delinquent dollar either charges off within two quarters or cures; one minus this is the estimated cure. About 80% charge off; it rose to 85% in 2009 and 88% in 2023. It needs the next two quarters, so it is known six months late. KeyBank includes nonaccrual, which it uses for cards where most banks do not.

30+ to charge-off, other consumer



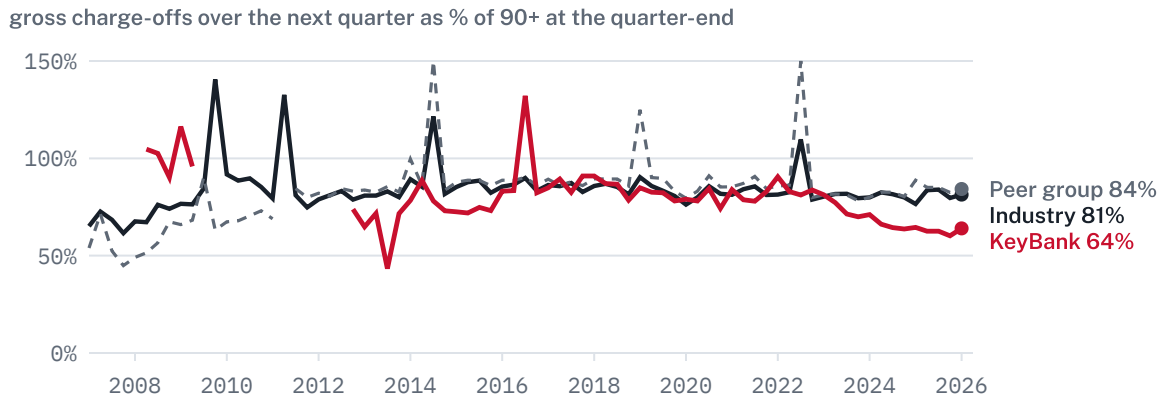
Lower and less steady: 24–50% before 2023, 64–81% since. Closed-end loans charge off at 120 days and open-end at 180, and the line mixes student, auto (until 2011), unsecured and secured loans; it breaks at 2011 and 2024Q4.

For other consumer there is no usable link from the roll ratio to charge-offs. A tracked cohort, internally, would separate rolling from curing directly.

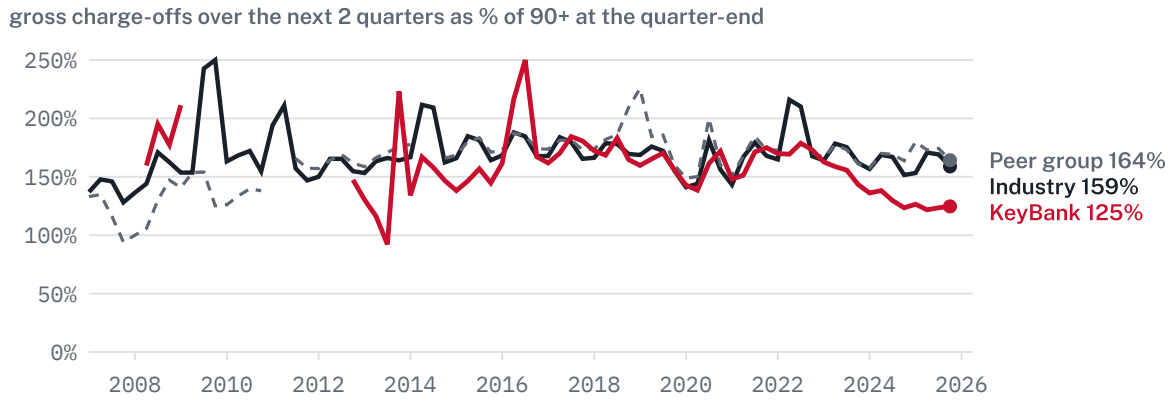
90+ to charge-off: one quarter and two quarters later

For each quarter-end: gross charge-offs over the next quarter, and over the next two quarters, as a share of the 90+ (including nonaccrual) dollars at that quarter-end. If everything in 90+ either charges off or cures within a quarter, the two lines sit close together; where the second runs well above the first, charge-offs are also coming from loans that were not yet 90+ at the quarter-end.

Card, next quarter



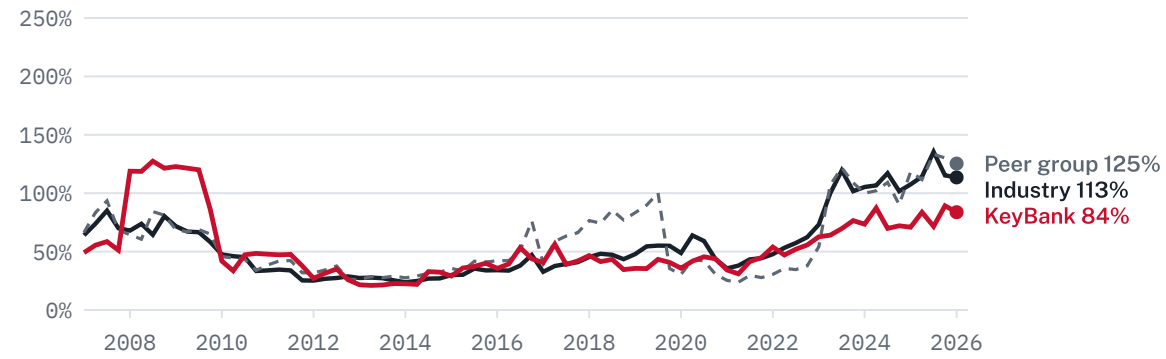
Card, next two quarters



Card charges off at 180 days, so the 90+ bucket (90–179 days) clears within one quarter: the one-quarter figure is the 90+ roll to charge-off (about 80% for the industry). The two-quarter figure adds the next wave, loans that were 30–89 at the quarter-end, so it runs about twice as high and is not a rate of the 90+ bucket.

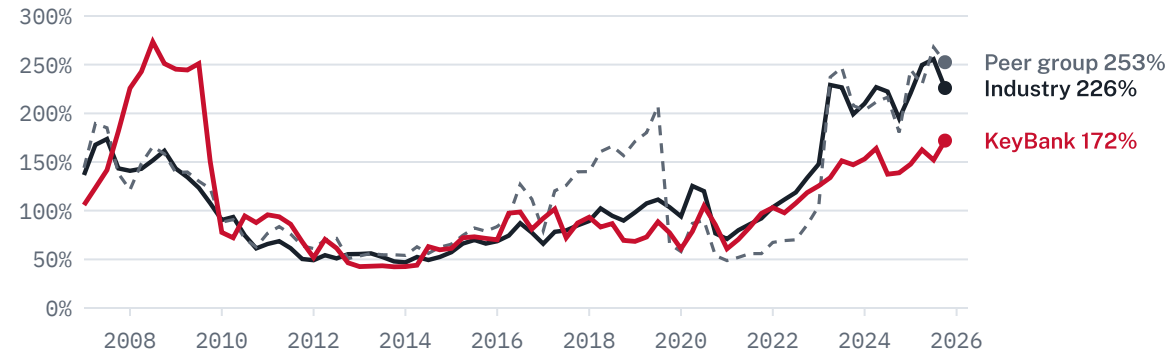
Other consumer, next quarter

gross charge-offs over the next quarter as % of 90+ at the quarter-end



Other consumer, next two quarters

gross charge-offs over the next 2 quarters as % of 90+ at the quarter-end



Closed-end loans charge off at 120 days, often inside a quarter without appearing in a quarter-end 90+ figure, so even the one-quarter figure can exceed 100%. It measures how much is charged off relative to the visible 90+ stock, not a roll rate.

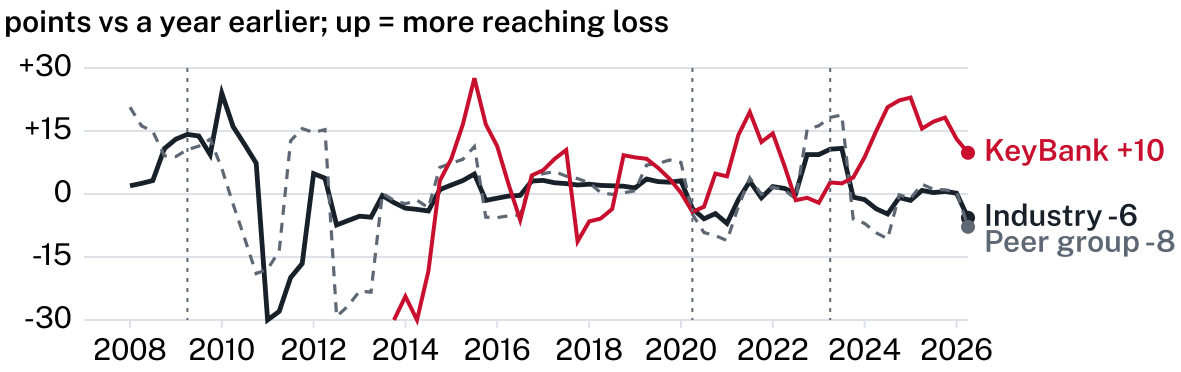
A2 · THE ROLL STEPS, MOVEMENT AND DIVERGENCE

Loss mitigation over time: is less delinquency turning into loss?

Each line is how much a step of the waterfall changed from a year earlier, after averaging it over four quarters. Up means more of the delinquent book is moving on toward loss (mitigation working less well); down means more is curing or being worked out. Levels above 100% (other consumer) do not matter here: only the movement is drawn. Dotted lines mark 2009Q2, 2020Q2 and 2023Q2.

Card

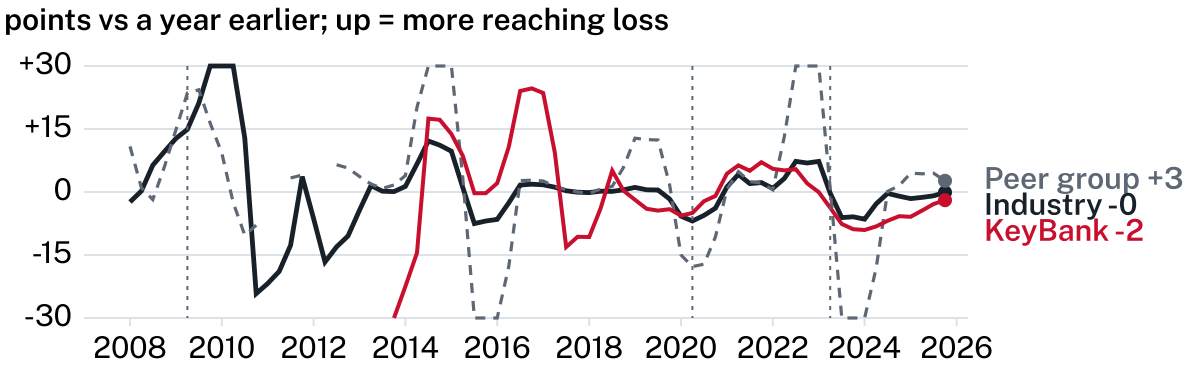
Step 1: 30–89 rolling to 90+



Step 2: 90+ to charge-off

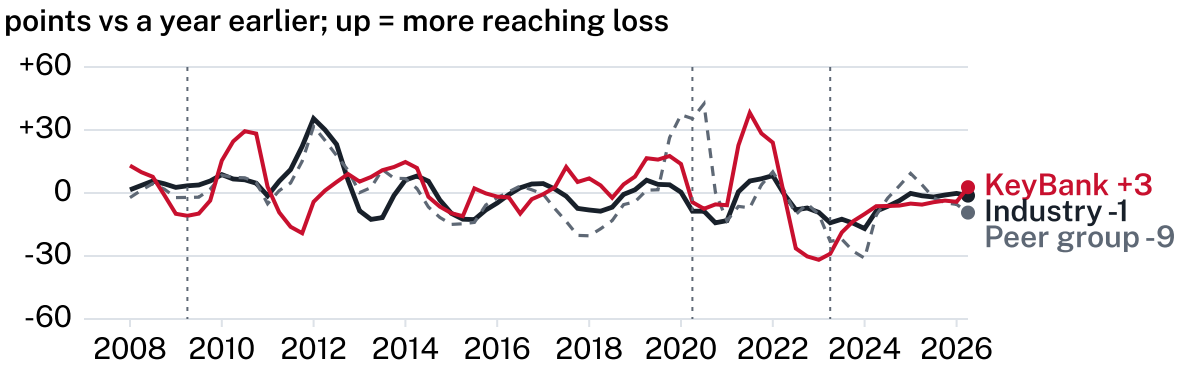


30+ to charge-off: share of delinquent dollars charged off

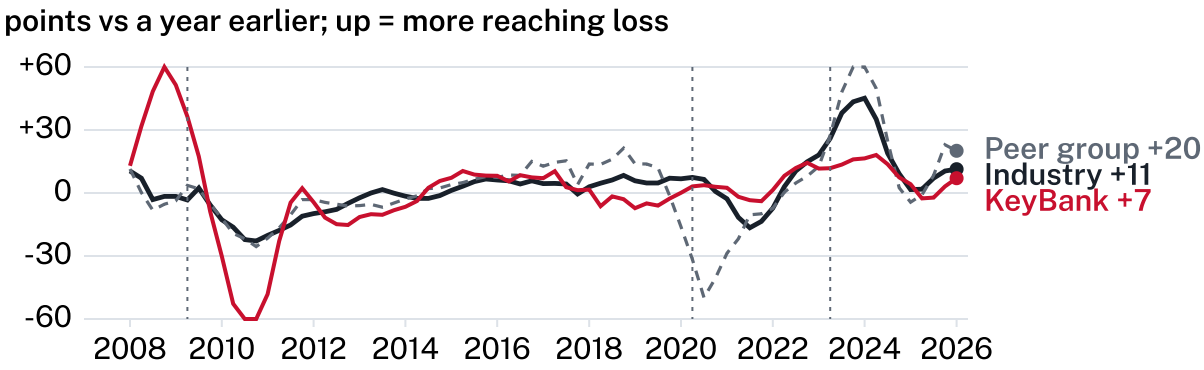


Other consumer

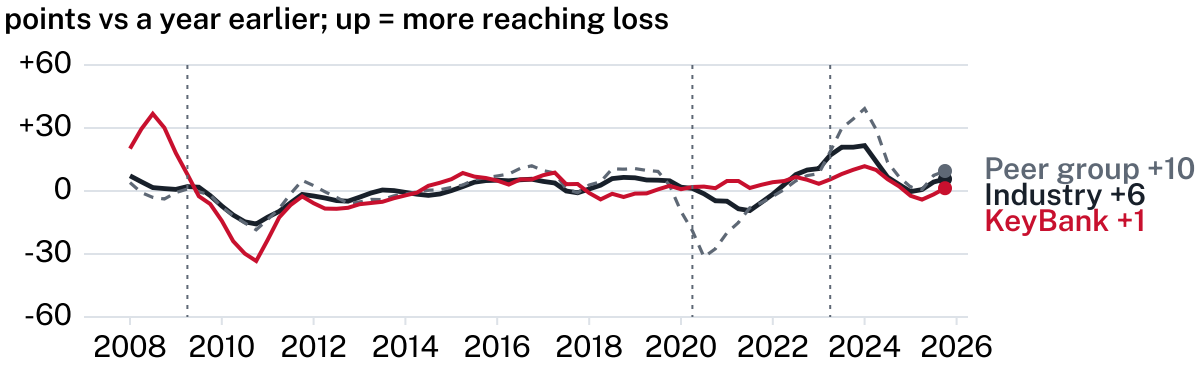
Step 1: 30–89 rolling to 90+



Step 2: 90+ to charge-off



30+ to charge-off: share of delinquent dollars charged off



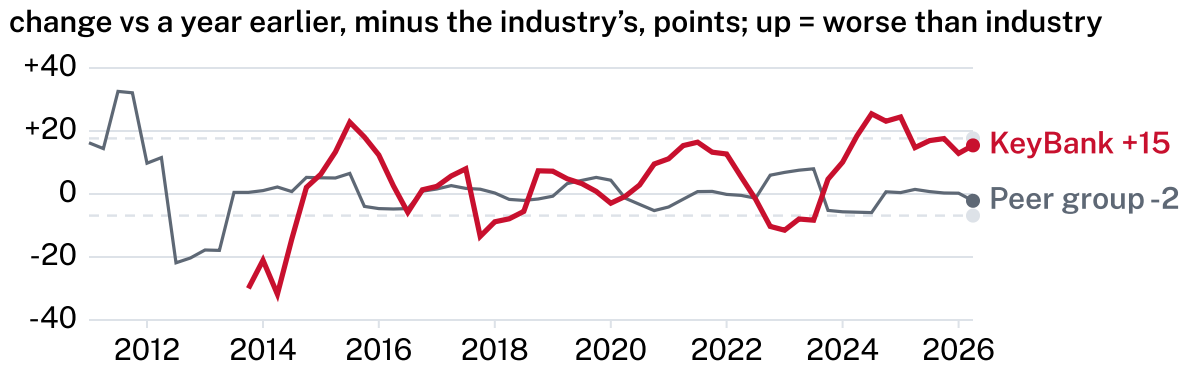
Read the three together: the first step (30–89 rolling to 90+) is where cures show first; the second (90+ to charge-off) is where workouts and settlements show; the third is the combined share of delinquent dollars that end up charged off (one minus the estimated cure). KeyBank's card figures include nonaccrual loans, and its book was small before 2013, so its early movements are large.

Roll-rate divergence: where KeyBank moves differently from the industry

For each step of the waterfall: KeyBank's change from a year earlier minus the industry's (both averaged over four quarters), and the same for the peer group. Common forces (charge-off rules, the economy, reporting) cancel out; what is left is the bank's own. Measured on movement, not level, so KeyBank's use of nonaccrual for cards and its small card book before 2013 do not create a gap by themselves. From 2011.

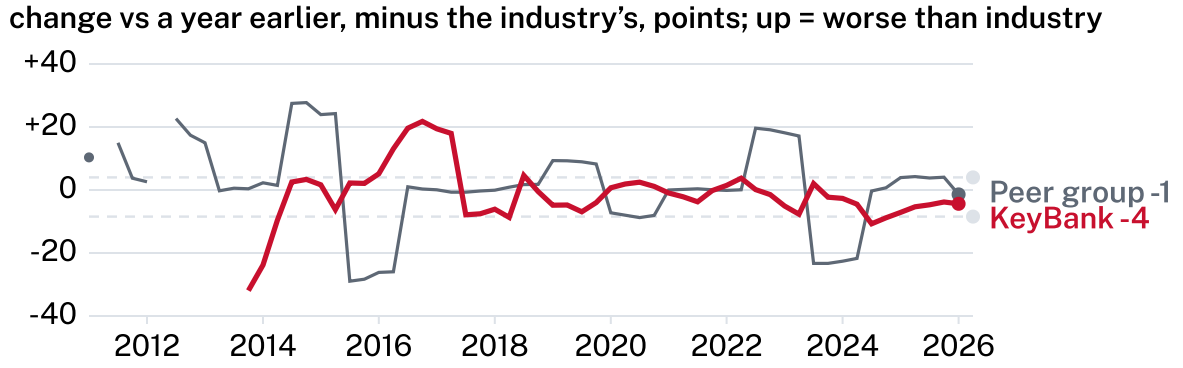
Card

Step 1: 30–89 rolling to 90+



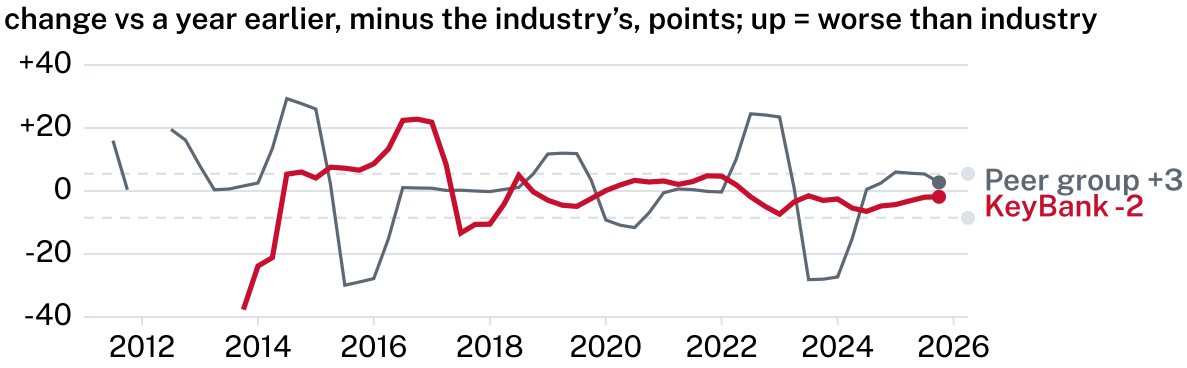
Dashed band: KeyBank's typical swing either side of its usual gap (1.4826 × median absolute deviation, 12.3 points). Now +15.4 points (2026Q2). Beyond twice that swing: 2013Q4, 2014Q1, 2014Q2.

Step 2: 90+ to charge-off



Dashed band: KeyBank's typical swing either side of its usual gap (1.4826 × median absolute deviation, 6.2 points). Now -4.4 points (2026Q1). Beyond twice that swing: 2013Q4, 2014Q1, 2016Q2, 2016Q3, 2016Q4, 2017Q1, 2017Q2.

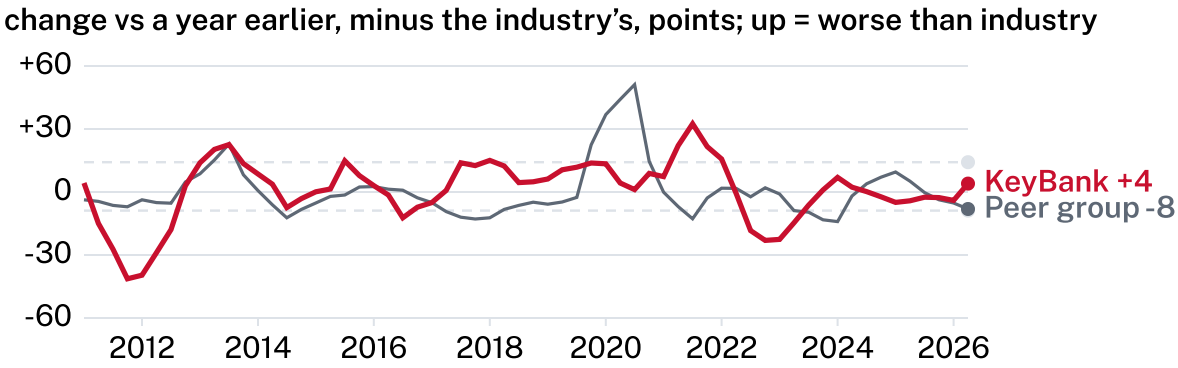
30+ to charge-off (all delinquent dollars charged off within two quarters)



Dashed band: KeyBank's typical swing either side of its usual gap (1.4826 × median absolute deviation, 7.0 points). Now -1.8 points (2025Q4). Beyond twice that swing: 2013Q4, 2014Q1, 2014Q2, 2016Q2, 2016Q3, 2016Q4, 2017Q1.

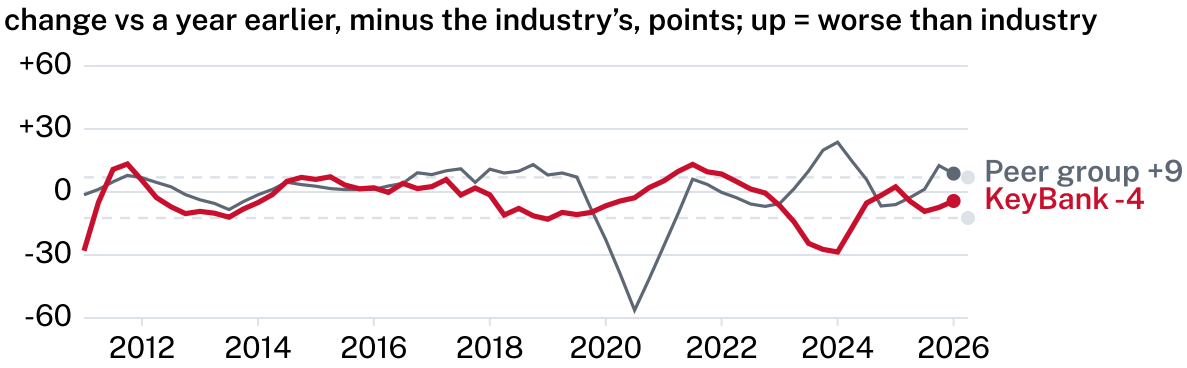
Other consumer

Step 1: 30–89 rolling to 90+



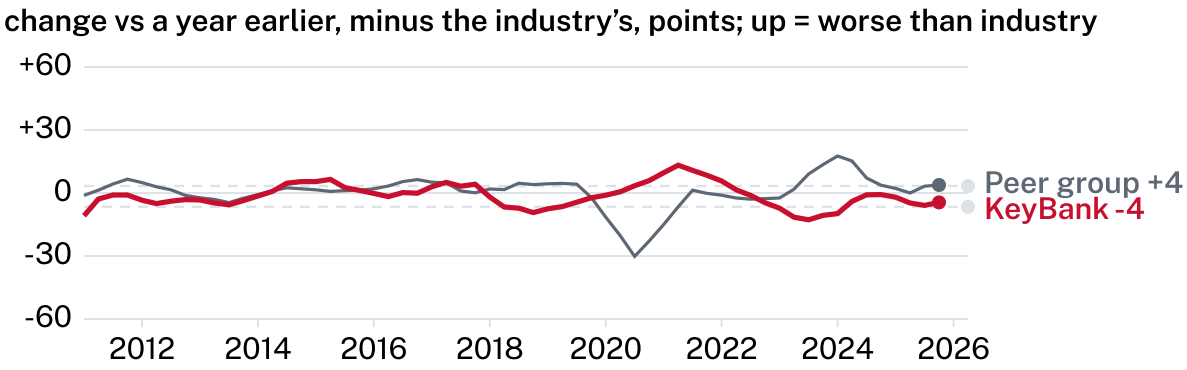
Dashed band: KeyBank's typical swing either side of its usual gap (1.4826 × median absolute deviation, 11.5 points). Now +4.0 points (2026Q2). Beyond twice that swing: 2011Q3, 2011Q4, 2012Q1, 2012Q2, 2021Q3, 2022Q4, 2023Q1.

Step 2: 90+ to charge-off



Dashed band: KeyBank's typical swing either side of its usual gap (1.4826 × median absolute deviation, 9.7 points). Now -4.3 points (2026Q1). Beyond twice that swing: 2011Q1, 2023Q3, 2023Q4, 2024Q1.

30+ to charge-off (all delinquent dollars charged off within two quarters)



Dashed band: KeyBank's typical swing either side of its usual gap (1.4826 × median absolute deviation, 4.9 points). Now -4.4 points (2025Q4). Beyond twice that swing: 2021Q1, 2021Q2, 2021Q3, 2021Q4, 2023Q2, 2023Q3.

How far is far enough to act on, and over how many quarters, are settings to choose; the band shown is only a reading of how much this gap usually moves.

A3 · BORROWER STRAIN: PAYING ONLY THE MINIMUM

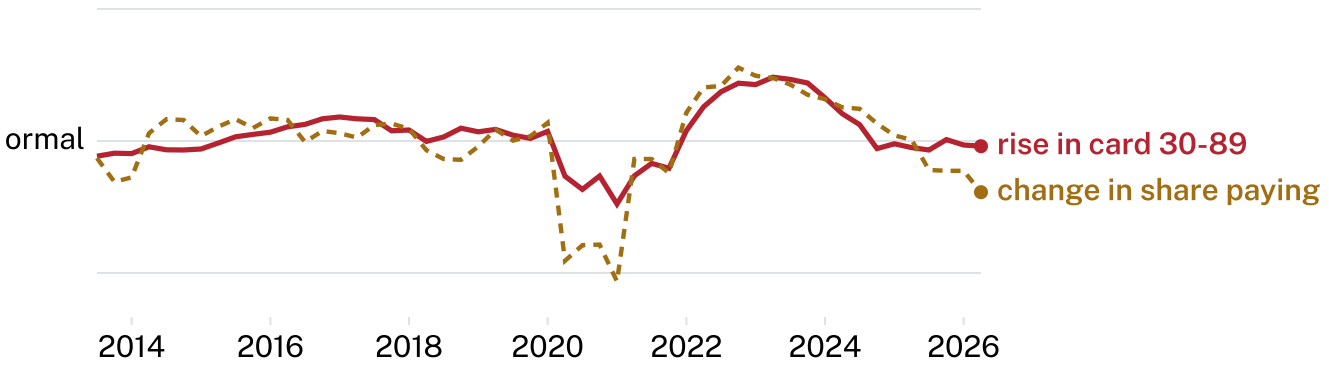
Card: strain on the existing book

Three measures of how cardholders are carrying their balances (large banks, 2012 on), each drawn as its change over a year, around its own normal, over the rise in card 30–89 days past due (solid red). Not moved: these turn at the same time as delinquency, like refreshed FICO, so they confirm stress rather than warn of it.

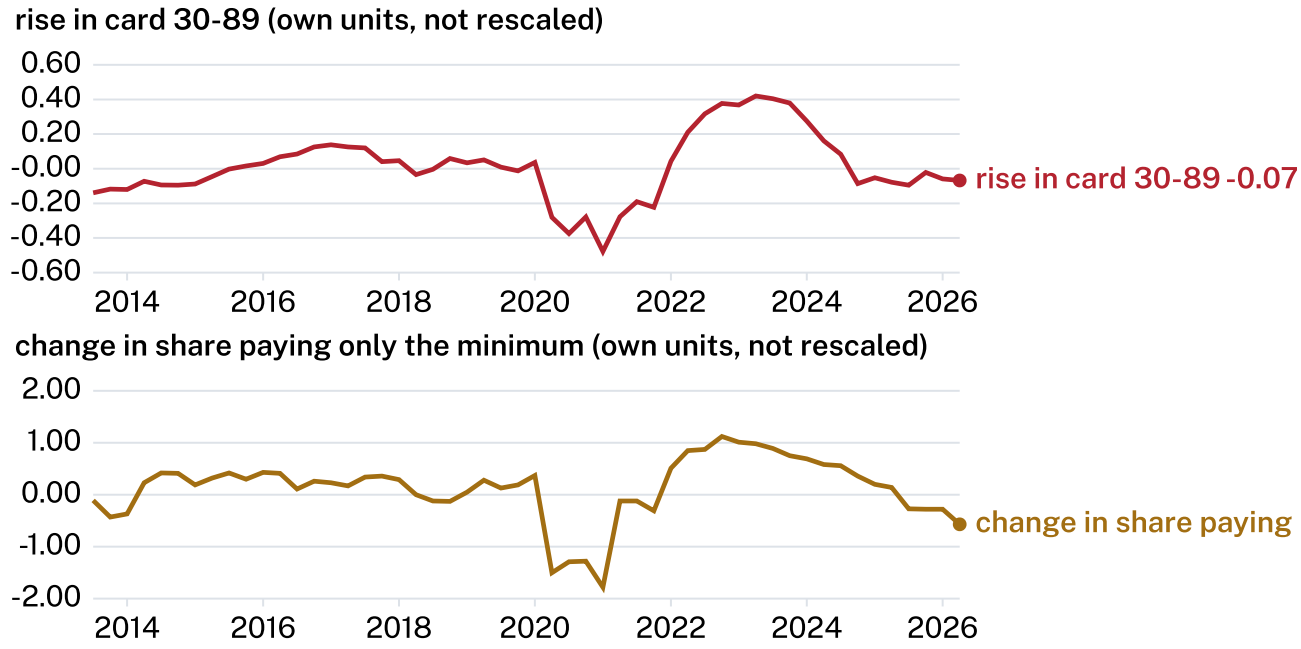
Share of accounts paying only the minimum

rise in card 30-89 change in share paying only the minimum

each line around its own normal



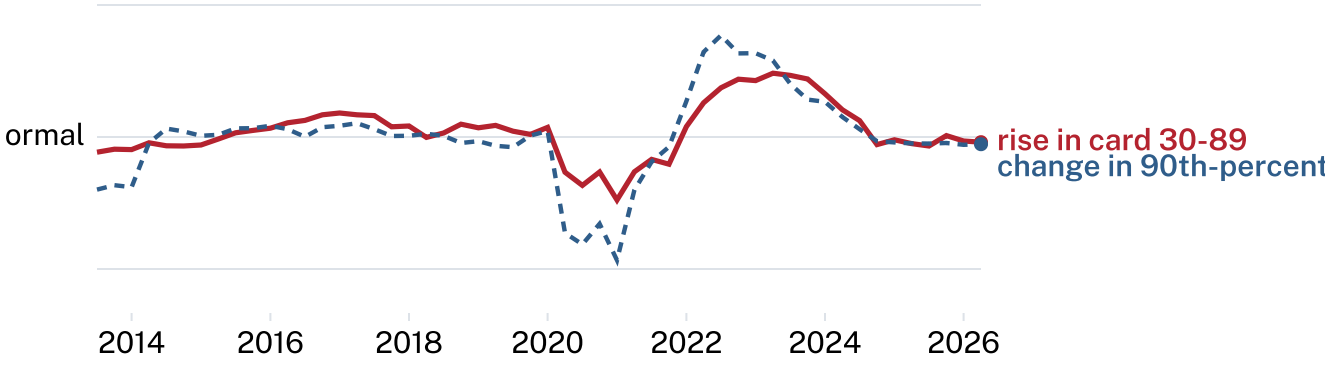
▼ The same two series in their own units



Utilization, 90th percentile (the most stretched tenth of active accounts)

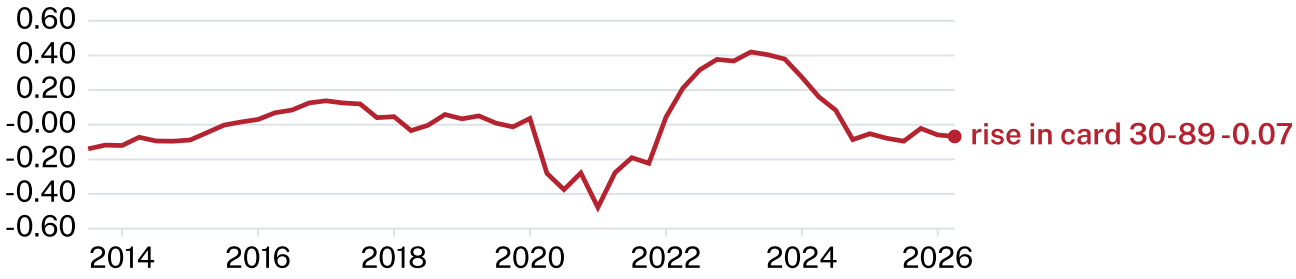
rise in card 30-89 change in 90th-percentile utilization

each line around its own normal

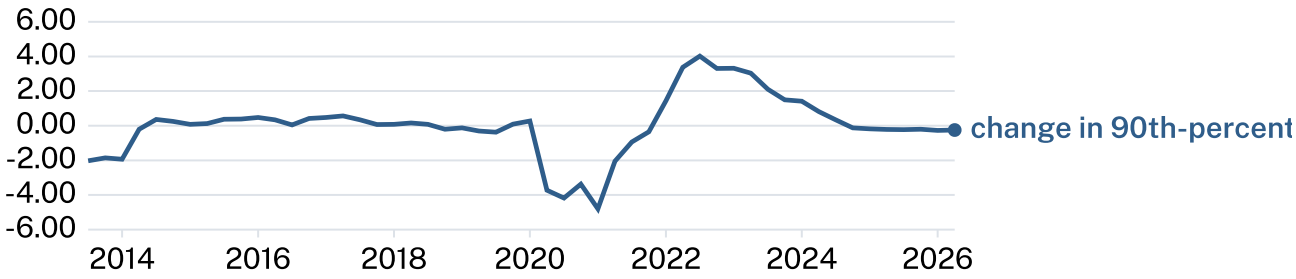


▼ The same two series in their own units

rise in card 30-89 (own units, not rescaled)



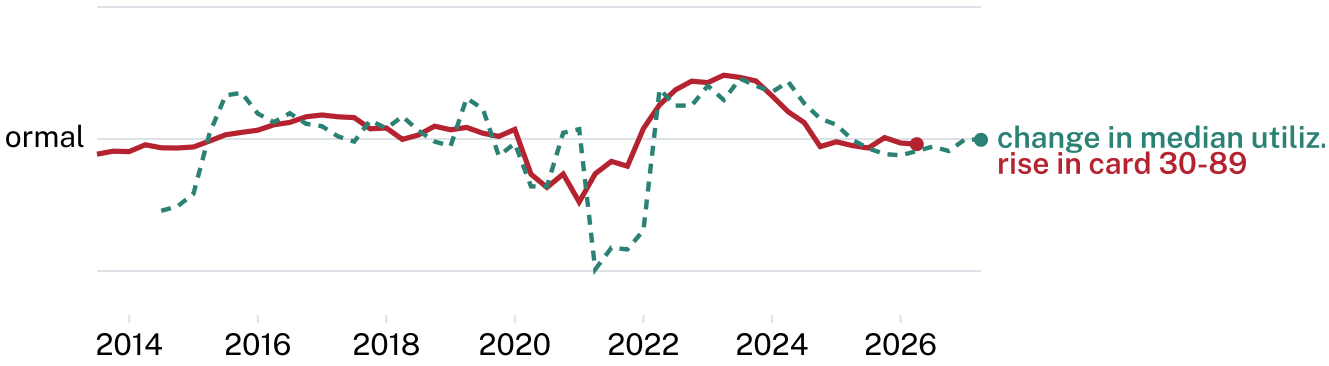
change in 90th-percentile utilization (own units, not rescaled)



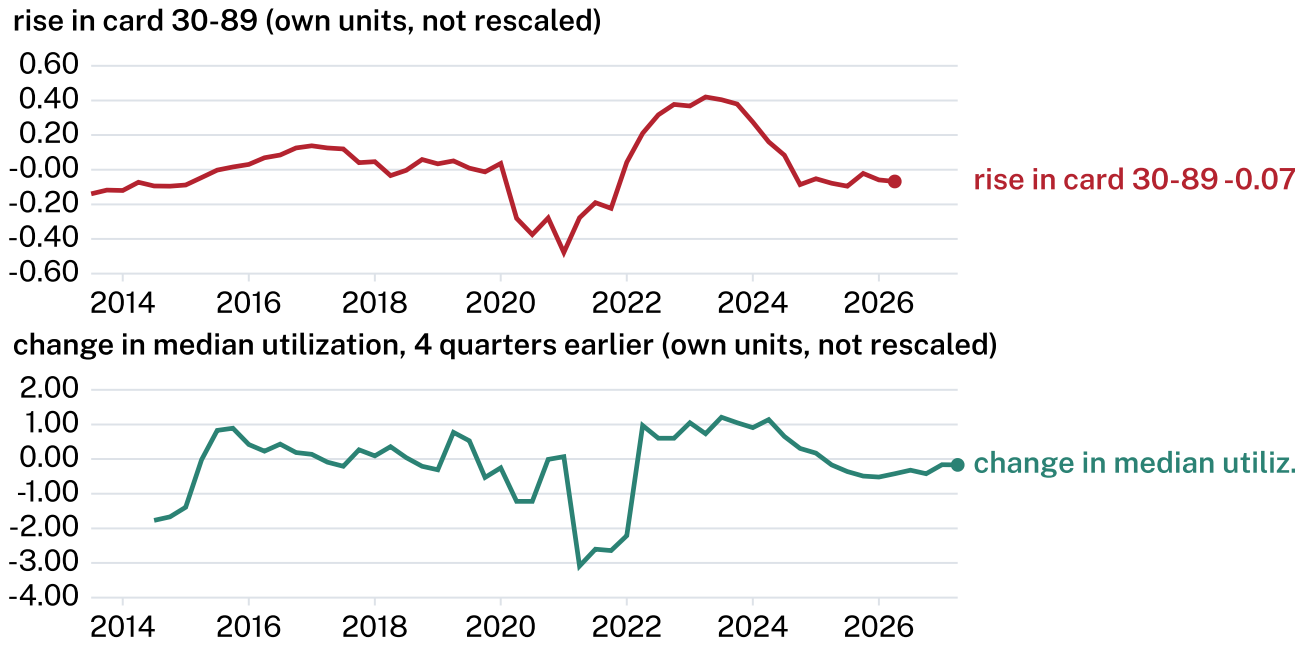
Utilization, median: the one with a small lead, moved 4 quarters later

rise in card 30-89 change in median utilization, 4 quarters earlier

each line around its own normal



▼ The same two series in their own units



Paying only the minimum and the most-stretched utilization match 30–89 best in the same quarter (0.85 and 0.90) and charge-offs about a year later. Median utilization matches 30–89 best 2 to 6 quarters later (0.66; 0.53 with 2020–2023 left out). Credit line cuts and increases are banks reacting and show no lead.

Minimum payments and utilization: one signal or two?

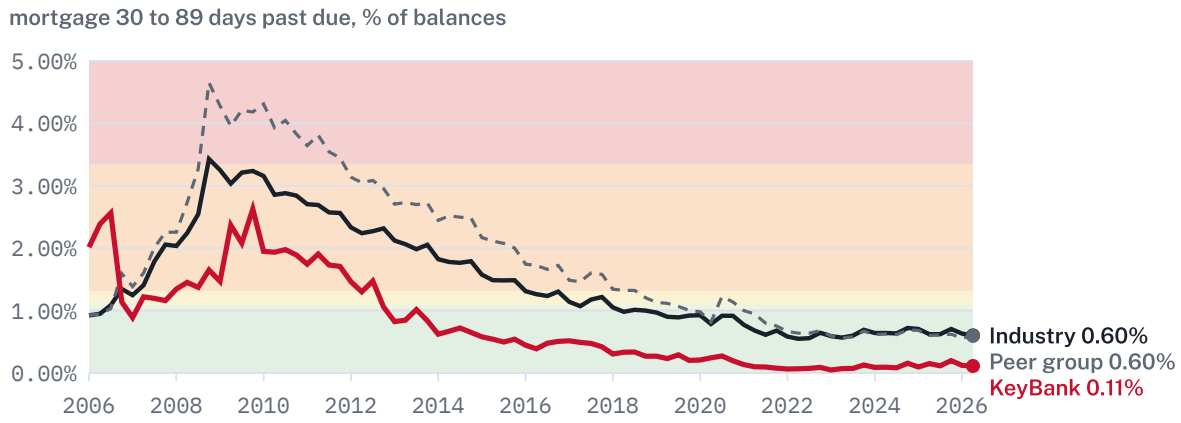
The share of accounts paying only the minimum and the most stretched tenth's utilization move together: their yearly changes match at 0.92 (0.70 with 2020–2023 left out). They are largely the same borrowers. Median utilization overlaps less (0.42 without 2020–2023) and is the one with a small lead on delinquency, so it carries something the minimum-payment share does not.

A4 · MORTGAGE ON DELINQUENCY

Mortgage delinquency on the book's lines

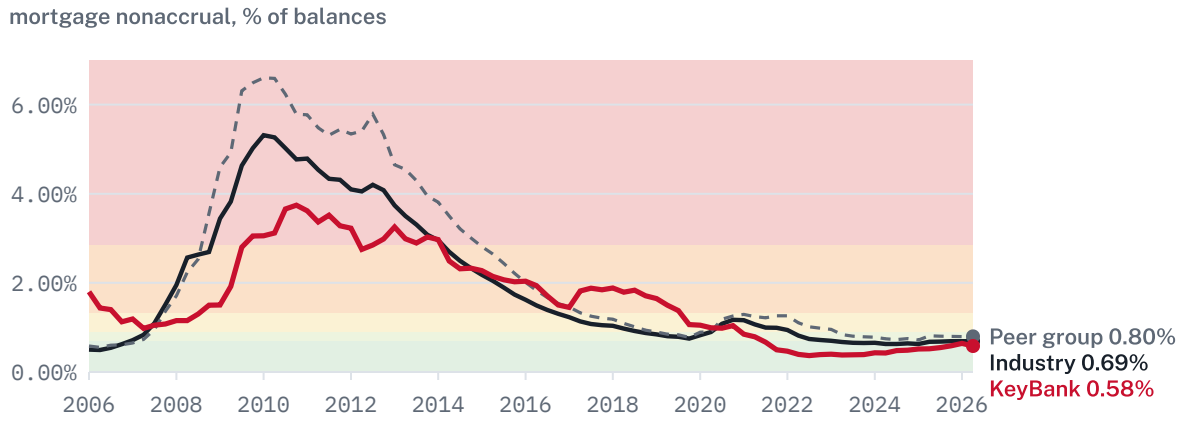
The book's industry series (solid black), peer group (dashed gray) and KeyBank (red), 2006 to 2026, with the five rating zones drawn from the industry's history at the agreed settings (width 0.5, Moderate-Low 1.0, z 2.5).

30 to 89 days past due



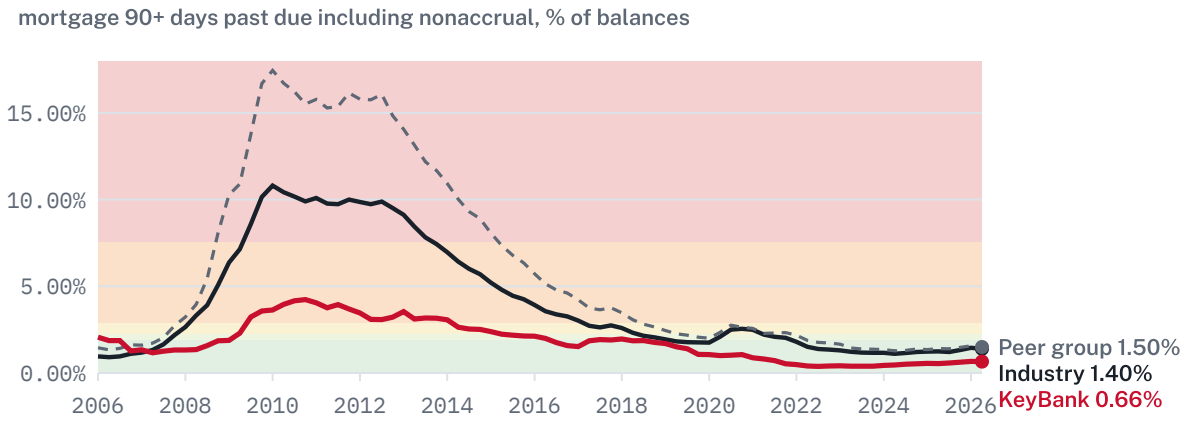
Zones from the industry's whole history (2006 on): 2 begins 1.03%, 3 1.13%, 4 1.33%, 5 High 3.35%. Since 2016 the industry rated 1 Low in 33 of 42 quarters. Industry now 0.60%, peer group 0.60%, KeyBank 0.11%. With lines from 2016 on only, the method refuses: High would begin beyond the worst quarter kept: nothing kept could rate High.

Nonaccrual



Zones from the industry's whole history (2006 on): 2 begins 0.69%, 3 0.90%, 4 1.33%, 5 High 2.86%. Since 2016 the industry rated 1 Low in 13 of 42 quarters. Industry now 0.69%, peer group 0.80%, KeyBank 0.58%. With lines from 2016 on only: 2 begins 0.68%, 3 0.76%, 4 0.91%, 5 High 1.43%; the industry now rates 2.

90+ days past due including nonaccrual



Zones from the industry's whole history (2006 on): 2 begins 1.90%, 3 2.24%, 4 2.90%, 5 High 7.58%. Since 2016 the industry rated 1 Low in 22 of 42 quarters. Industry now 1.40%, peer group 1.50%, KeyBank 0.66%. With lines from 2016 on only, the method refuses: High would begin beyond the worst quarter kept: nothing kept could rate High.

The whole-history lines are set by 2008 and its long tail. They still separate quarters on nonaccrual (Low in 13 of 42 quarters since 2016) and on 90+ (22 of 42), but 30–89 has read Low in 33 of 42. Lines drawn only from 2016 on fit nonaccrual and refuse 30–89 and 90+, because nothing since 2016 is far enough from normal to rate High. Which window the lines come from is a setting on Decide (periods left out), not a change to the data.

A5 · UNDERWRITING AND ORIGINATION QUALITY, AS CONTEXT

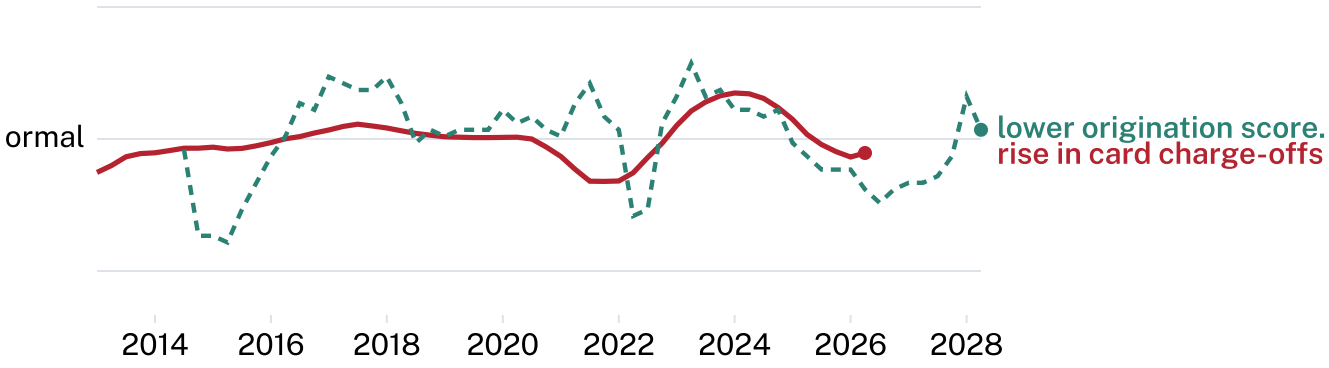
Card: who got new accounts, and what followed

The score of the bottom tenth of new card accounts each quarter (lower is riskier), and the share of new accounts going to scores under 660, each moved later by its best delay and laid over the rise in card charge-offs (solid red).

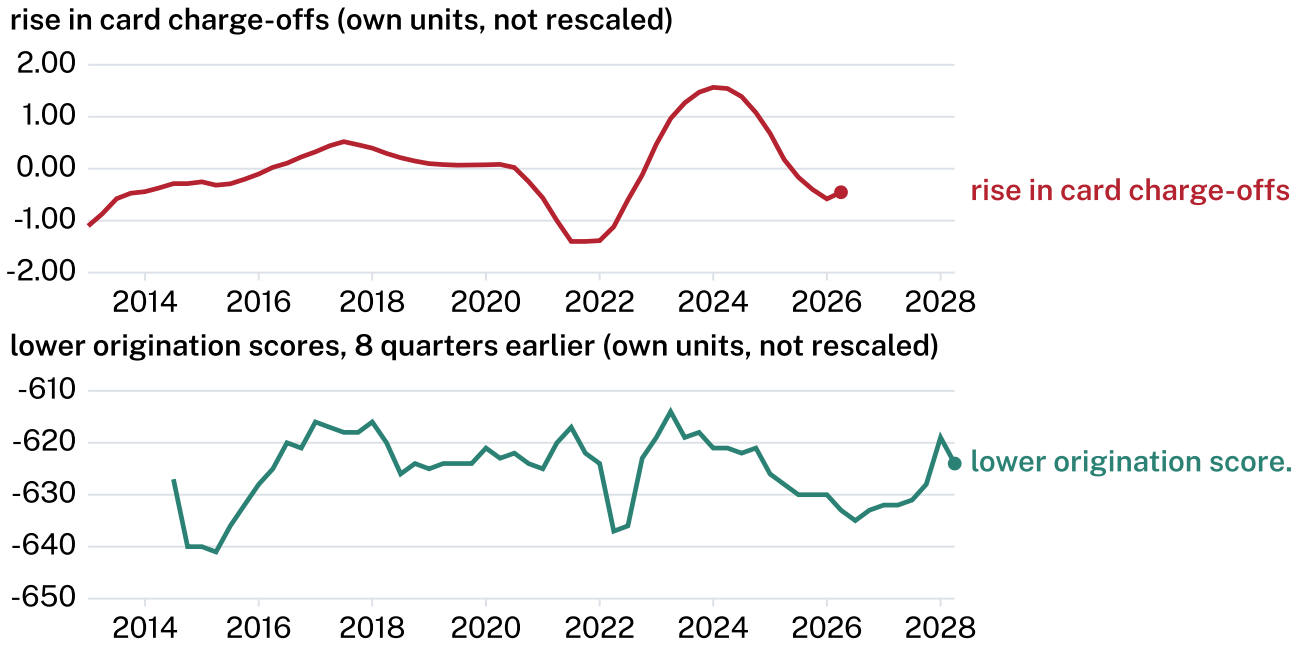
Score at origination, 10th percentile (drawn so lower is up), moved 8 quarters later

rise in card charge-offs lower origination scores, 8 quarters earlier

each line around its own normal



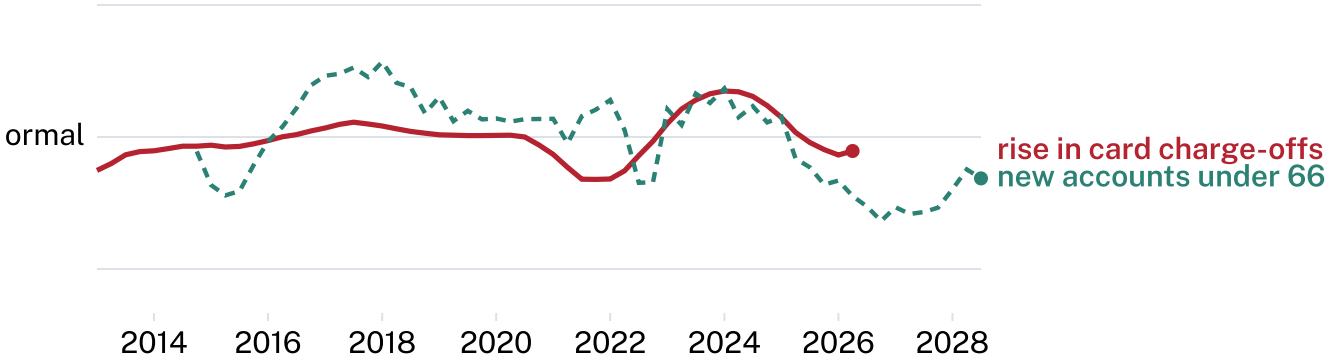
▼ The same two series in their own units



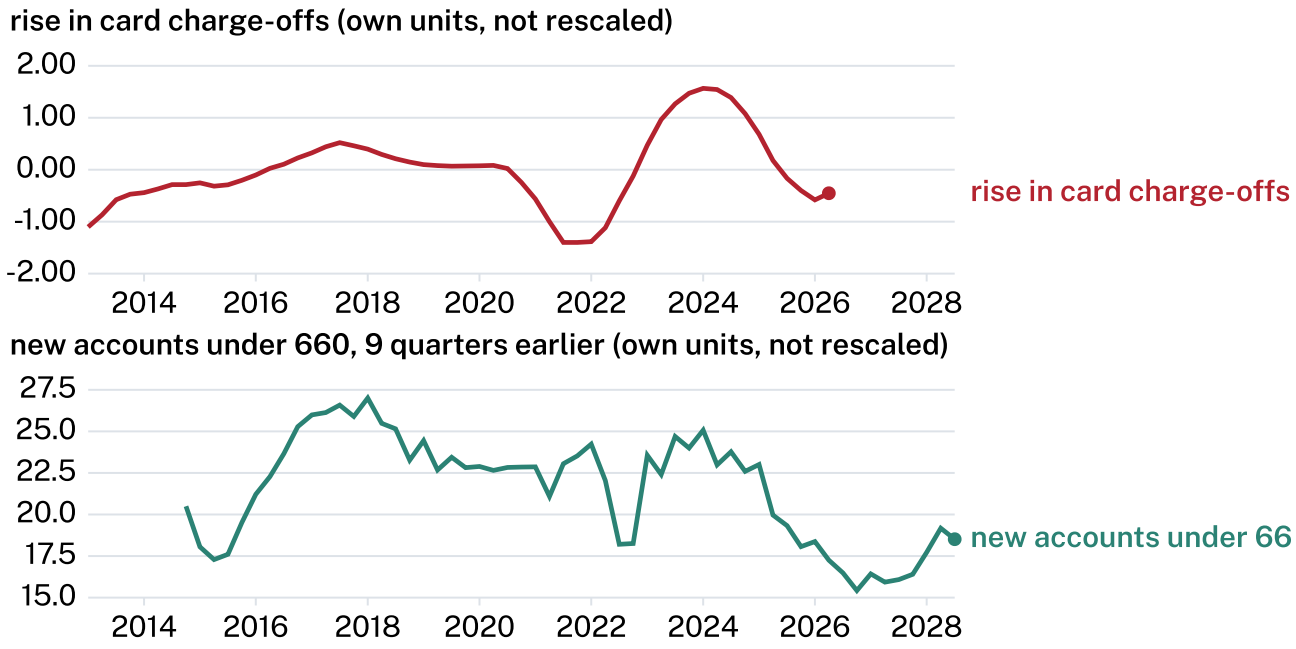
Share of new accounts under 660, moved 9 quarters later

rise in card charge-offs new accounts under 660, 9 quarters earlier

each line around its own normal



▼ The same two series in their own units

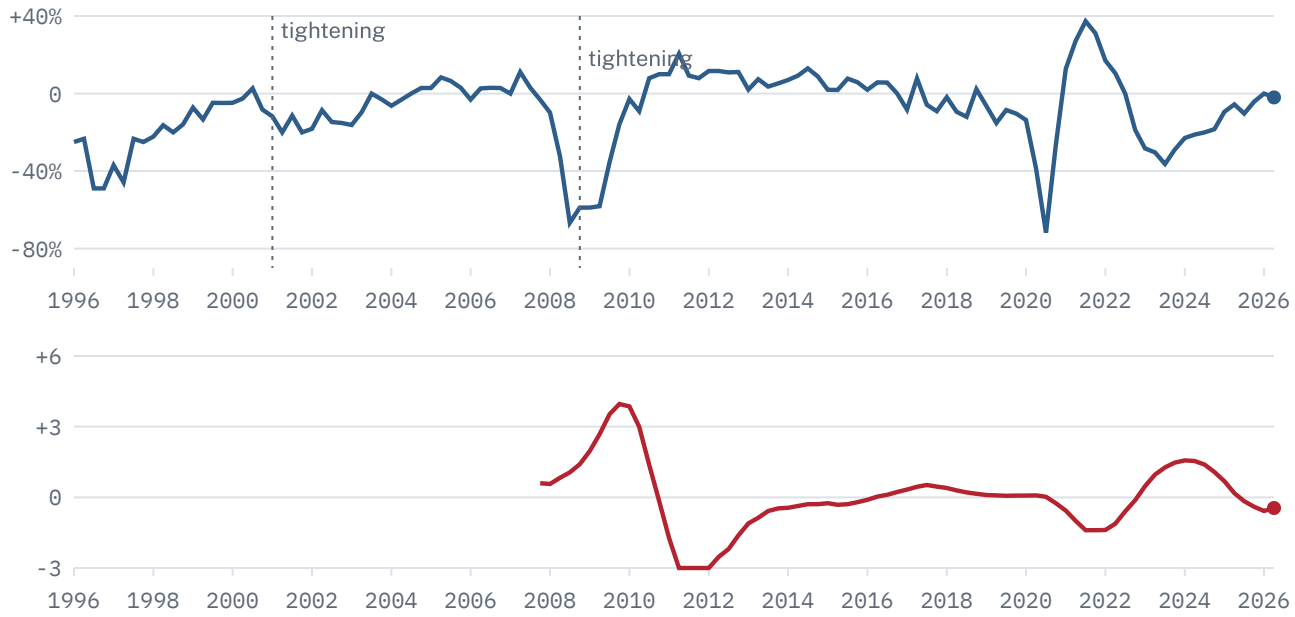


Both lead charge-offs by about two years: 0.85 to 0.96 with 2020–2023 left out, 0.40 to 0.42 with them in. The 2021 low (bottom tenth at 614) sits over the 2023 rise. Now: bottom tenth 624 (a year earlier 632); share under 660 18.5% (a year earlier 16.1%).

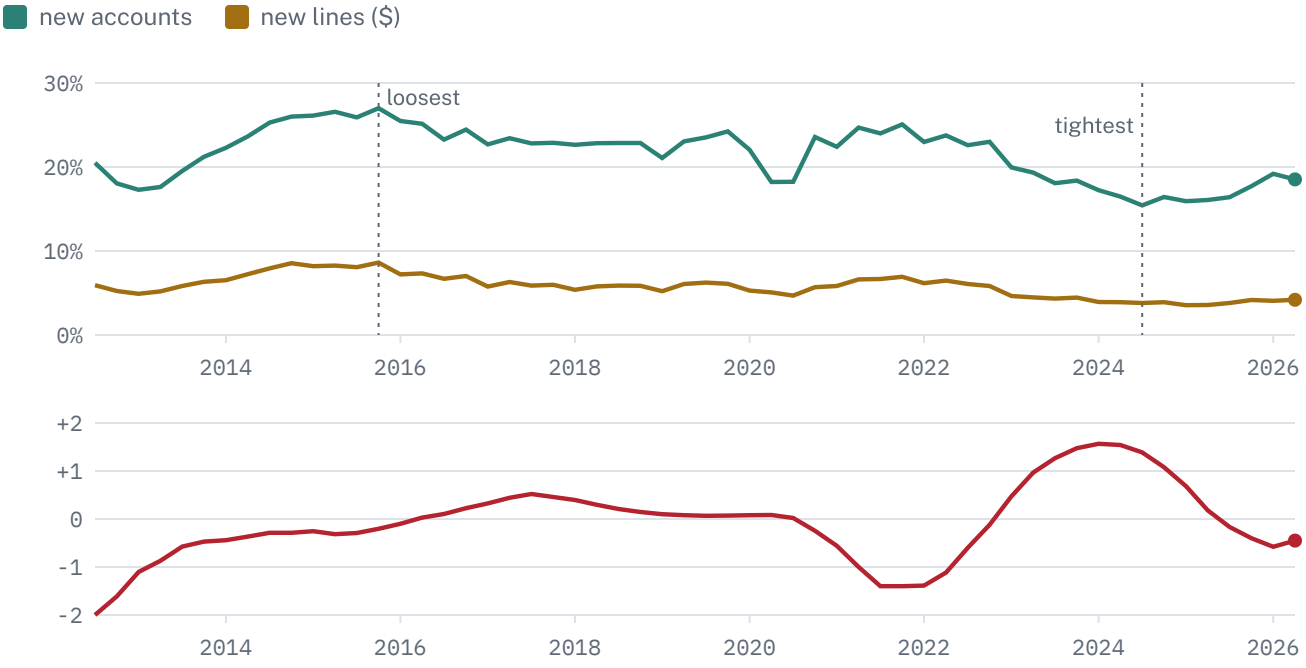
Underwriting: does loosening come first?

Refreshed FICO moved with delinquency. Underwriting sits earlier in the chain: who is approved, and on what terms, comes before anyone goes delinquent. Two public card measures: the Federal Reserve's Senior Loan Officer Survey (the net share of banks easing card standards; easing is up), from 1996; and the share of new card accounts, and of new credit lines in dollars, going to scores under 660 at the large banks (Philadelphia Fed), from 2012.

Banks easing card standards (1996 on), and the rise in card charge-offs (2007 on)

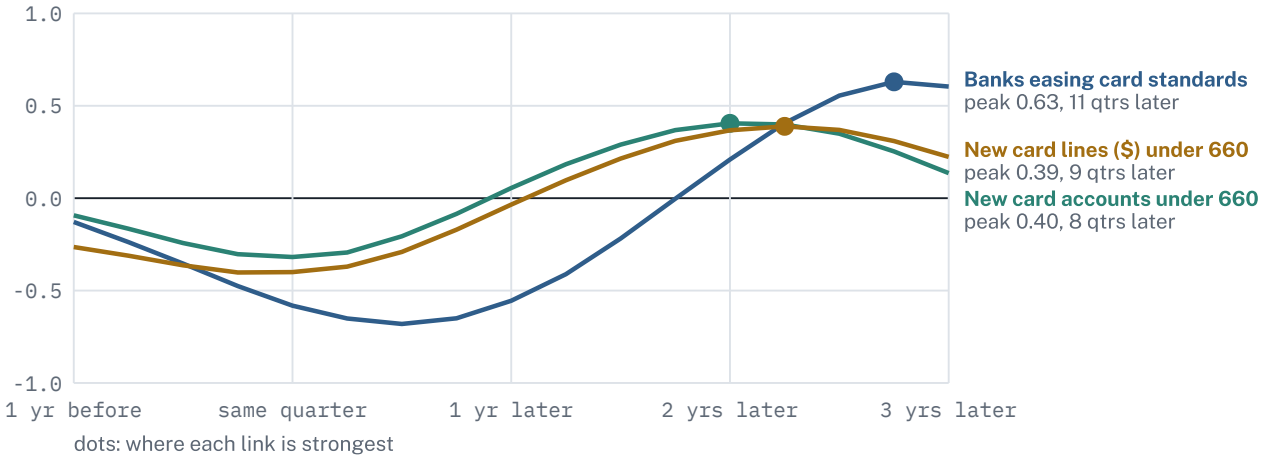


New card accounts and lines going to scores under 660, and the rise in card charge-offs

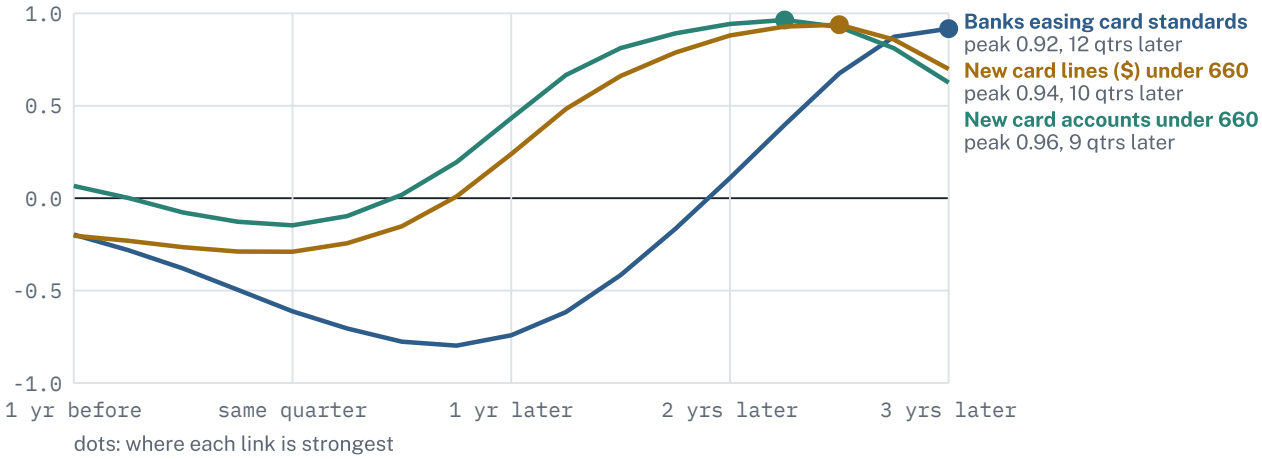


How strongly looser underwriting goes with a later rise in charge-offs

All quarters, then 2020–2023 left out. Below zero in the first year is banks reacting: they tighten while losses climb.

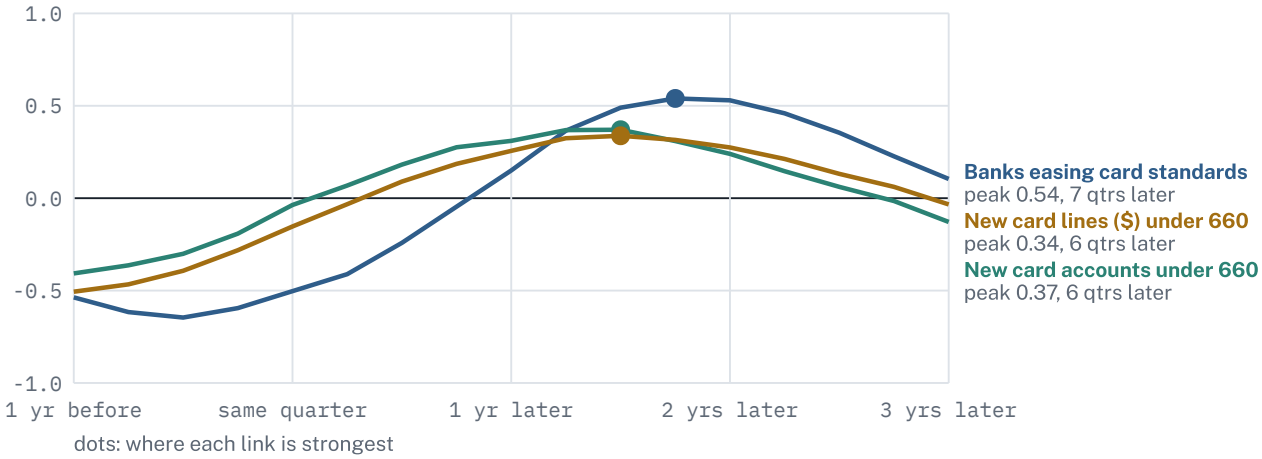


dots: where each link is strongest



dots: where each link is strongest

The same against a later rise in 30–89 days past due



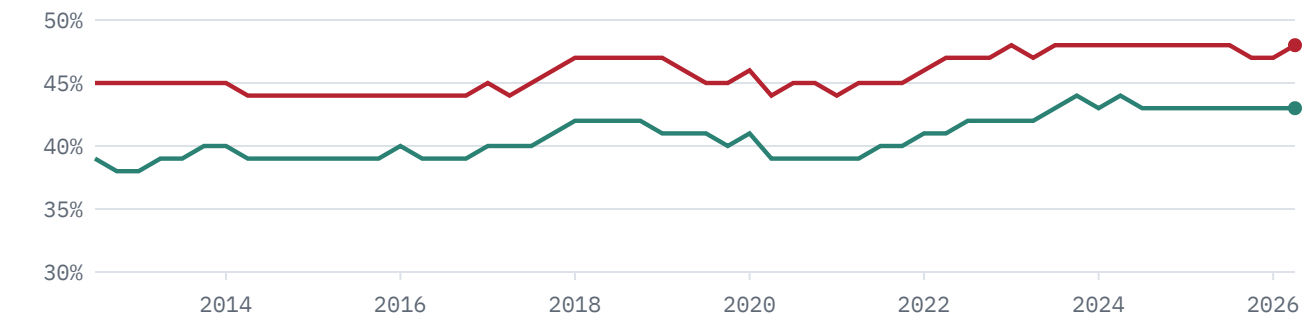
dots: where each link is strongest

Unlike refreshed FICO, these come before delinquency, not with it: easier underwriting is followed by rising 30–89 about one and a half to two years later and by rising charge-offs two to three years later. The left-out view rests on 20 to 37 quarters, and the survey before 2006 has no card loss series here to test against.

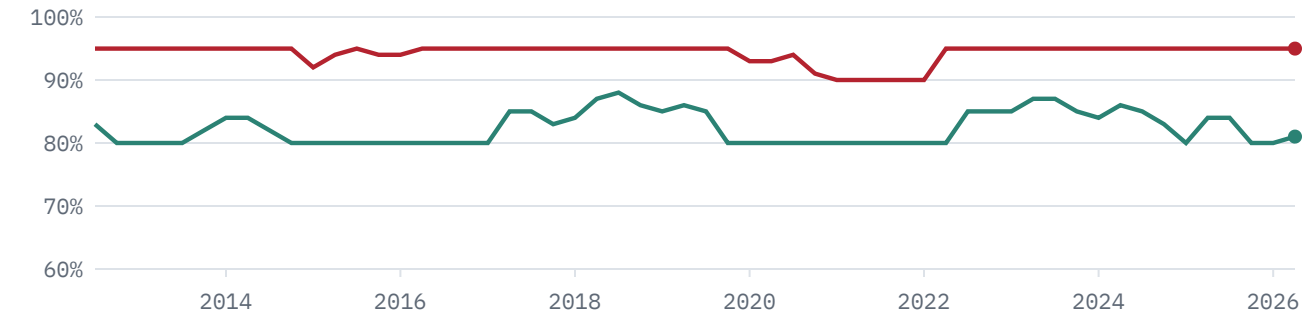
Mortgage terms at origination (large banks, 2012 on)

75th percentile 90th percentile

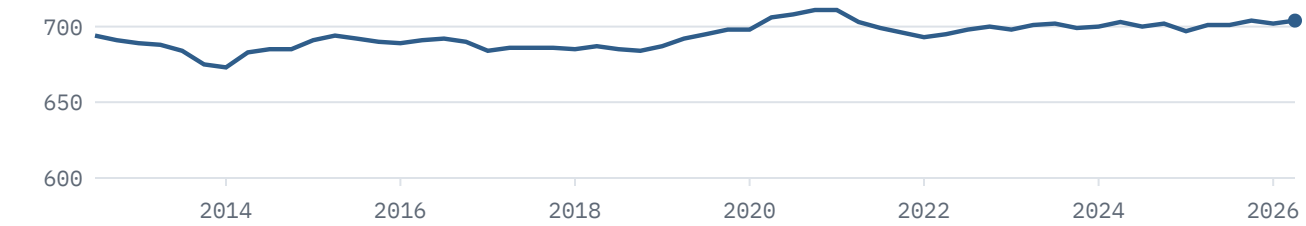
Back-end debt-to-income



Loan-to-value



Credit score, 10th percentile



Mortgage losses have been near zero since 2014, so these cannot be tested against losses here; they show how terms have moved. The survey's mortgage questions were redrawn in 2015, so its mortgage history is in two pieces.

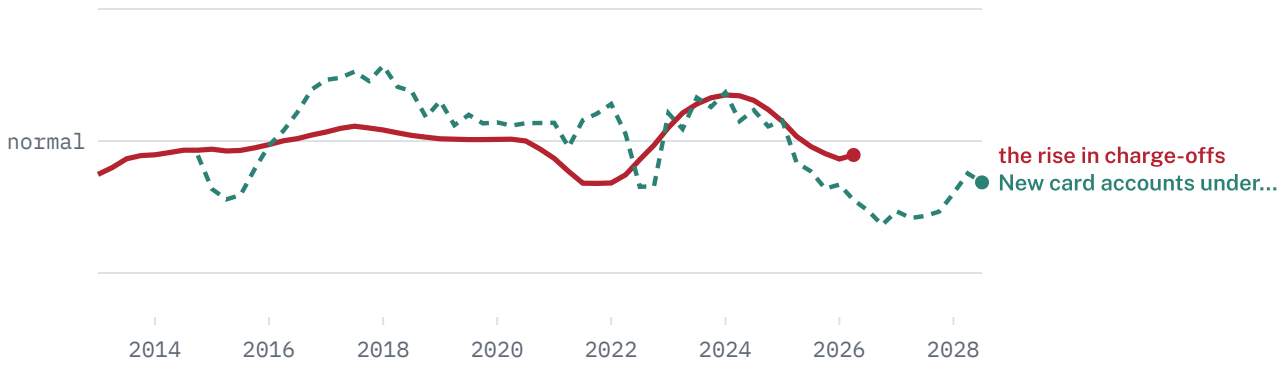
Read it this way: slide underwriting forward and lay it over losses

The red line is how much card charge-offs (or 30–89 days past due) rose over the past year. The dashed line is an underwriting measure, moved later by the delay where it matched best, so a loosening in 2015 is drawn in 2017. Both are drawn on their own scale around their own average (the middle line), because they are in different units: read the shape, not the height. Where the dashed line rises, the red one should rise with it.

New card accounts under 660, moved 9 quarters later, over the rise in charge-offs

rise in card charge-offs new accounts under 660, 9 quarters earlier

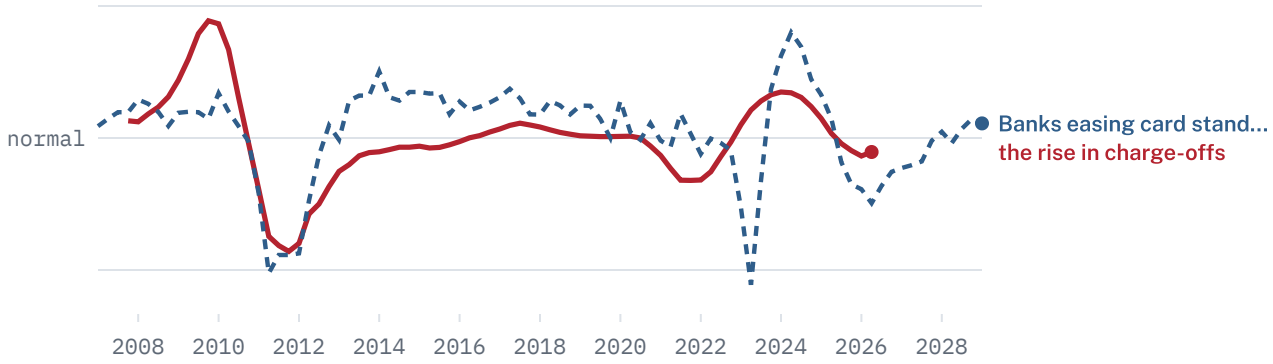
each line around its own normal



Banks easing card standards, moved 11 quarters later, over the rise in charge-offs

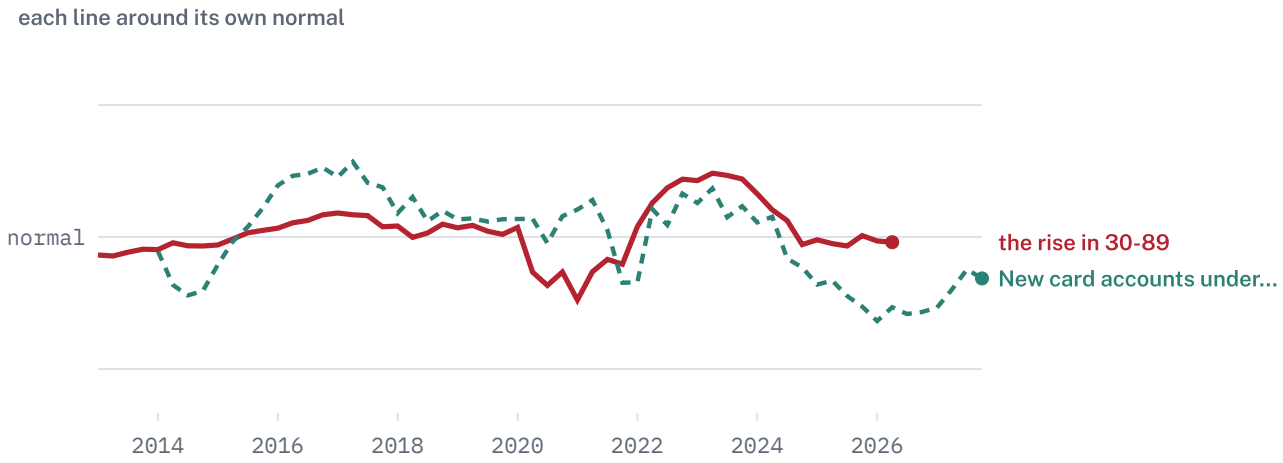
rise in card charge-offs banks easing, 11 quarters earlier

each line around its own normal



New card accounts under 660, moved 6 quarters later, over the rise in 30–89 days past due

rise in card 30–89 new accounts under 660, 6 quarters earlier

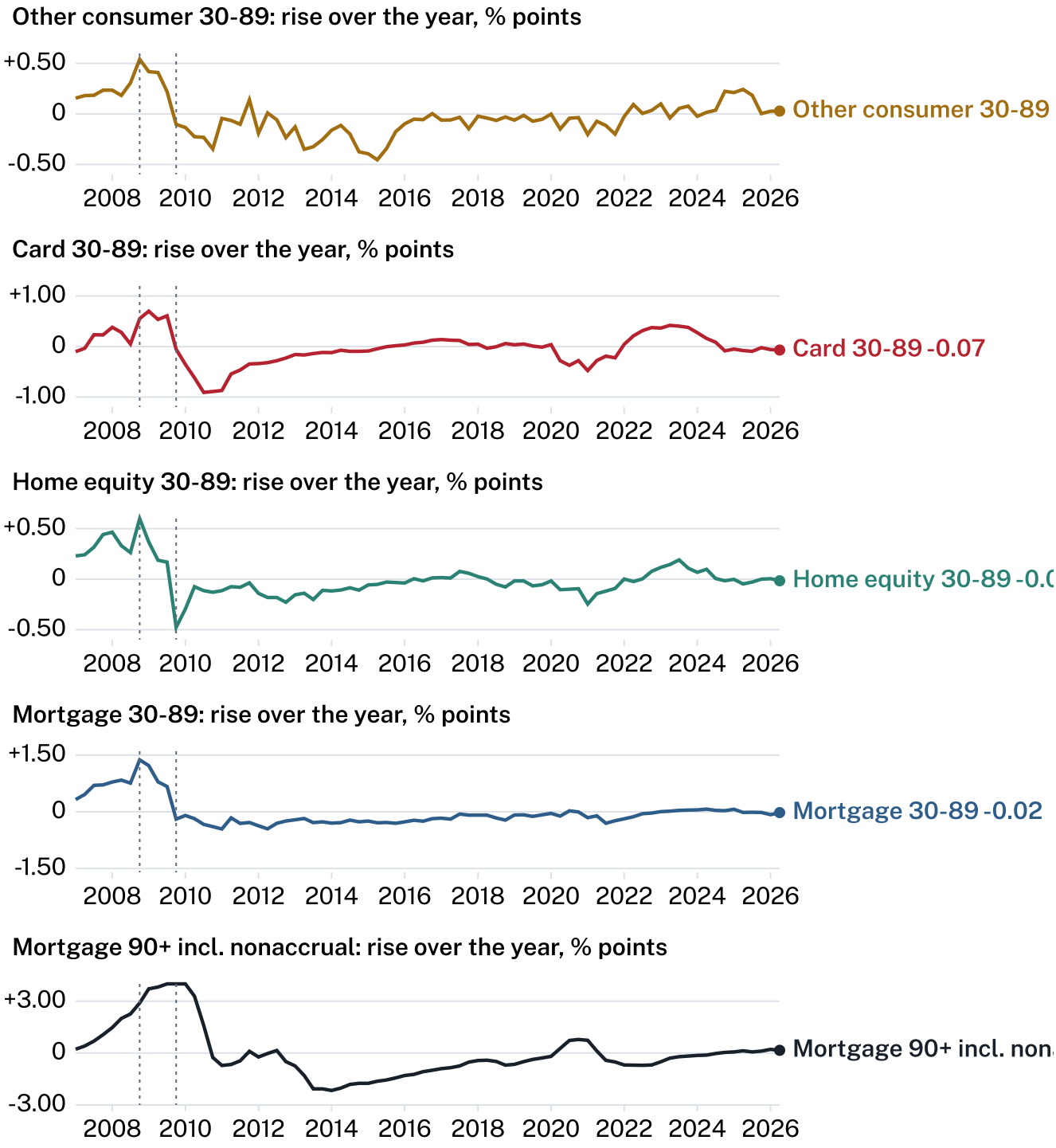


The dashed line runs past 2026 where it has already happened: that stretch is what the measure says is coming.

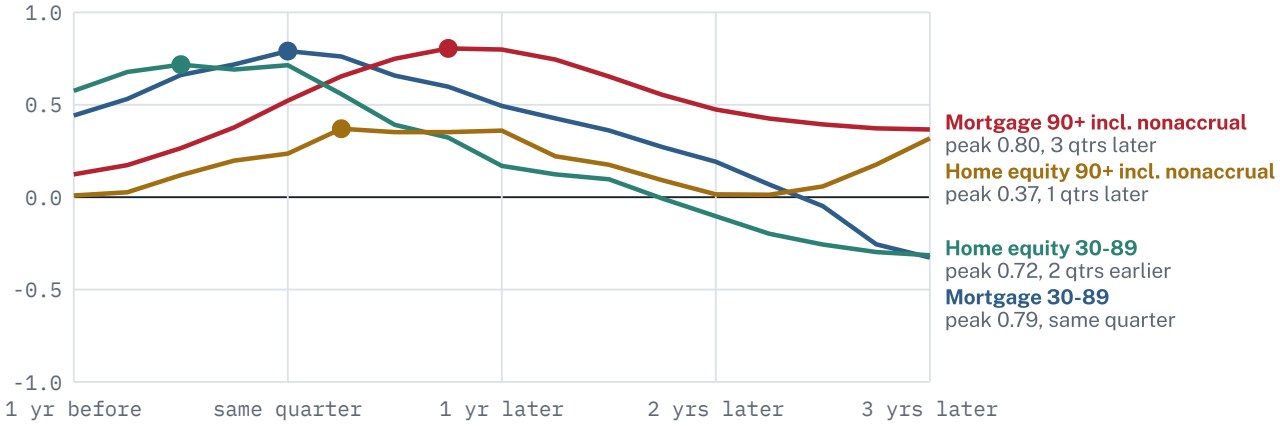
A6 · ACROSS PRODUCTS

Across products: does consumer delinquency come before mortgage?

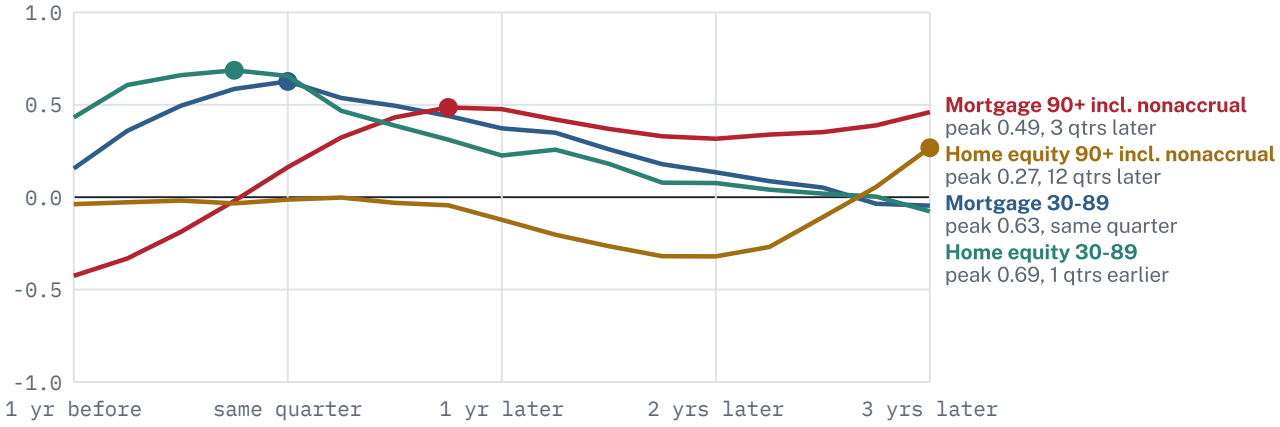
The book's industry series, every bank, 2007 to 2026, each drawn as its rise over the past year. Stacked on the same quarters so the turns can be compared; dotted lines mark 2008Q4 and 2009Q4.



How strongly a rise in other consumer 30–89 goes with a rise in each, that many quarters later



The same for card 30–89



Early delinquency rises across products in the same quarter, with home equity 30–89 about two quarters ahead of card and other consumer. Other consumer 30–89 comes about a year before mortgage 90+ and nonaccrual (0.80; 0.82 with 2020–2023 left out); card 30–89 less clearly (0.48). Mortgage's own 30–89 turns at the same time, so the other products are not earlier than mortgage's own early delinquency. Mortgage 90+ includes government-guaranteed loans that banks hold at no loss (the book's note); other consumer breaks in 2011 (auto split out) and 2024Q4 (Call Report revision).

B. What was set aside, and the evidence for that

Each of these was tested the same way. The charts show why it adds nothing the kept measures do not, or measures something else.

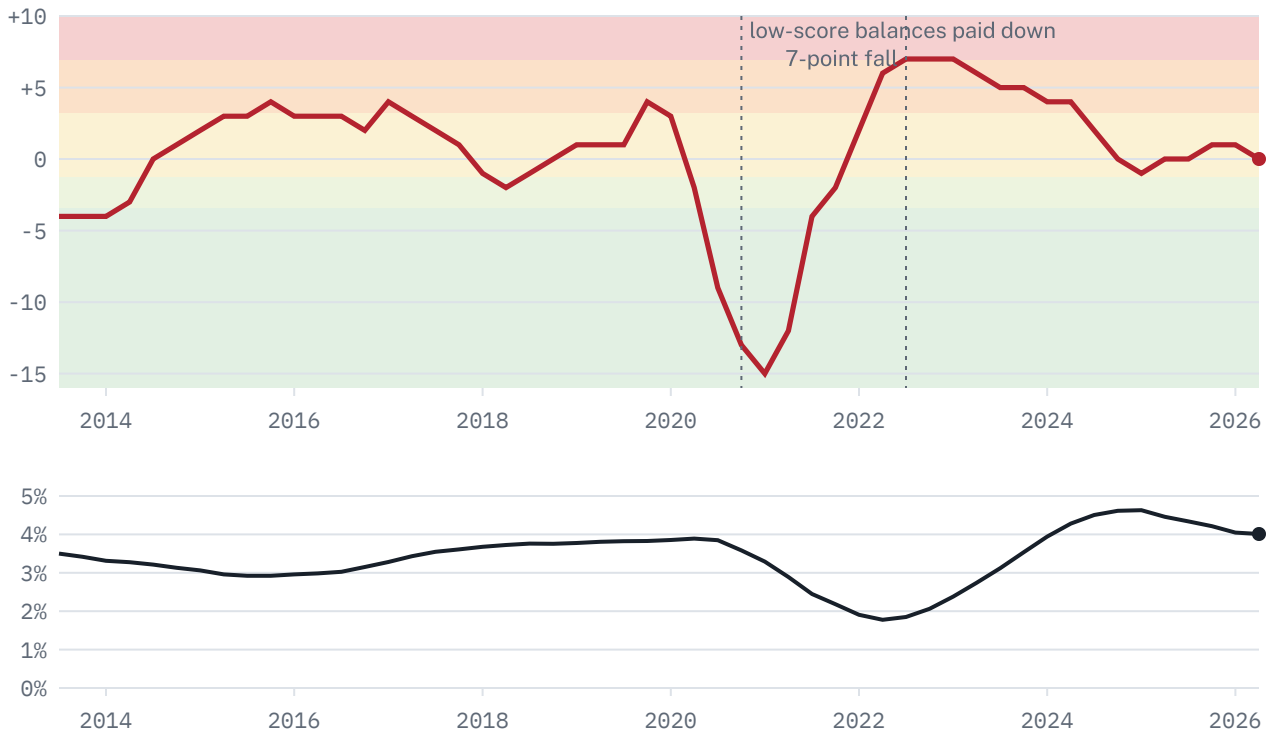
B1 · REFRESHED FICO

Change in FICO: the early-warning measure

The level of a score trends with lending policy, but its change over a year moved ahead of card losses. The series is the score of the weakest tenth of card accounts at the large banks, on outstandings, so it falls for either of two reasons the data cannot separate: existing cardholders' scores fall, or more low-score borrowers carry balances and so make up more of the accounts. Both lead to losses. Over 12 separate years, the large-bank card 10th percentile's change and the next year's change in industry card net charge-offs correlate at −0.91; measured in the same year, −0.36. Shown as a fall, so higher is worse, like the loss series.

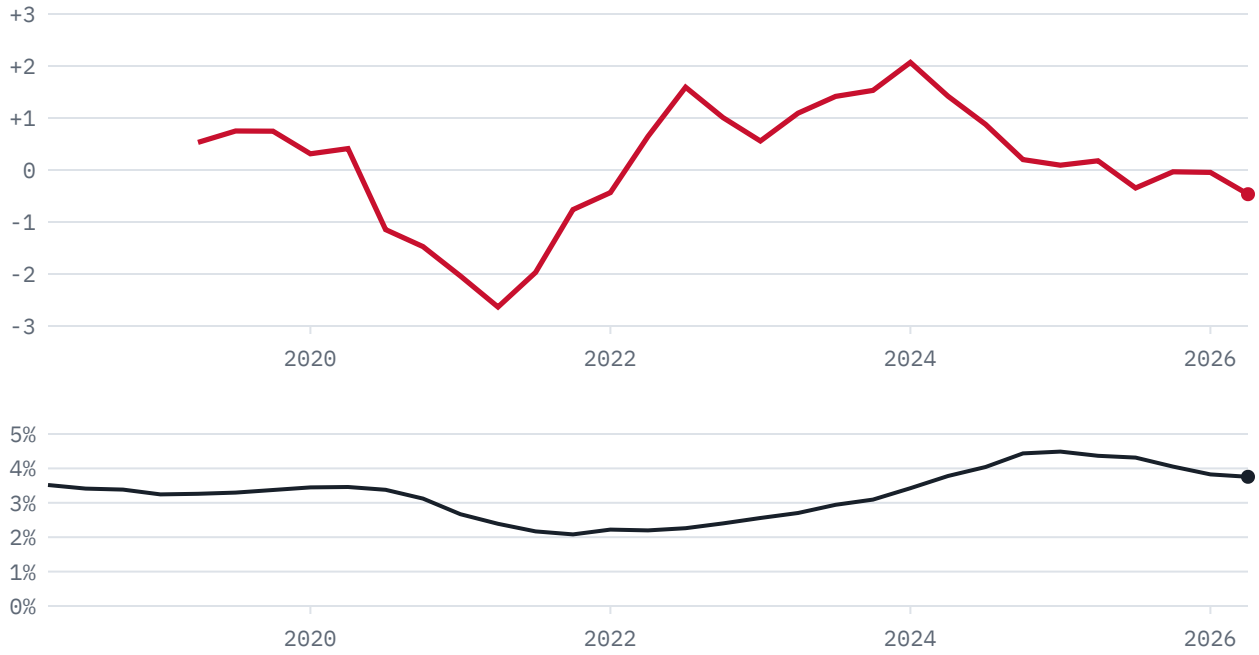
1 Low | 2 Moderate-Low | 3 Moderate | 4 Moderate-High | 5 High

Large banks, card: fall in the score of the weakest tenth of accounts over four quarters (points)



Zones from this history with width 0.5, Moderate-Low 1.0 and z 2.0, with no floor at zero (a change can be negative). At z 2.5 it is refused; it fits at z ≤ 2.02. High begins at a 6.9-point fall: 2022Q3, 2022Q4, 2023Q1, before industry card losses rose from 1.8% (2022Q3) to 4.6% (2025Q1). Below: industry card net charge-offs, same quarters.

KeyCorp, card: rise in share under 660 over four quarters (percentage points)

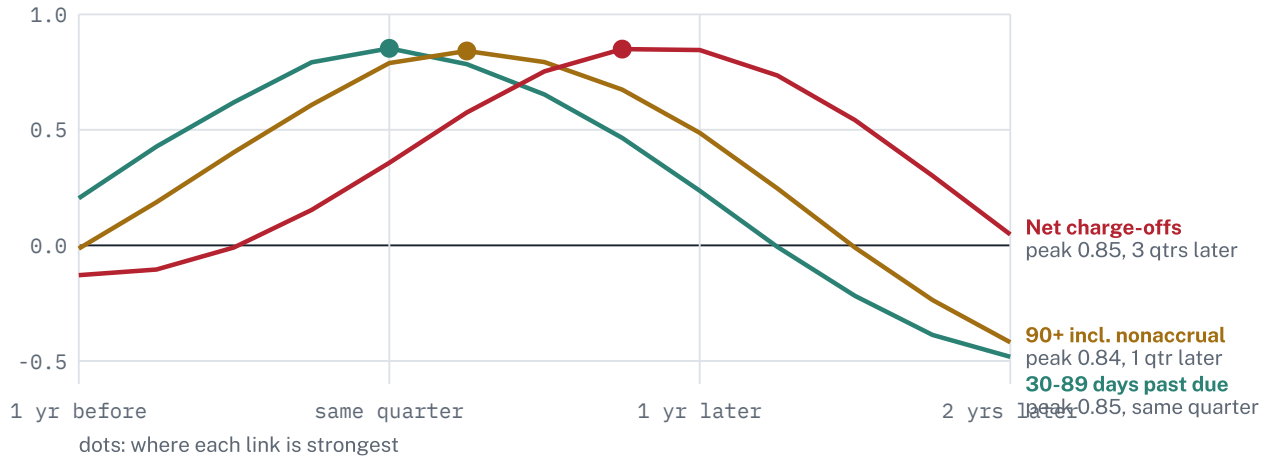


From KeyCorp's 10-Qs, 2019Q2 on (2021Q1 not read). Its change goes with KeyBank's card losses in the same year (correlation 0.67, 27 quarters) and the next (0.59, 23). Lines from its own history are refused at z 2.0; it fits at z ≤ 1.71. Below: KeyBank card net charge-offs.

When does the score move: before delinquency, or with it?

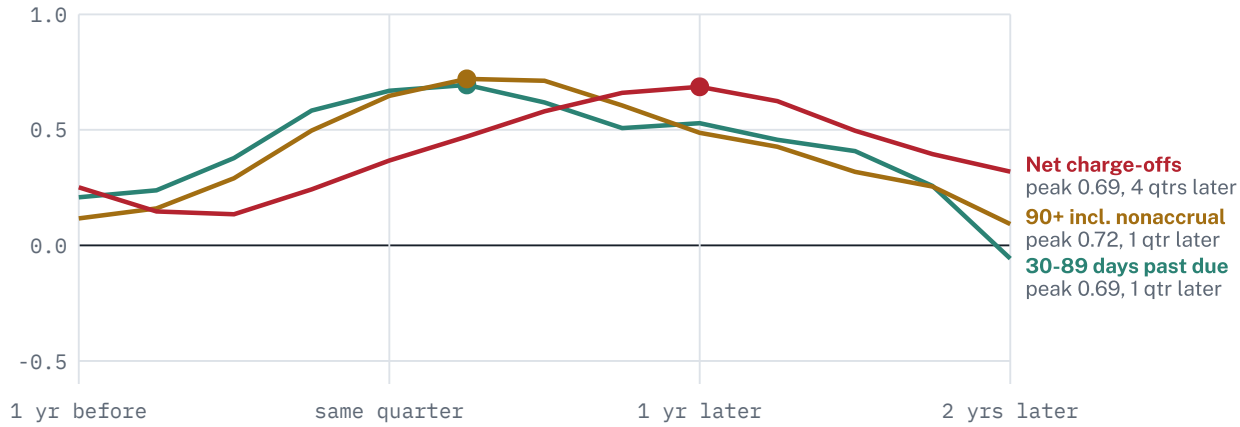
Scores fall when people run up their cards and when they go delinquent, so the score should move with delinquency, and charge-offs come later (cards charge off at 180 days past due). Each line is how strongly a fall in the weakest tenth's score over a year goes with a rise in that measure over a year, measured that many quarters later (1 means they move together exactly; 0, no link). All three are the book's industry card series.

All quarters



dots: where each link is strongest

2020–2023 left out

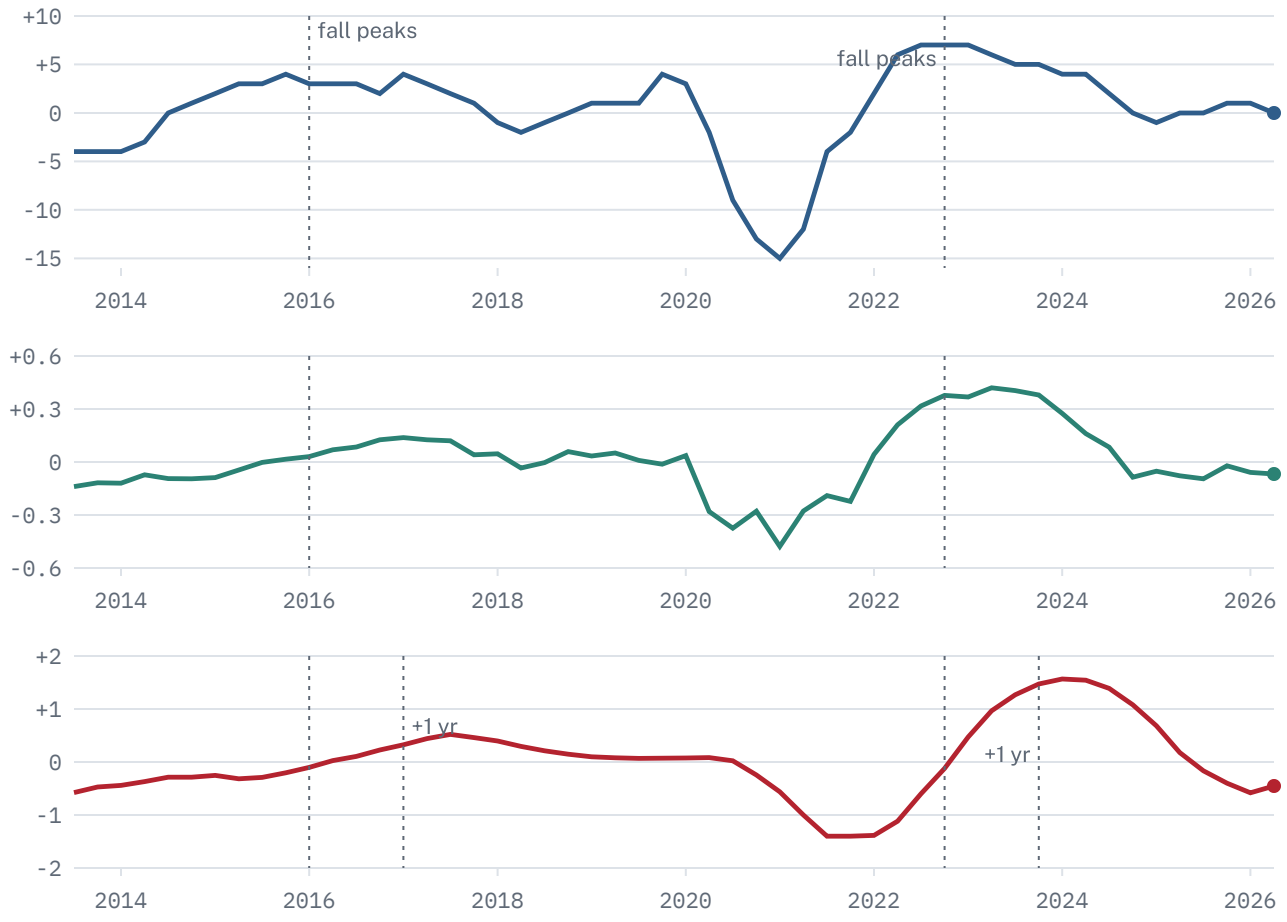


dots: where each link is strongest

The score moves within a quarter of 30–89 days past due, one to two quarters ahead of 90+ and nonaccrual, and about a year ahead of charge-offs. Its lead on charge-offs is the time delinquency takes to become a loss; against 30–89 it is not ahead.

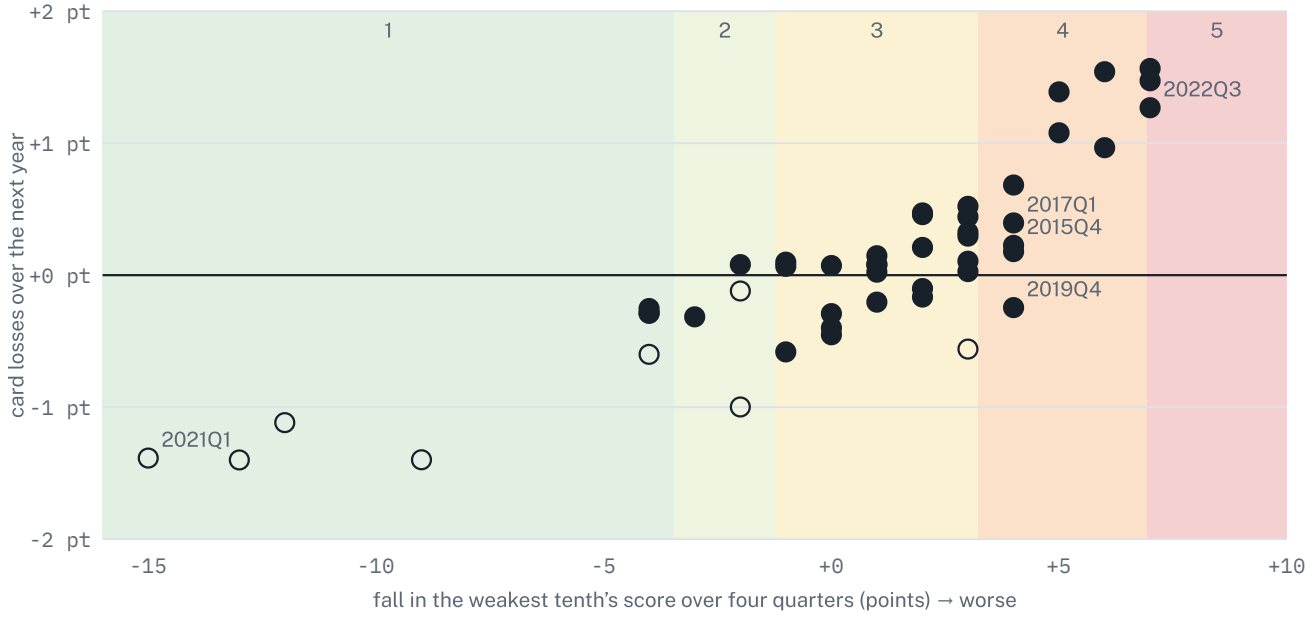
The same thing on one timeline

Top: how far the weakest tenth's score fell over the past year (points; up is worse). Middle: how much card 30–89 days past due rose over the past year (percentage points). Bottom: how much card net charge-offs rose over the past year. Same quarters on each. The dotted lines mark where the score's fall peaked; in the middle chart delinquency rises at the same time, and in the bottom chart charge-offs peak about a year later (the second pair of lines).

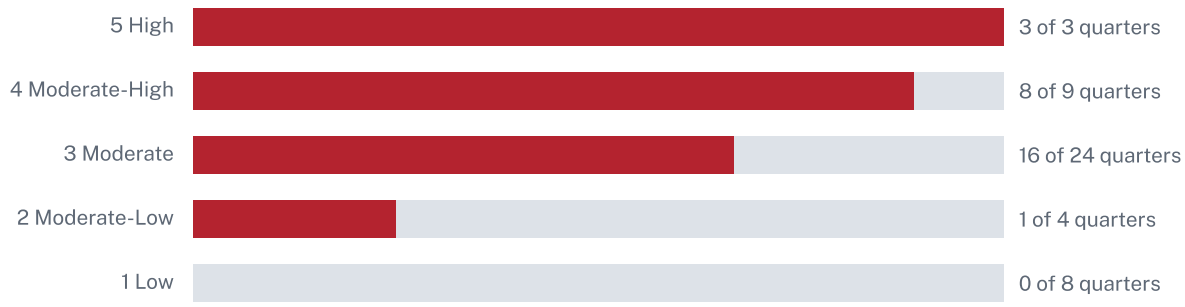


Should FICO change be a rating? The evidence, drawn

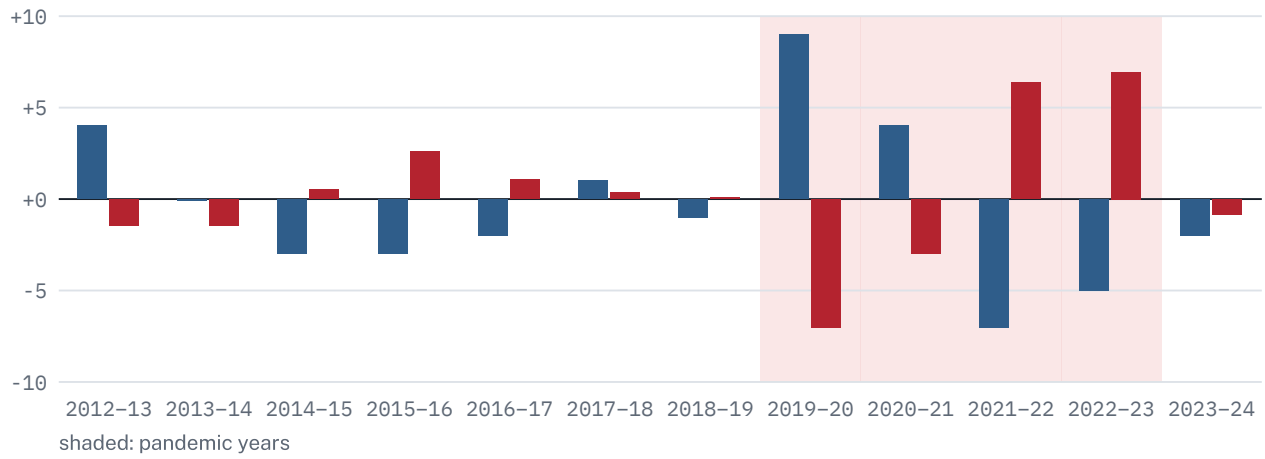
Each dot is a quarter: how far the weakest tenth's score fell over the past year (left to right, worse to the right), and what card losses did over the next year (up is worse). Background bands are the five ratings the book's method draws from this history (width 0.5, Moderate-Low 1.0, z 2.0, no floor). Hollow dots are 2020–21. If the measure works, dots in bands 4 and 5 sit above the line.



At each rating, how often card losses rose over the next year



Year by year: the score change (blue) and the next year's change in losses (red, ×5 to share the scale)

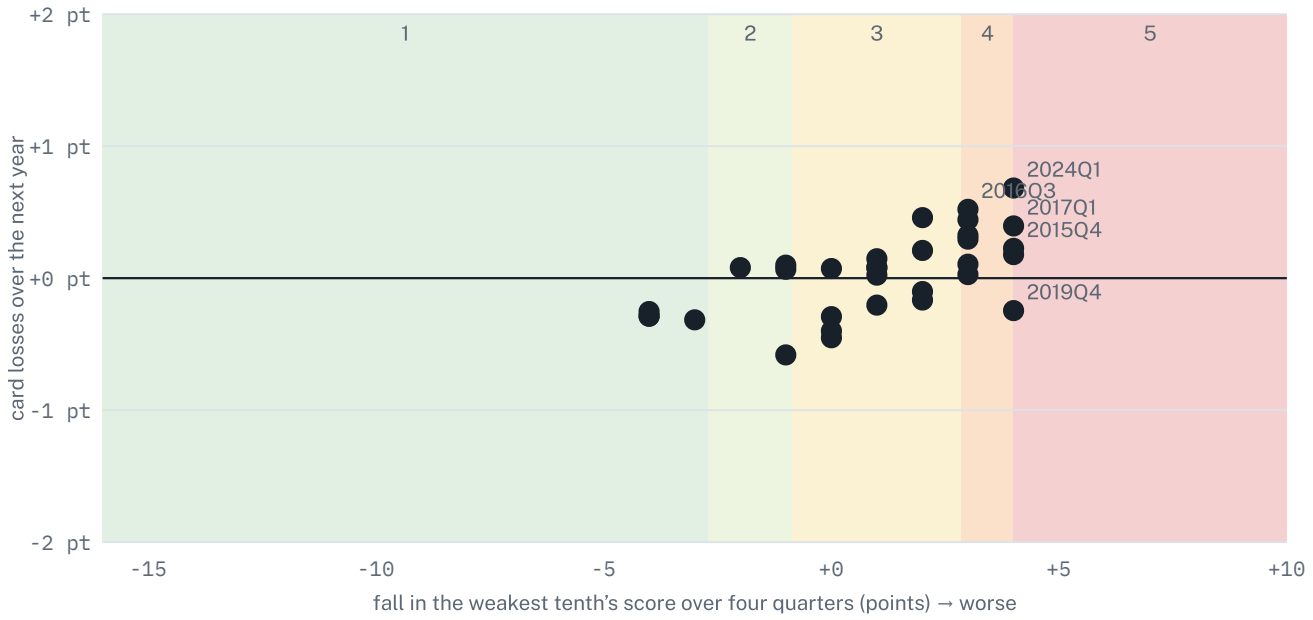


shaded: pandemic years

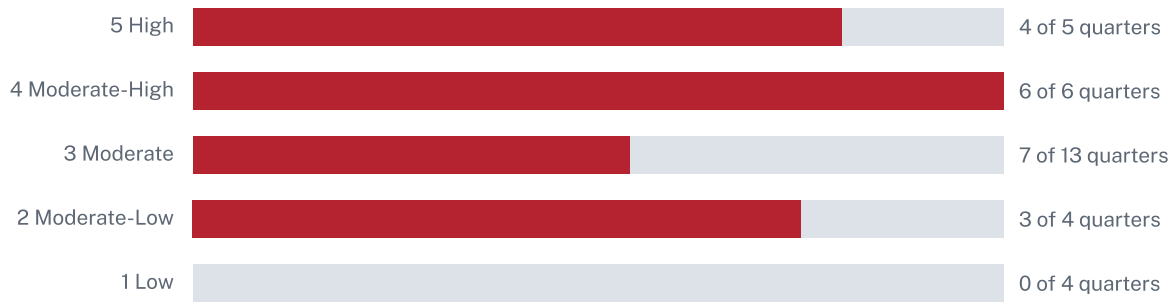
Read the blue bar against the red one: a falling score (blue below zero) should be followed by rising losses (red above). Outside the shaded years the moves are small: 2014–16 is the only clear one, a 3-point fall twice, then losses up 0.5. The lines here are drawn from the same history they rate, so the dots flatter the measure; a test that sets the lines only from earlier quarters is still to do.

The same test with 2020–2023 left out

The pandemic years raised the score and cut losses, then reversed both, so they are left out here as the book would leave out a period, with that reason. Lines are redrawn from the 36 quarters that remain. At z 2.0 the method refuses (nothing left reaches High); shown at z 1.01, the highest that fits. Quarters after 2023 are rated too.



At each rating, how often card losses rose over the next year (2020–2023 left out)



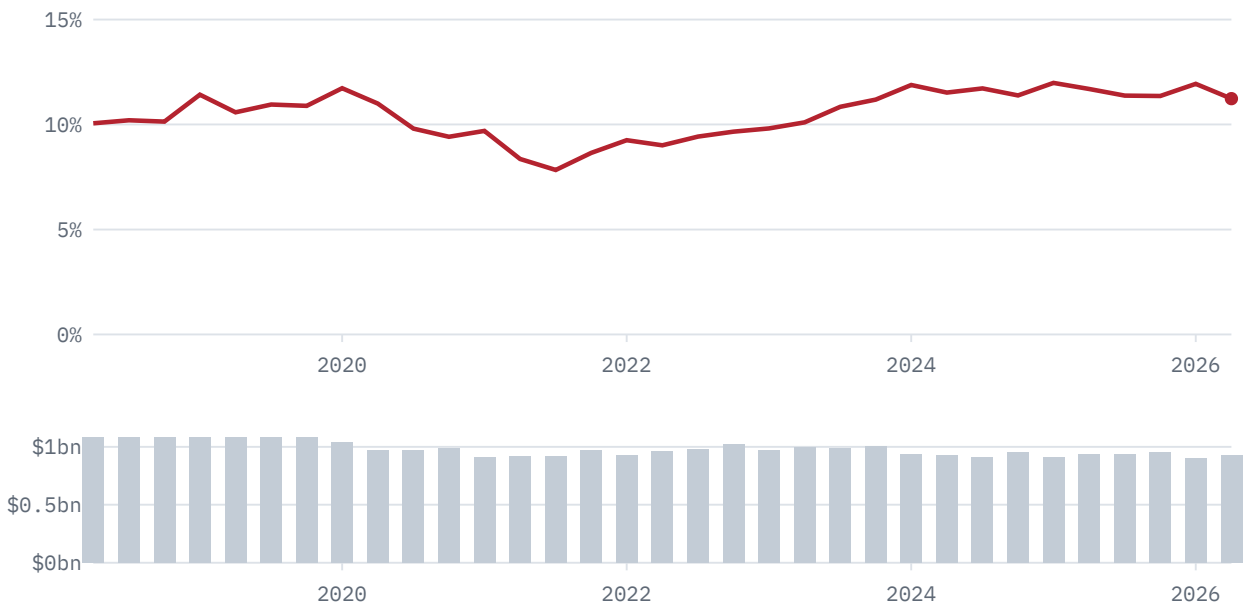
What is left is 32 quarters of small moves, 2013 to 2025: the score's fall ranged from -4 to +4 points and losses moved by at most 0.68 points a year.

KeyCorp, refreshed FICO by product, from its own 10-Qs and 10-Ks

2018Q2 to 2026Q2 from 33 filings (2021Q1 read from the printed table: that filing has no tagged data). Share of scored balances with a refreshed score under 660, with the rating zones the book's method draws from each history, and the scored balance beneath.

1 Low | 2 Moderate-Low | 3 Moderate | 4 Moderate-High | 5 High

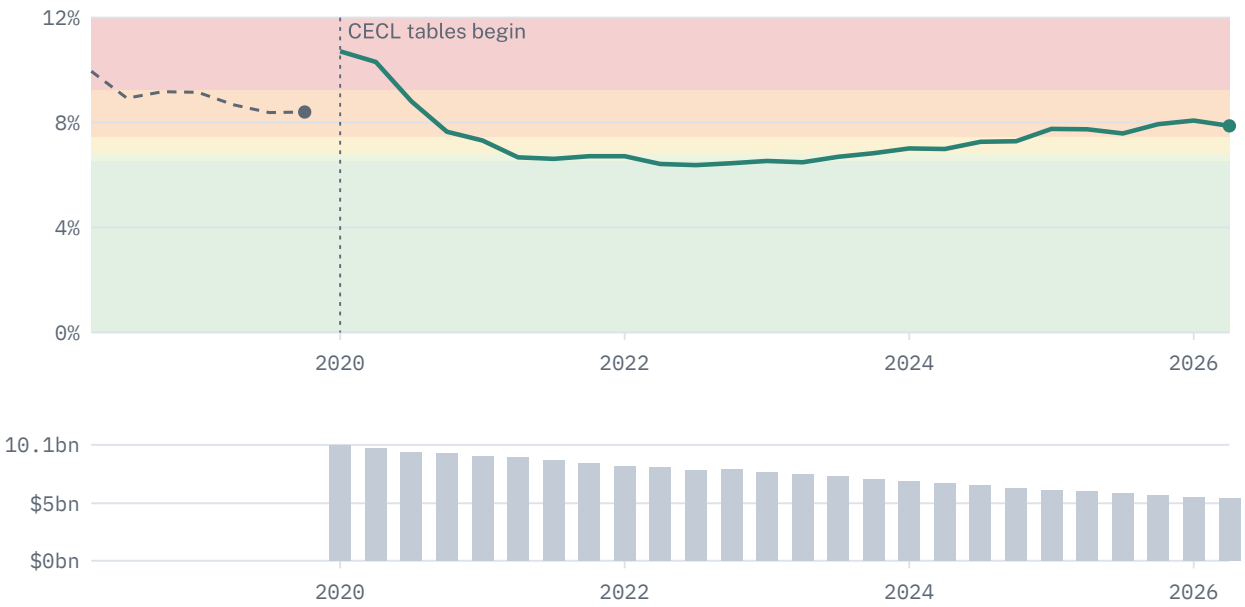
Card



Share of scored balances with a refreshed FICO under 660, 2018Q2 to 2026Q2 (top; shading is the five rating zones the book's method draws from this history at the agreed settings); scored balance (bottom).

Refused at z 2.5 over its 33 quarters: no quarter reaches High. Fits at z ≤ 0.90. Latest 11.2%.

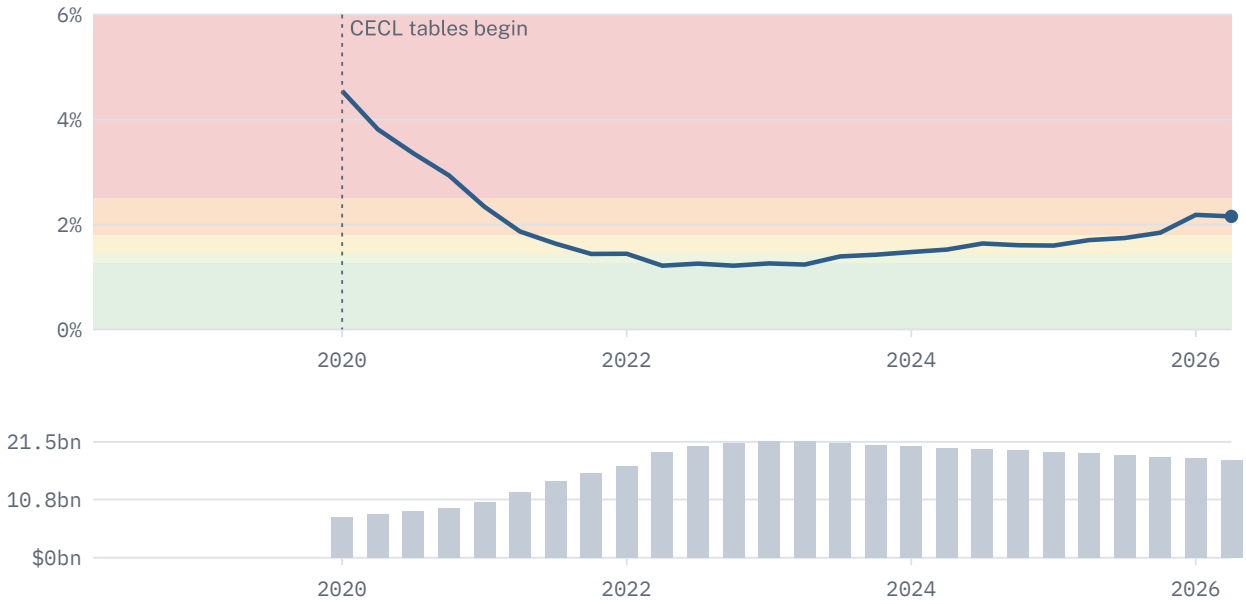
Home equity



Share of scored balances with a refreshed FICO under 660, 2020Q1 to 2026Q2 (top; shading is the five rating zones the book's method draws from this history at the agreed settings); scored balance (bottom). The dashed gray line before 2020 is KeyCorp's "Residential, prime" class, mortgage and home equity together, as it reported them then.

Lines from its own 26 quarters: 2 begins 6.53%, 3 6.84%, 4 7.45%, **5 High 9.23%**. Latest 7.9%.

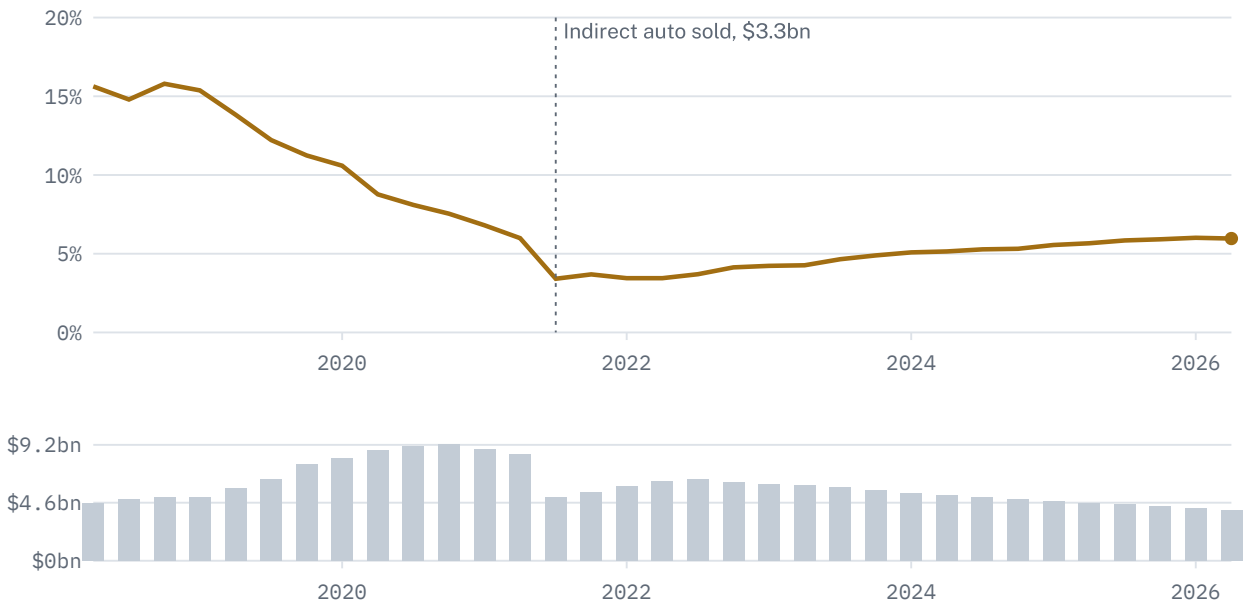
Mortgage



Share of scored balances with a refreshed FICO under 660, 2020Q1 to 2026Q2 (top; shading is the five rating zones the book's method draws from this history at the agreed settings); scored balance (bottom).

Lines from its own 26 quarters: 2 begins 1.27%, 3 1.45%, 4 1.79%, **5 High 2.50%**. Latest 2.2%.

Other consumer



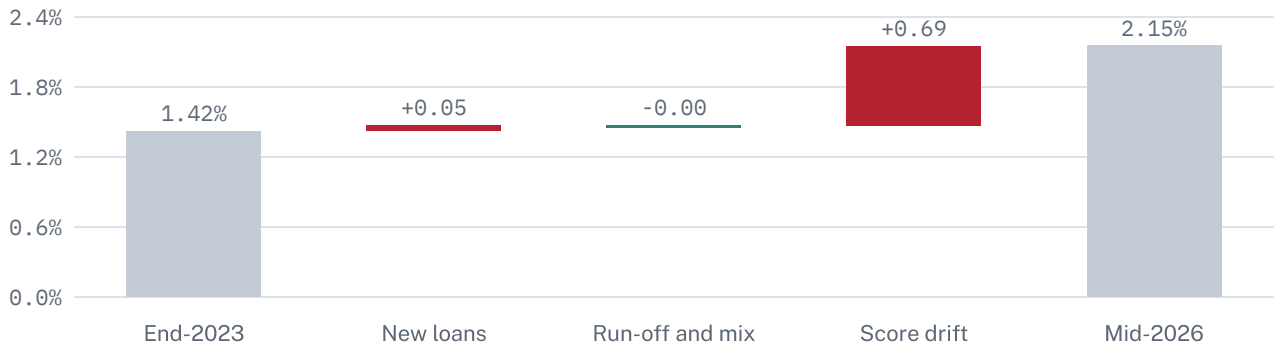
Share of scored balances with a refreshed FICO under 660, 2018Q2 to 2026Q2 (top; shading is the five rating zones the book's method draws from this history at the agreed settings); scored balance (bottom).

Refused at z 2.5 over its 19 quarters: no quarter reaches High. Fits at z ≤ 0.76. Latest 6.0%. Lines from 2021Q4 on: the quarters before the 2021Q3 sale are left out, as the book's jump rule would leave them (scored balance $8.5bn to $5.0bn, -41%).

Why KeyCorp's share under 660 rose: the waterfall

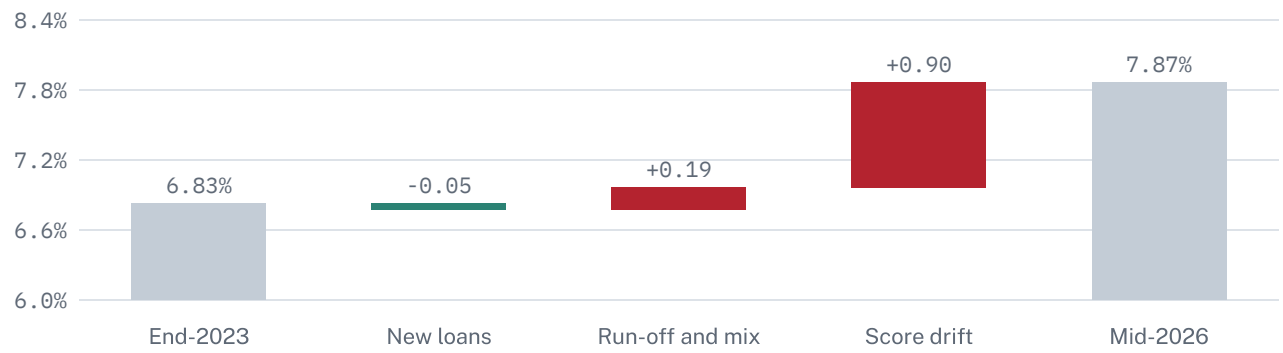
Since 2020 each bank's FICO table is split by the year the loans were made, which separates new loans, run-off and mix, and the same loans' scores moving. KeyCorp, end-2023 to mid-2026.

Mortgage



1.42% to 2.15% under 660; scored balance $20.9bn to $18.2bn.

Home equity

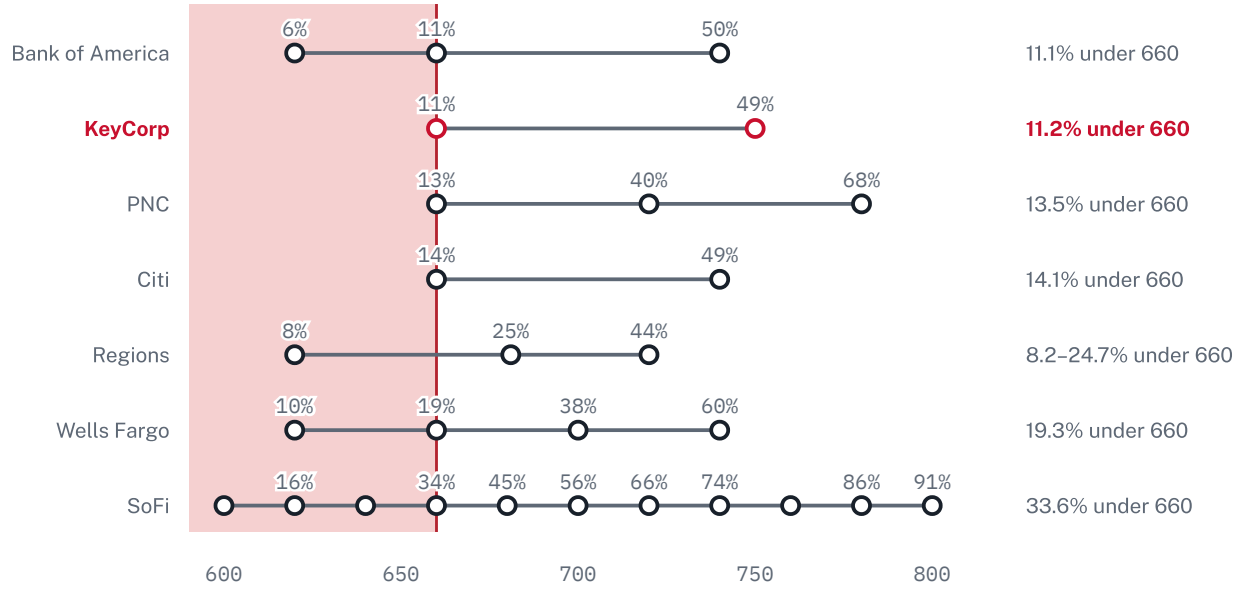


6.83% to 7.87% under 660; scored balance $7.1bn to $5.4bn.

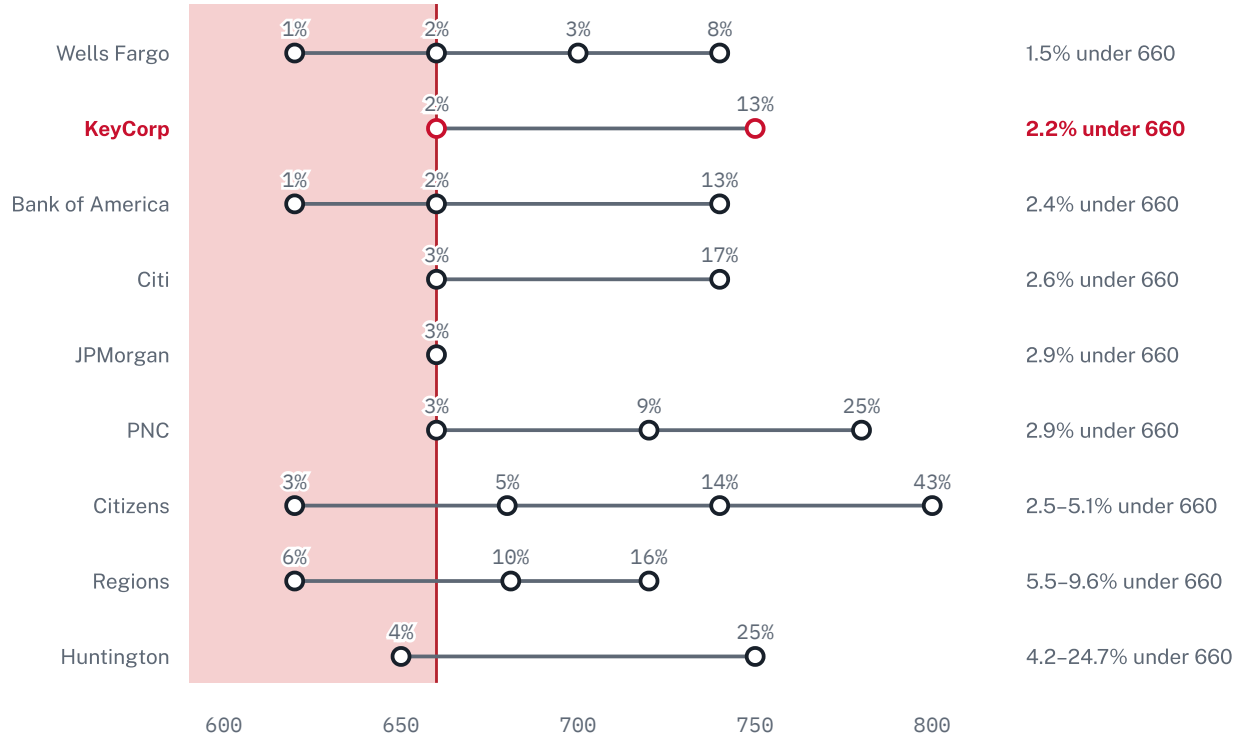
Competitors on the same ruler

Each bank's 10-Q prints its consumer loans in FICO bands, cut wherever that bank chooses. Each dot is one of its cuts, with the share of its scored balances under it. Where a bank cuts at 660 the share under 660 is exact; where it does not, the share must lie between its nearest cuts, and that proven range is what is shown. KeyCorp in red.

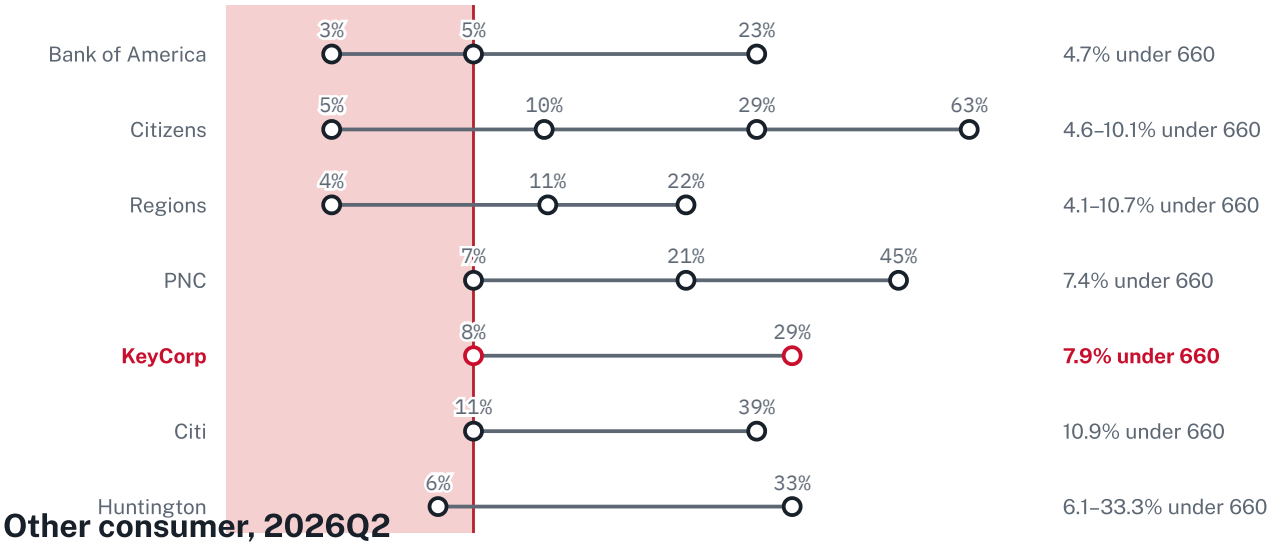
Card, 2026Q2



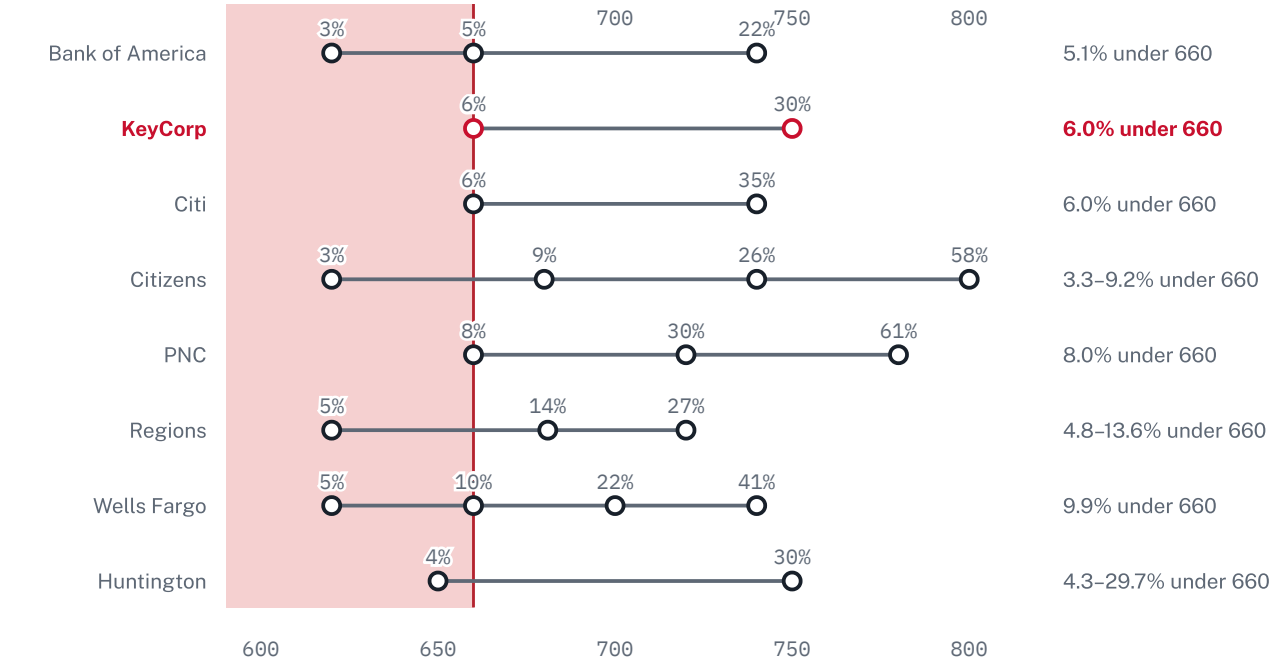
Mortgage, 2026Q2



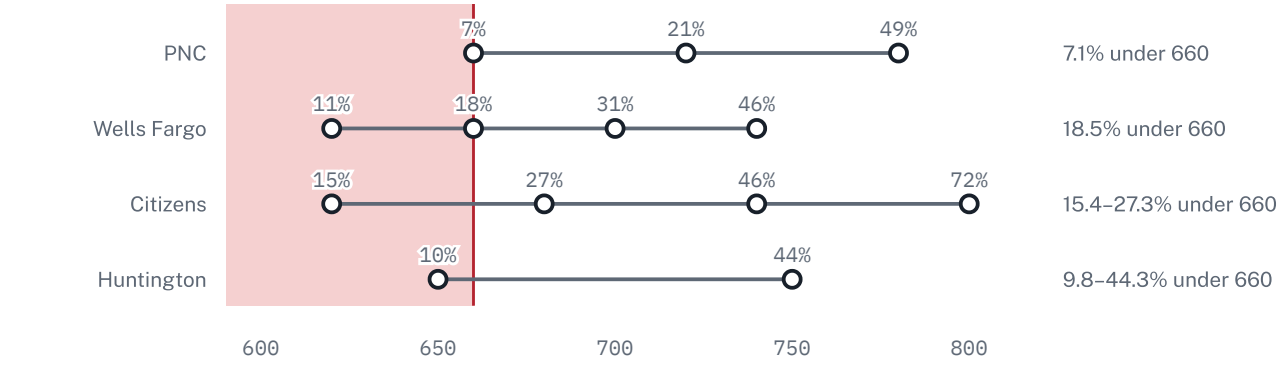
Home equity, 2026Q2



Other consumer, 2026Q2



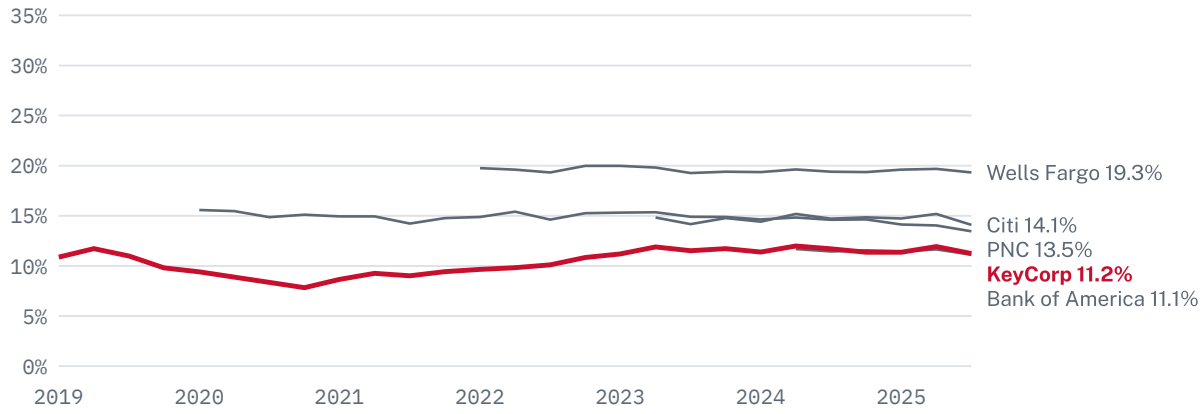
Auto, 2026Q2



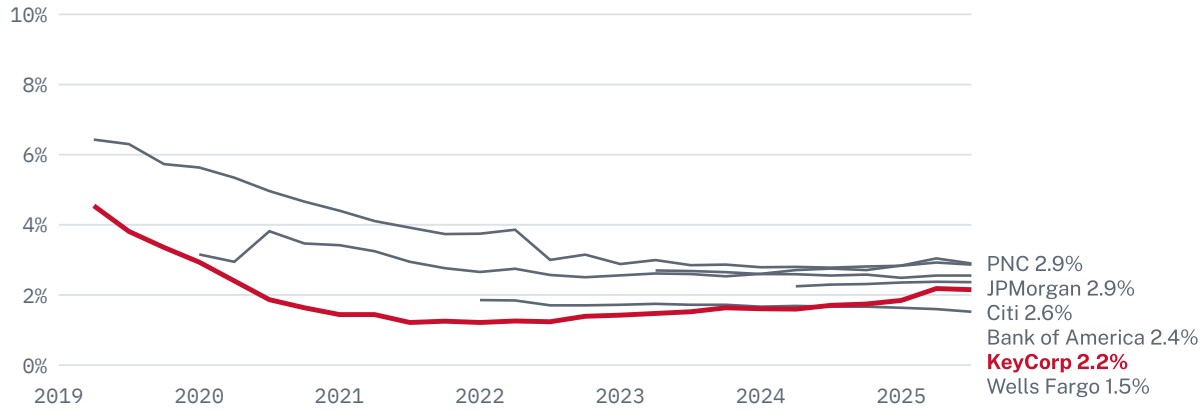
Share under 660 since 2019

Only exact figures: a bank appears in the quarters it cut at 660. Banks on other cuts, and quarters before a bank moved to 660, are ranges, read on the rulers above. KeyCorp in red.

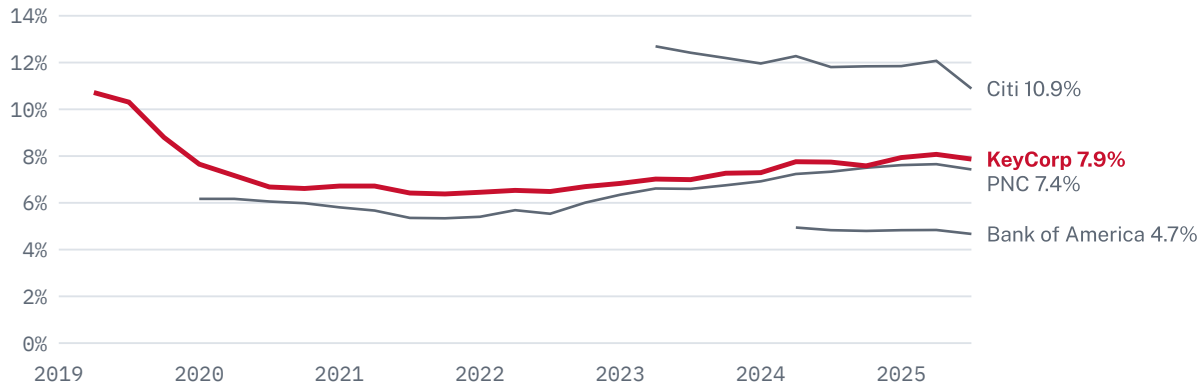
Card



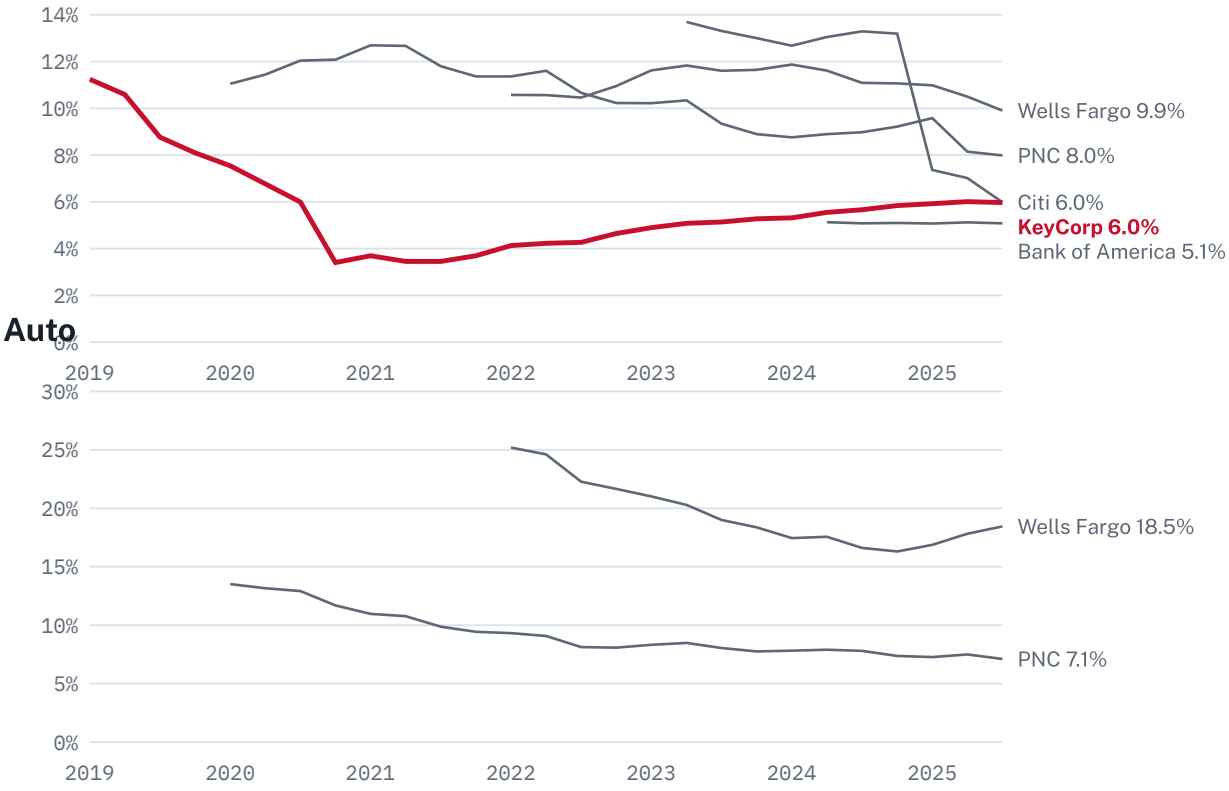
Mortgage



Home equity



Other consumer


Auto
14%
12%
10%
8%
6%
4%
2%
0%
2019
2020
2021
2022
2023
2024
2025
Wells Fargo 9.9%
PNC 8.0%
Citi 6.0%
KeyCorp 6.0%
Bank of America 5.1%
30%
25%
20%
15%
10%
5%
0%
Wells Fargo 18.5%
PNC 7.1%

What was inferred, and how

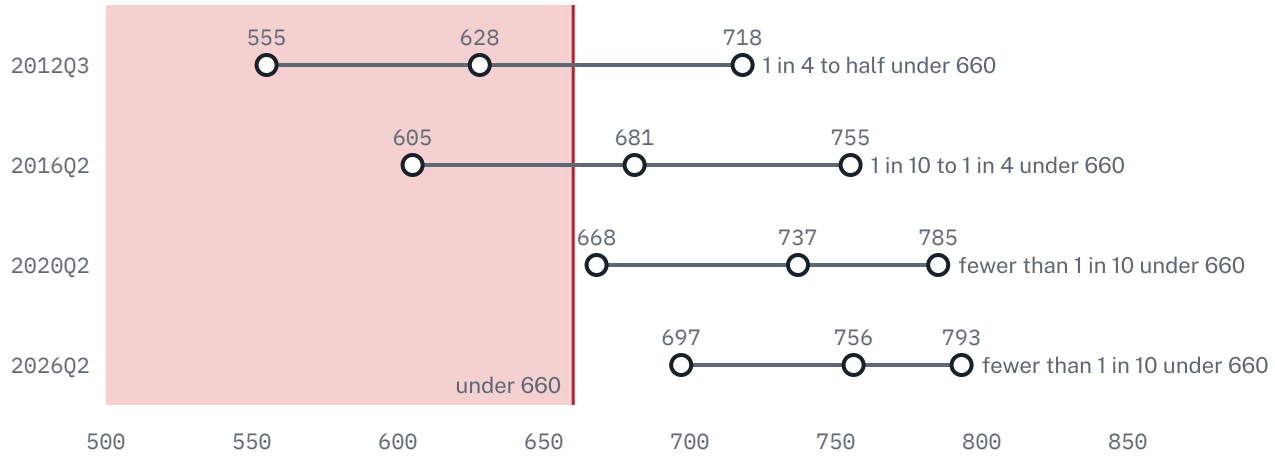
- **Every bank.** Under 660 is exact where the bank publishes a 660 edge. Elsewhere it is a range between the bank's nearest published edges below and above 660, which the filing proves. Balances with no score are left out of every share, with their dollars stated.
- **Band edges.** Each band's top is read as the lower of its own stated top plus one and the next band's bottom: KeyCorp's tag says 'greater than 750' where its table prints '750 and above'.
- **Bank of America.** Home equity is its total home equity class (lines, loans converted to term and reverse mortgages are parts of it); before 2022Q4 that total is tagged with no class member, and equals its parts ($30.5bn at 2021Q2). Other consumer is 'direct and indirect', which includes auto loans; most of it, securities-based lending, has no score.
- **Wells Fargo.** Mortgage excludes government-insured loans. Until 2022 it split first and second mortgages and cut every 40 points (600 to 800); home equity is its second mortgages until then, after which they sit inside residential mortgage and home equity has no Wells figure. Other consumer is mostly securities-based lending with no score; the share is of the scored part.
- **Citi.** U.S. classes only. Card is its 'credit card receivables' class; it also tags a slightly smaller 'credit card' class, not used. Until 2024 it cut at 680 and 760 only, so its share under 660 there is known only to be at most its share under 680.
- **JPMorgan.** Residential real estate, which also holds home equity, summed across its loan-to-value splits; it publishes only the 660 edge.
- **Regions.** Home equity is its lines and its loans added together. Edges 620, 680 and 720.
- **Citizens.** Other consumer is its other retail loans plus student loans, as the Call Report counts them. Edges 620, 680, 740, 800.
- **Huntington.** Other consumer is RV and marine plus other consumer. Edges 650 and 750, so under 660 is a wide range. Whether its scores are refreshed is not stated in the filing read.
- **PNC.** Edges 660, 720, 780. Card, auto and other consumer are its 'using FICO' rows.
- **SoFi.** Card only, held at fair value; fine bands every 20 points.
- **Capital One.** Auto only, and scored at origination, not refreshed: shown apart, never in a peer range.
- **KeyCorp.** Before 2020, purchased credit-impaired and other loans added together; its 'consumer direct' and 'consumer indirect' classes are other consumer until 2023, when they became one class.
- **Cuts that changed.** A bank that moved its cuts is exact on one side of the change and a proven range on the other: Bank of America card from 2025Q1 (620/680/740 to 620/660/740); Wells Fargo card from 2022Q4 (600/640/680/720/760/800 to 620/660/700/740); PNC card from 2020Q4 (620/650/720 to 660/720/780); Citi card from 2024Q1 (680/761 to 660/740); SoFi card from 2022Q4 (640/660/680/700/720/740/760/780/800 to 600/620/640/660/680/700/720/740/760/780/800); Bank of America other consumer from 2025Q1 (620/680/740 to 620/660/740); Wells Fargo other consumer from 2022Q4 (600/640/680/720/760/800 to 620/660/700/740); PNC other consumer from 2020Q4 (620/650/720 to 660/720/780); Citi other consumer from 2024Q1 (680/761 to 660/740); Bank of America mortgage from 2025Q1 (620/680/740 to 620/660/740); Wells Fargo mortgage from 2022Q4 (600/640/680/720/760/800 to 620/660/700/740); PNC mortgage from 2020Q4 (661 to 660/720/780); Citi mortgage from 2024Q1 (680/761 to 660/740); Bank of America home equity from 2025Q1 (620/680/740 to 620/660/740); PNC home equity from 2020Q4 (661 to 660/720/780); Citi home equity from 2024Q1

(680/761 to 660/740); Wells Fargo auto from 2022Q4 (600/640/680/720/760/800 to 620/660/700/740); PNC auto from 2020Q4 (620/650/720 to 660/720/780).

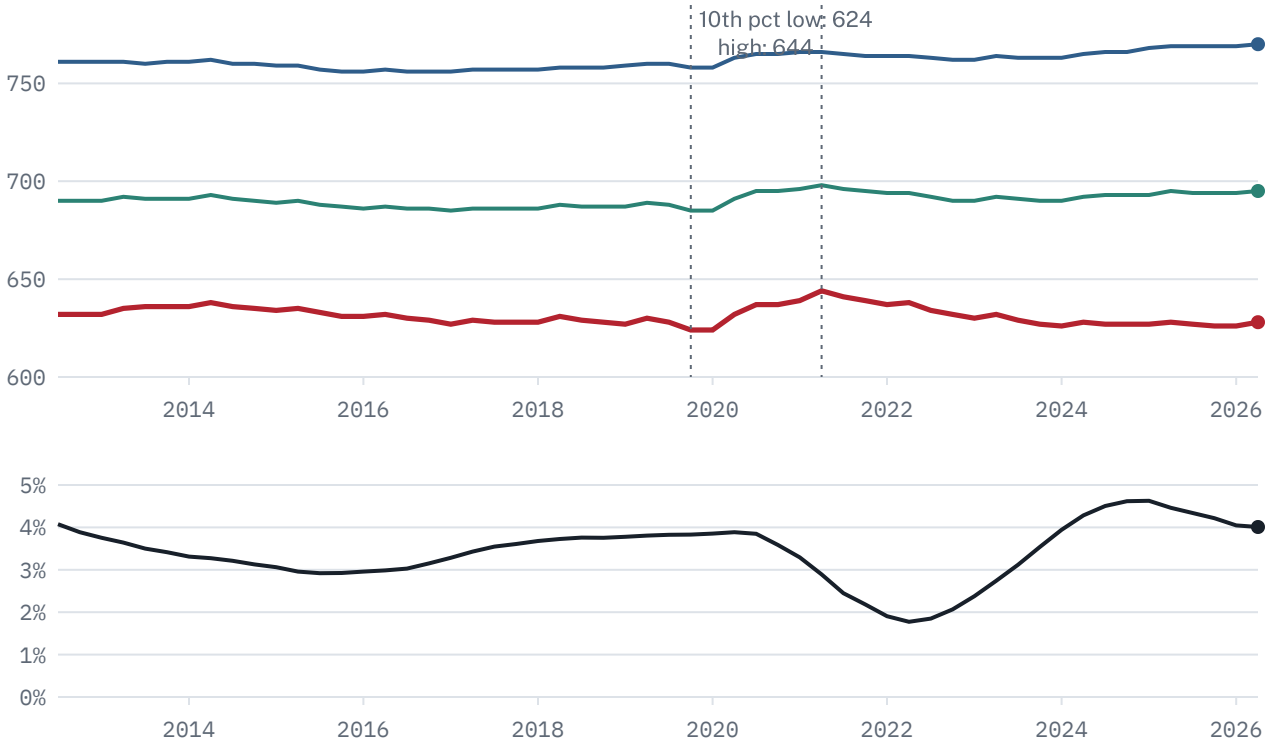
- **SoFi** is on the card ruler but not the history: its book is $0.1bn to $0.5bn, held at fair value, and its early quarters jump from 3% to 30% under 660.

Large-bank card and mortgage scores (Philadelphia Fed), as a ruler and over time

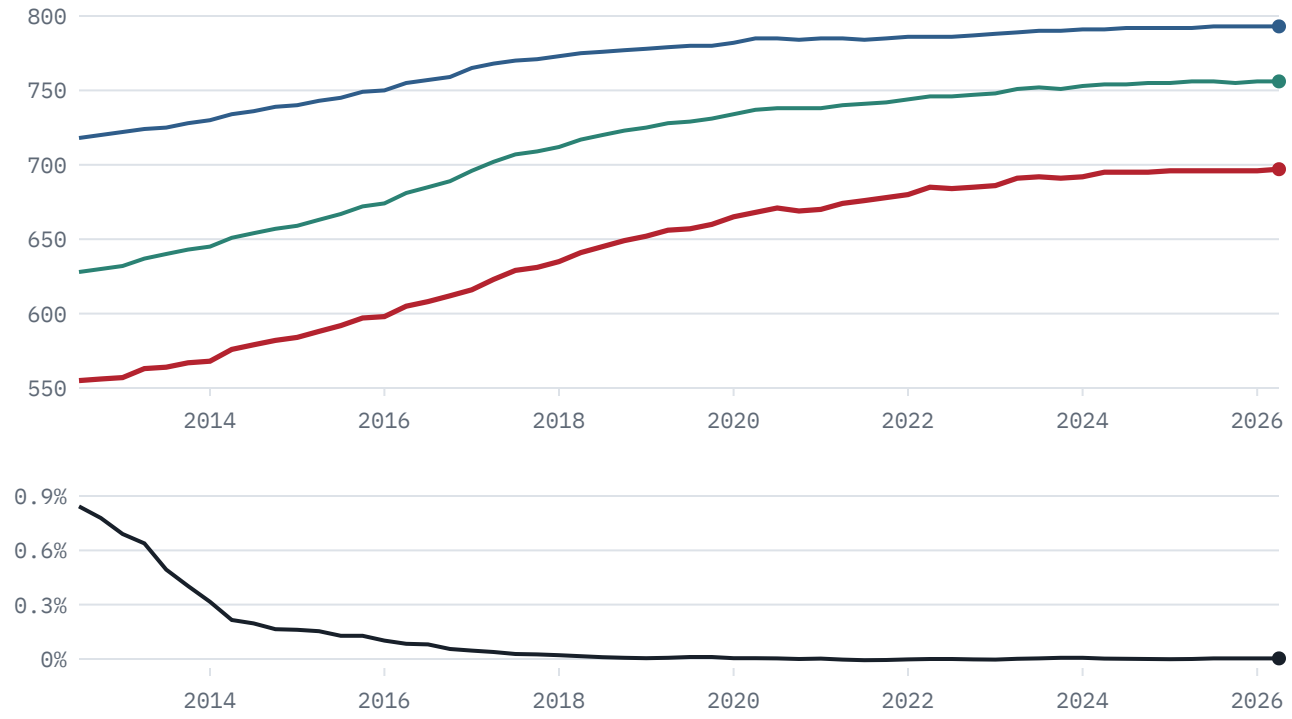
Read the ruler: line up every account from lowest score to highest; the 10th percentile is the score one in ten accounts fall at or below, the 25th one in four, the median half. Where a dot sits in the red zone, at least that share scores under 660.



Card accounts, current score percentiles, and industry card net charge-offs



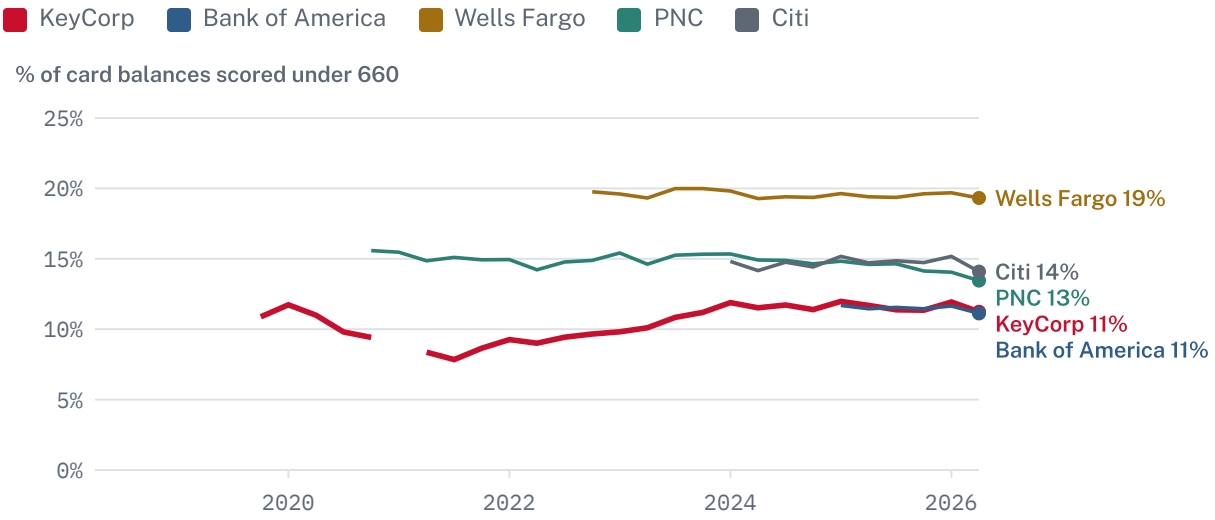
First-lien mortgage accounts, current score percentiles, and industry mortgage net charge-offs



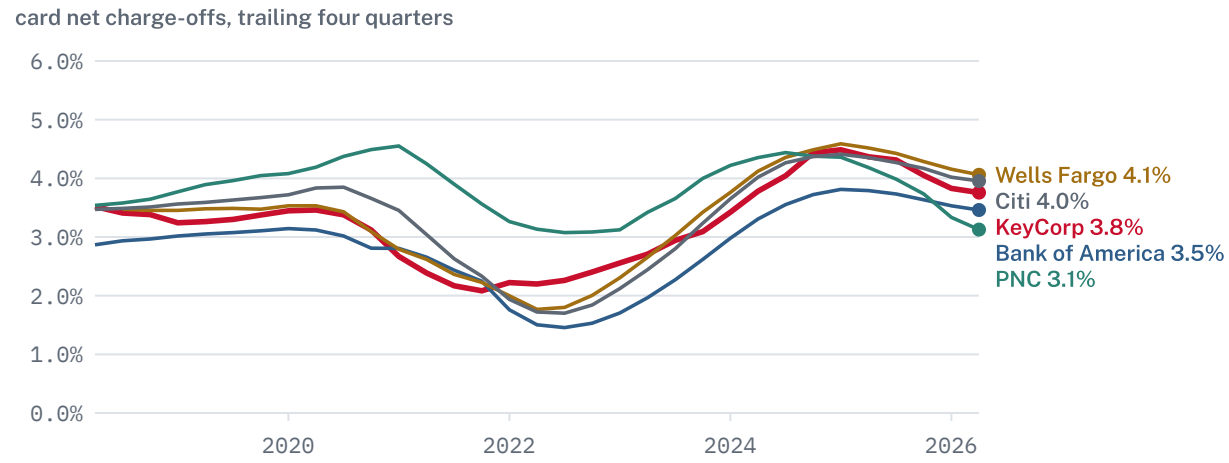
Card: refused at z 2.5 at every percentile; nothing since 2012 is far enough from normal. Mortgage: the 10th percentile rose from 555 to 697 and never fell back, so lines from it rate a trend, not a cycle.

B2 · CARD BALANCES UNDER 660 AS A RATING

Card: balances under 660, KeyBank against peers



Card net charge-offs, same banks and quarters (trailing four quarters, each bank's Call Report)



Each bank-quarter: balances under 660 against net charge-offs

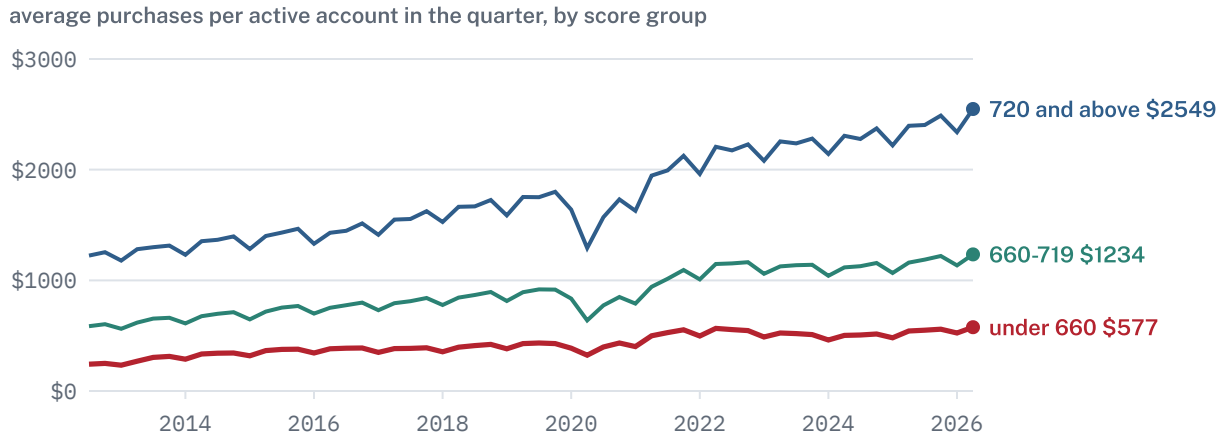


From each bank’s 10-Q: the share of its scored card balances with a refreshed score under 660, drawn only for the quarters the bank cuts at 660 (KeyCorp from 2018, PNC from 2020Q4, Wells Fargo from 2022Q4, Citi from 2024Q1, Bank of America from 2025Q1). No public series gives this for the whole industry. KeyCorp now 11.2%; its low was 7.8% (2021Q3). Its yearly rise went with KeyBank’s card losses the same year (0.67) and the next (0.59).

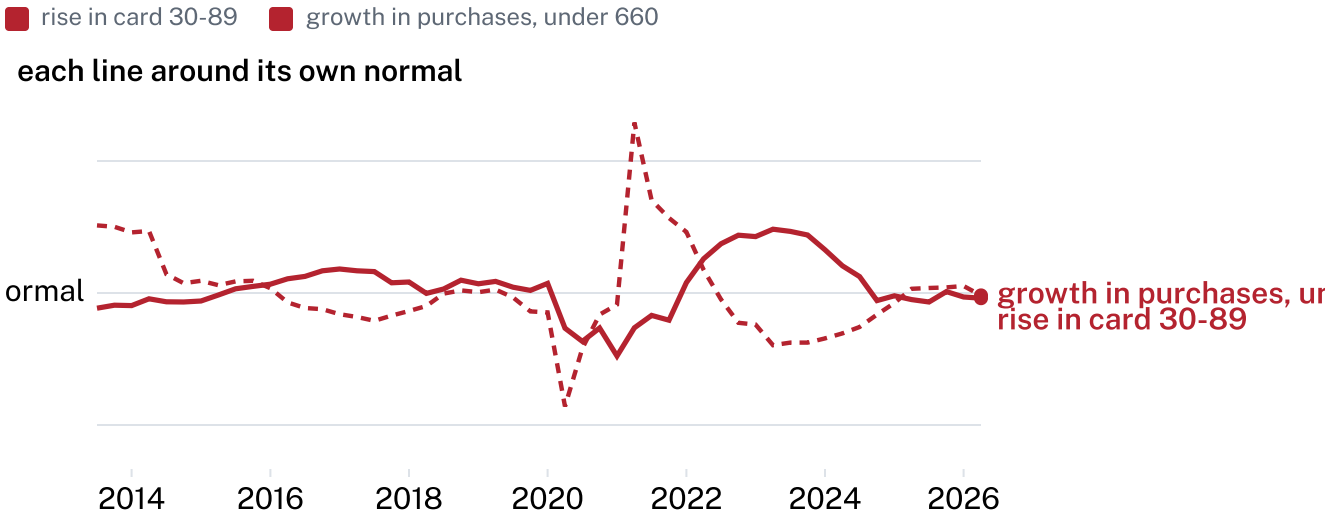
B3 · PURCHASE VOLUME BY SCORE

Card: are low-score borrowers spending more before losses rise?

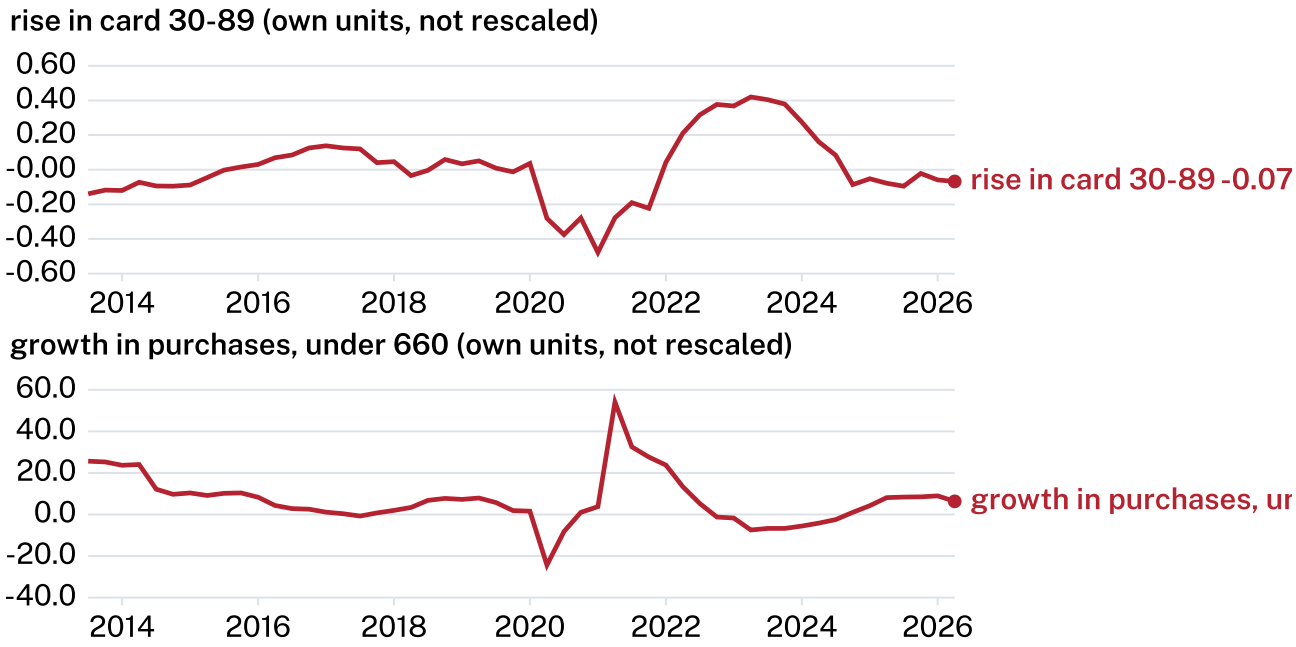
Large banks (Philadelphia Fed), 2012 on: average purchases per active card account in the quarter, by score group.



Growth in purchases by accounts under 660 (moved 0 quarters) over the rise in card 30–89



▼ The same two series in their own units



No. When delinquency rises, accounts under 660 spend less in the same quarter (−0.7 to −0.85): they are at their limits, have had lines cut, or are delinquent. It does not move ahead of losses. This is purchases, not balances; no public series gives revolving balances by score group, so the weakest borrowers drawing their lines is best read from the share paying only the minimum and the most stretched tenth's utilization.

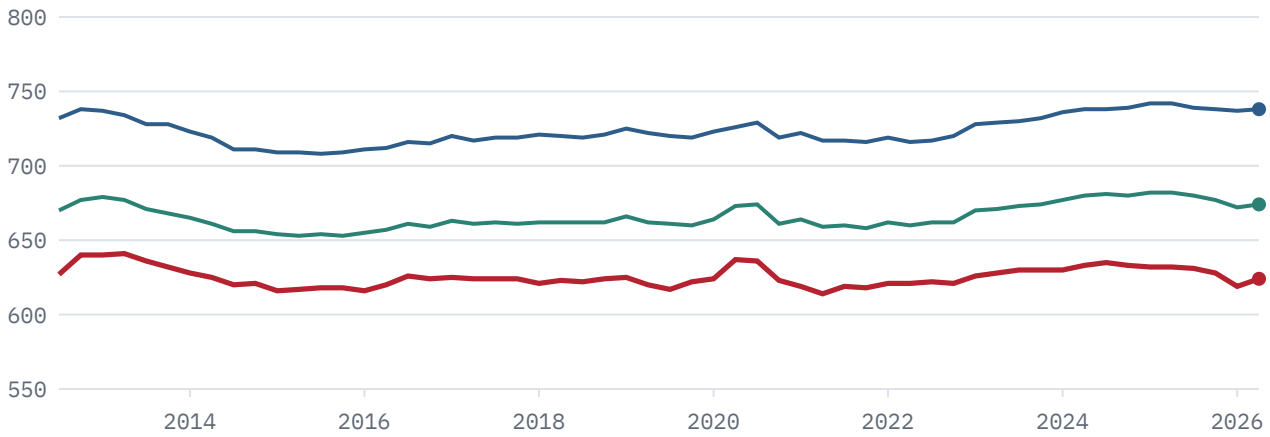
B4 · MORTGAGE ORIGINATION TERMS AND CURRENT LTV

Who got new loans: scores at origination

Read literally: in each quarter, the large banks' new loans went to borrowers with these scores. One in ten new accounts scored at or below the red line; one in four at or below the green; half at or below the blue. A falling red line means banks were lending deeper into lower scores.

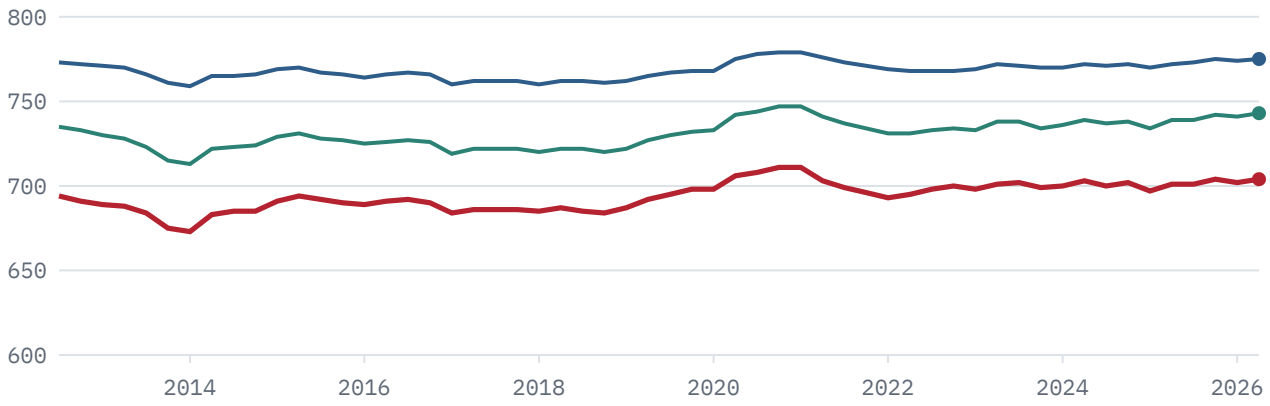
10th percentile 25th Median

New card accounts



Latest 10th percentile 624, against 632 a year earlier and 614 at its lowest (2021Q2).

New first-lien mortgages

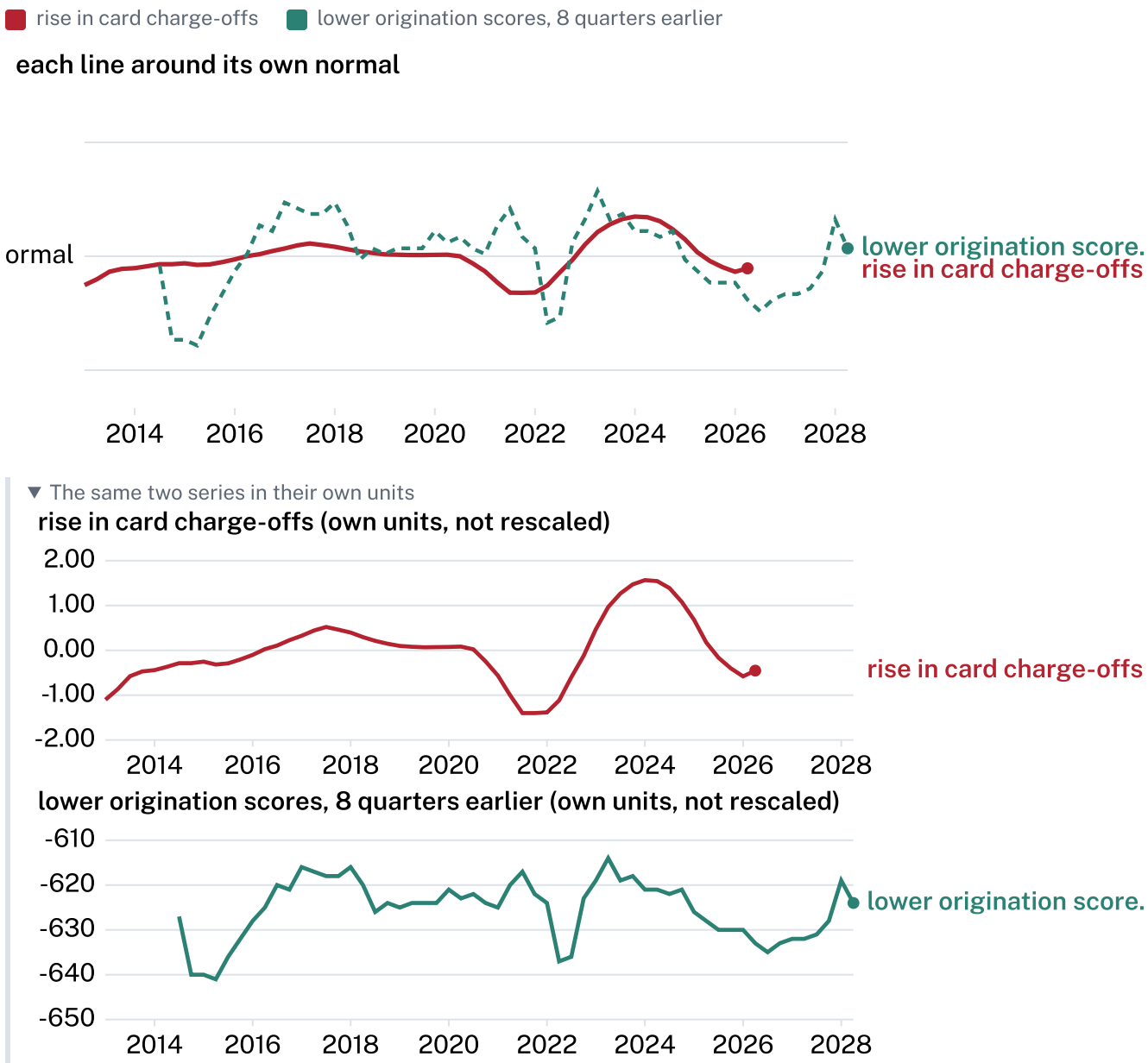


Philadelphia Fed, FR Y-14M banks, 2012Q3 to 2026Q2: the score at origination of the primary borrower, a commercially available score, not necessarily FICO. Debt-to-income and loan-to-value at origination exist for mortgages only (in the underwriting section below); there is no public card equivalent.

Literal terms at origination: which ones come first?

Each chart lays an origination term (dashed) over what later happened to the book (solid red), both drawn around their own normal so the shapes can be compared. Where the dashed line is moved later, it is by the delay where it matched best.

Card: score at origination (10th percentile, drawn so lower scores are up), moved 8 quarters later, over the rise in card charge-offs

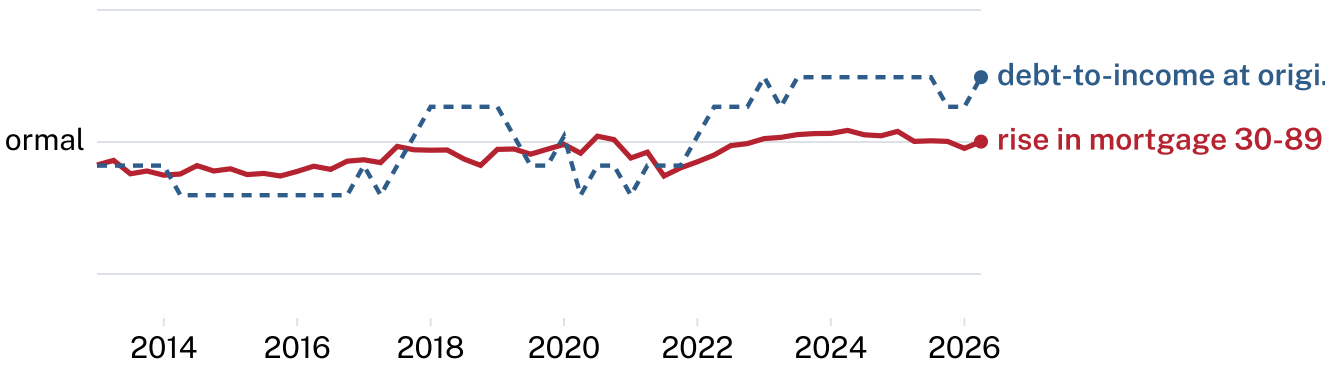


This one leads. When the bottom tenth of new card accounts scored lower, charge-offs rose about two years later (the 2021 dip to 614 sits over the 2023 rise). With 2020–2023 left out the match is 0.85 to 0.87; with them in, 0.42.

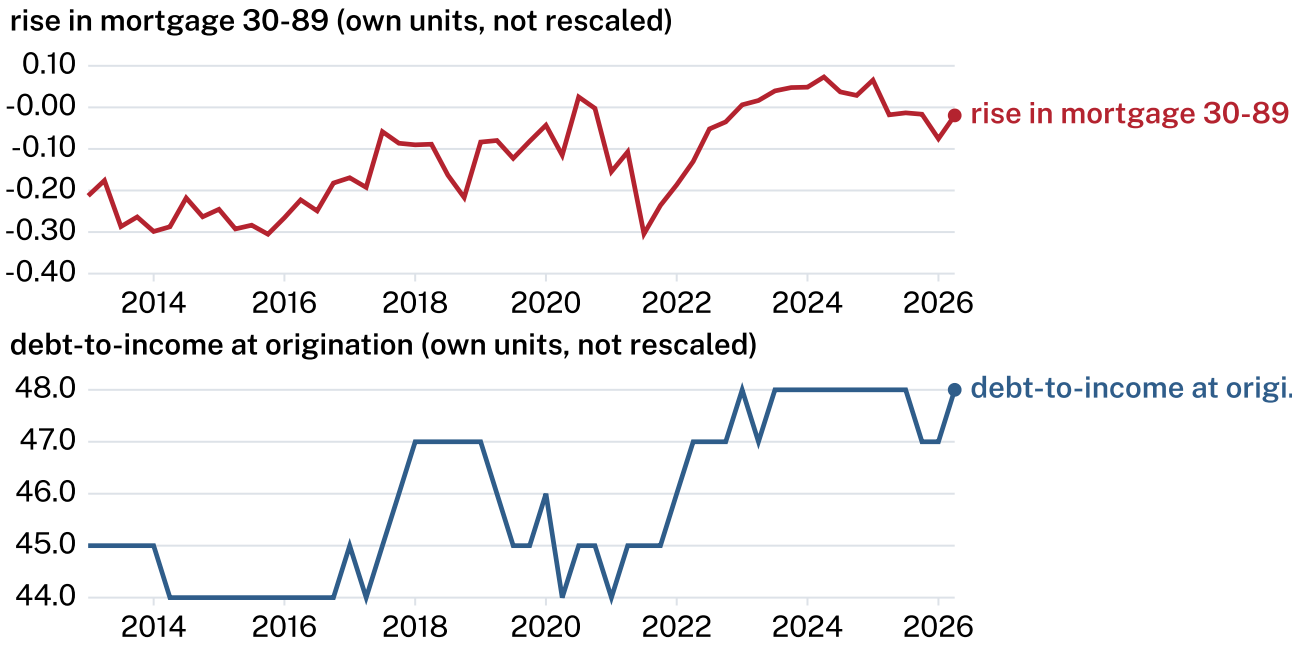
Mortgage: debt-to-income at origination (90th percentile), not moved, over the rise in mortgage 30–89



each line around its own normal



▼ The same two series in their own units

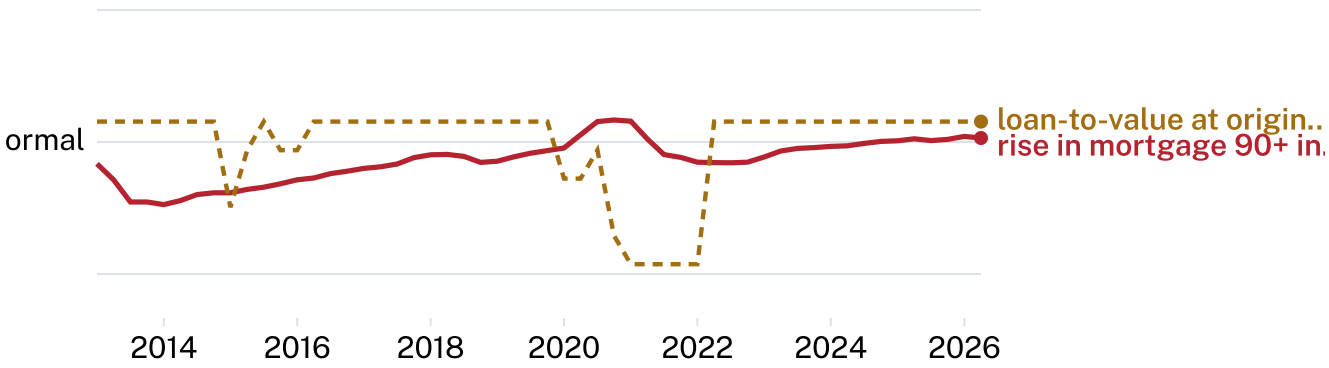


This one moves at the same time, not before: the match is strongest with no delay (0.78 to 0.85) and fades as the delay grows. Both rose and fell with interest rates and refinancing waves, so it is not evidence of a lead.

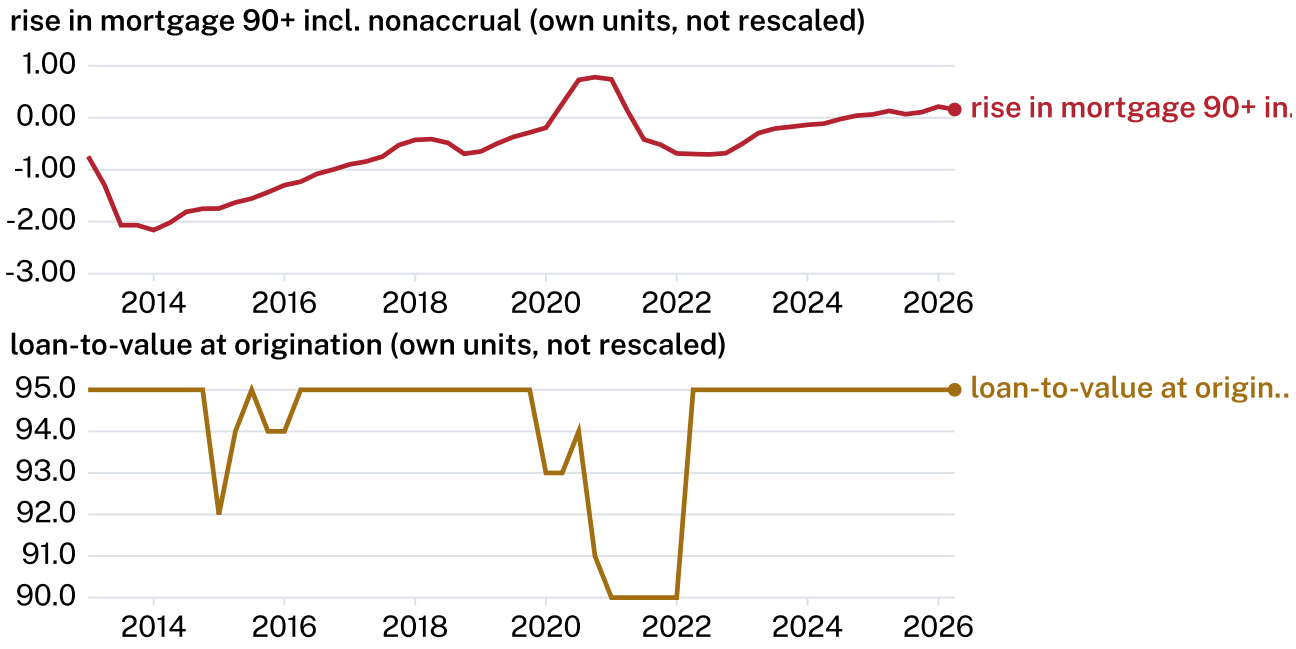
Mortgage: loan-to-value at origination (90th percentile), not moved, over the rise in mortgage 90+ and nonaccrual

rise in mortgage 90+ incl. nonaccrual loan-to-value at origination

each line around its own normal



▼ The same two series in their own units



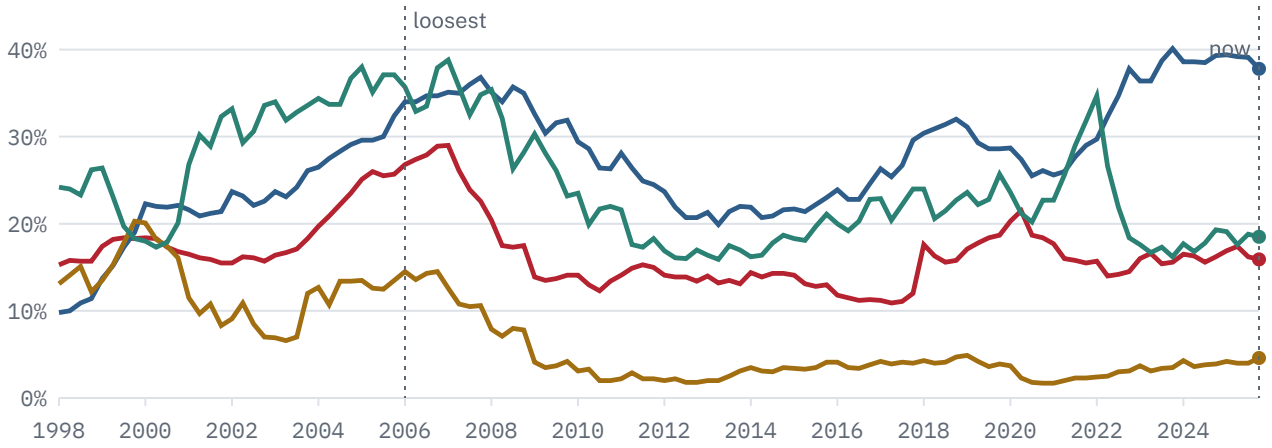
No consistent link either way. Mortgage delinquency in 2012–2026 was driven by the pre-2008 loans running off and by pandemic forbearance, and there has been no mortgage downturn since 2012 to test these terms against.

Mortgage: terms at origination, tested against 2008

FHFA's National Mortgage Database: a 5% sample of every first-lien mortgage in the U.S., all lenders, terms at origination by quarter from 1998 and performance of the whole book from 2002. Unlike the large-bank data (2012 on), it covers the run-up to 2008.

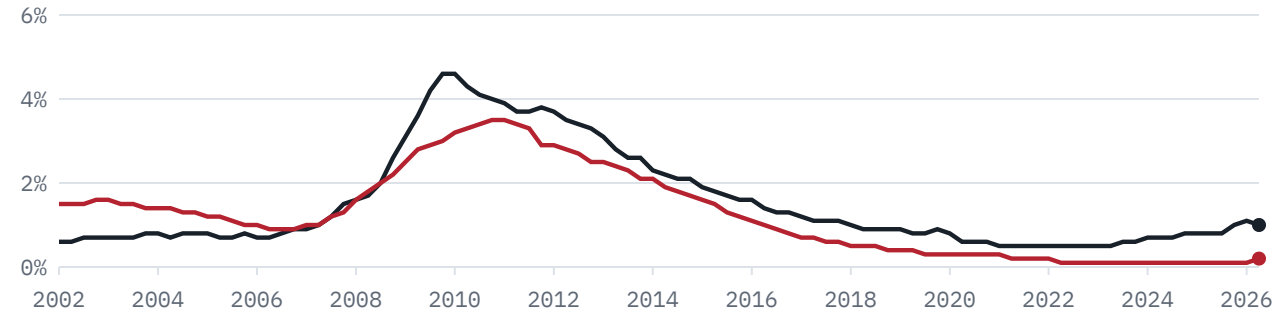
purchase loans with combined LTV over 97% · purchase loans with DTI 44% or more · loans to poor or very poor scores · cash-out share of all loans

Share of new mortgages on each risky term



90+ days past due · in foreclosure

What the whole book did

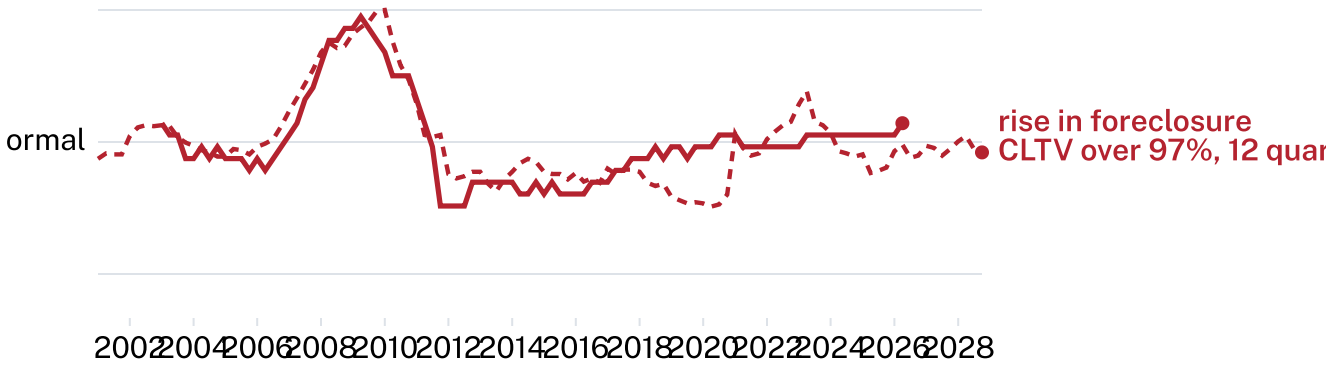


Slid forward and laid over the rise in foreclosure

Purchase loans with combined LTV over 97%, moved 12 quarters later (match 0.85):

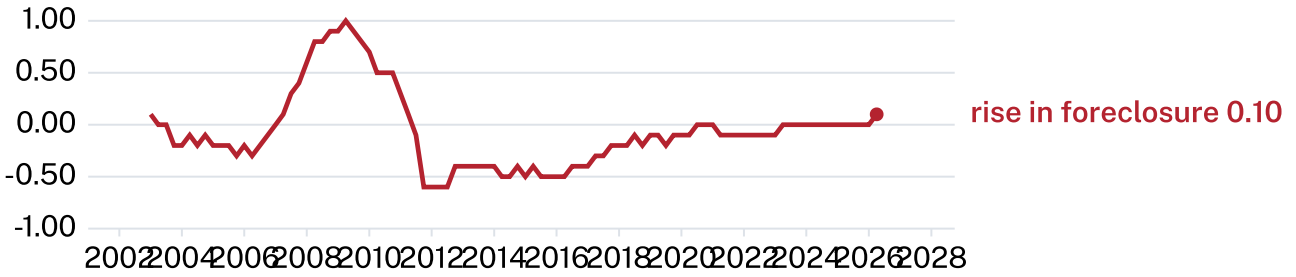
rise in foreclosure · CLTV over 97%, 12 quarters earlier

each line around its own normal



▼ The same two series in their own units

rise in foreclosure (own units, not rescaled)



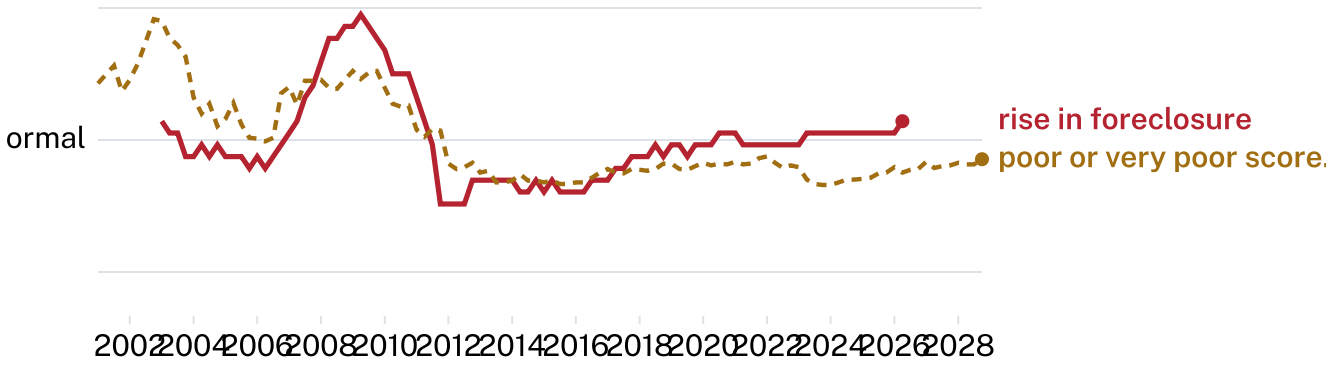
CLTV over 97%, 12 quarters earlier (own units, not rescaled)



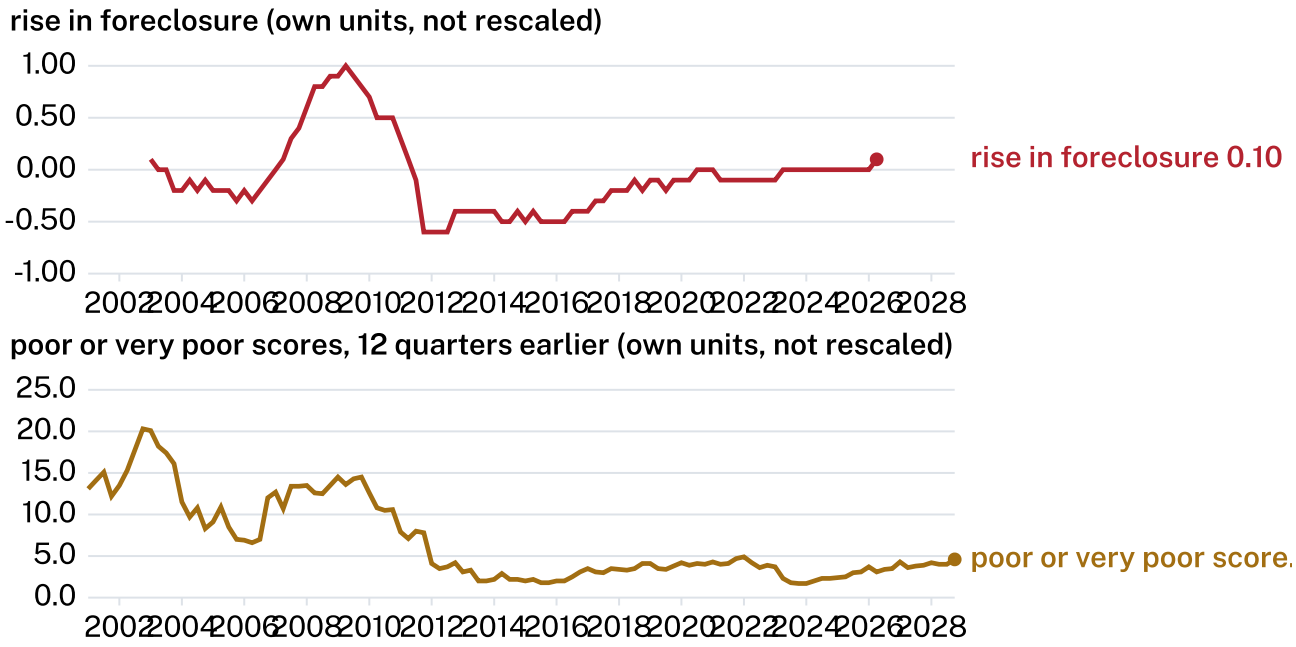
Loans to poor or very poor scores, moved 12 quarters later (match 0.80):

rise in foreclosure poor or very poor scores, 12 quarters earlier

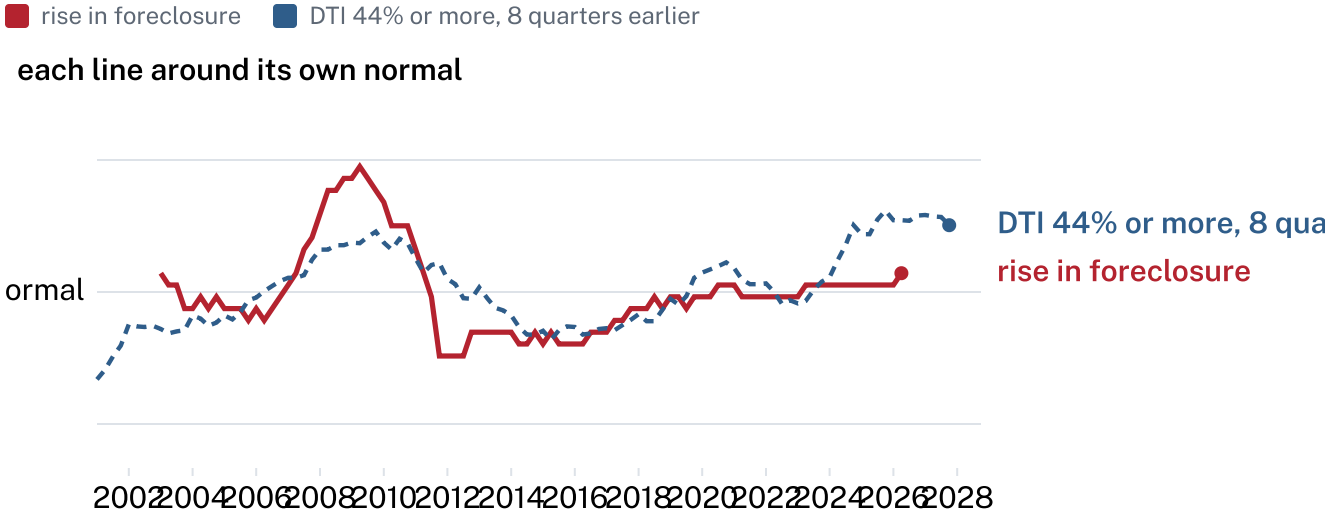
each line around its own normal



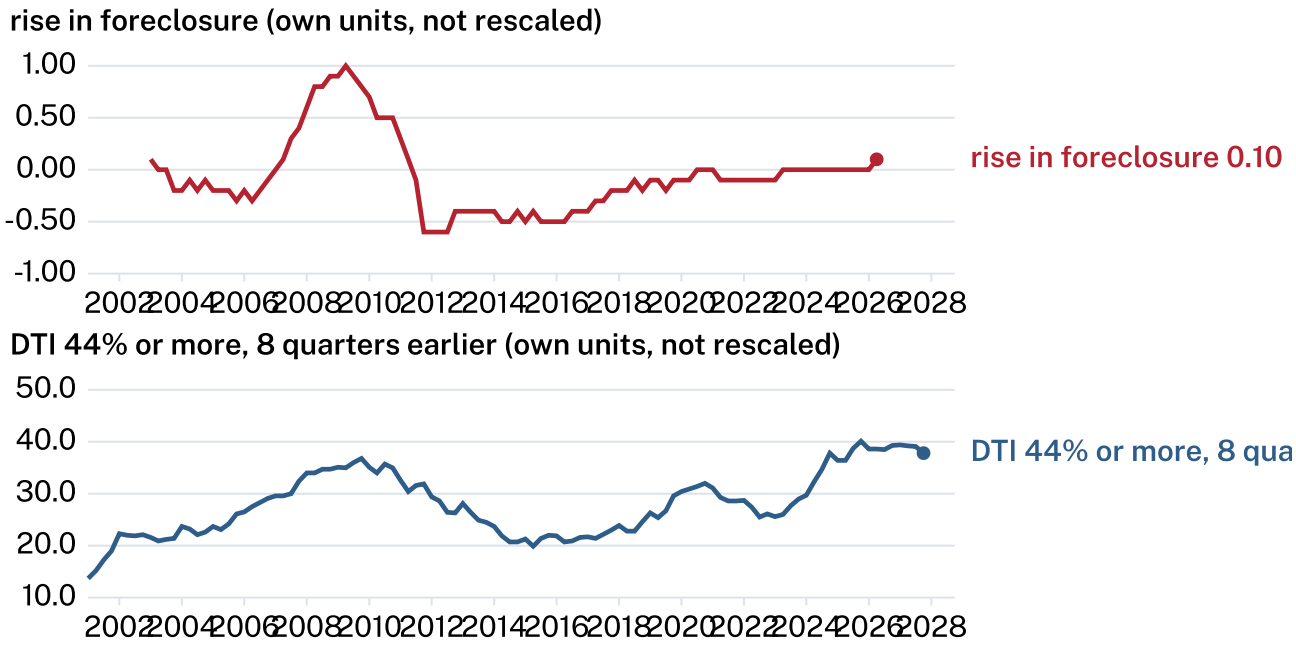
▼ The same two series in their own units



Purchase loans with DTI 44% or more, moved 8 quarters later (match 0.70):



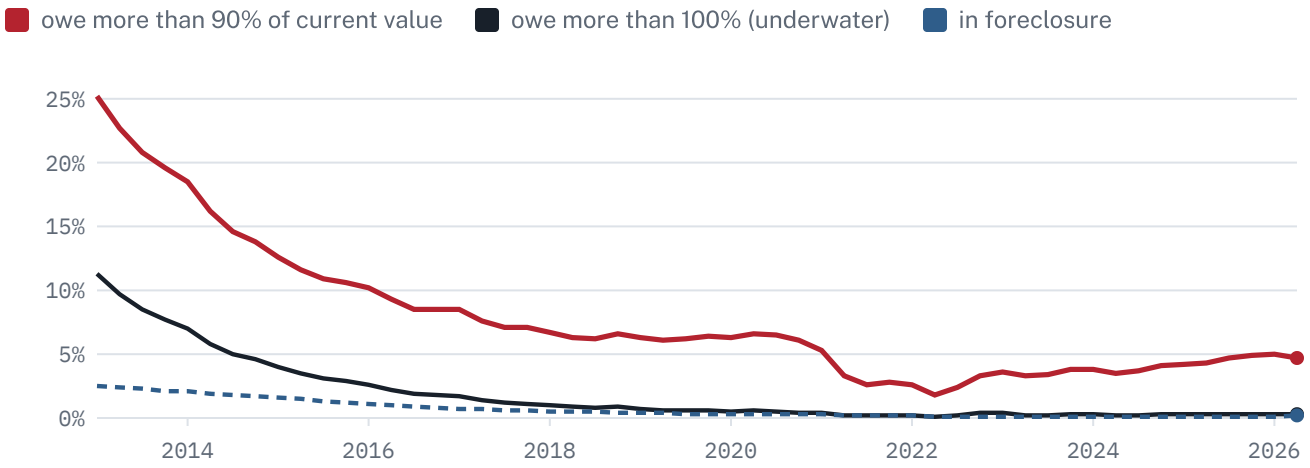
▼ The same two series in their own units



One episode carries these matches: terms loosened through 2005–2007 and serious delinquency and foreclosure peaked in 2009–2010. Now: purchase loans with DTI 44% or more are 37.8%, above early 2008's 35.1% (the series high was 40.1% in 2023Q4); combined LTV over 97% is 15.9% (2005: 25.1%); poor or very poor scores 4.6% (2005: 13.5%). The DTI rise partly reflects higher rates raising payments, not only looser lending. Scores here are VantageScore, from the NMDB's own credit data.

Mortgage: why high DTI is not turning into foreclosures

The same database values every outstanding mortgage against today's home price (current, or mark-to-market, loan-to-value), from 2013. A borrower with equity who cannot pay can sell; one who owes more than the house is worth cannot.



2013Q1: 25.2% owed more than 90% of their home's value and 11.3% were underwater, with foreclosure still high. 2026Q2: 4.7% and 0.3%; the average outstanding mortgage owes 45% of current value. That cushion, from the 2020–2022 price rise, is why the high share of high-DTI loans has not produced foreclosures. The point to worry is when this line climbs: home prices falling, or new loans at high combined LTV piling up. The series starts in 2013, after the crisis, so it cannot show the run-up to 2008.

C. The kept measures, as they sit in the book

Each row is laid out as a row on Decide: what is rated, the history its lines come from, the settings, the rating now, the chart with its zones, and where each rating begins. Built into the book on 18 Oct 2026.

INHERENT RISK: HOW MUCH DELINQUENCY TURNS INTO LOSS

Credit card · 30+ to charge-off

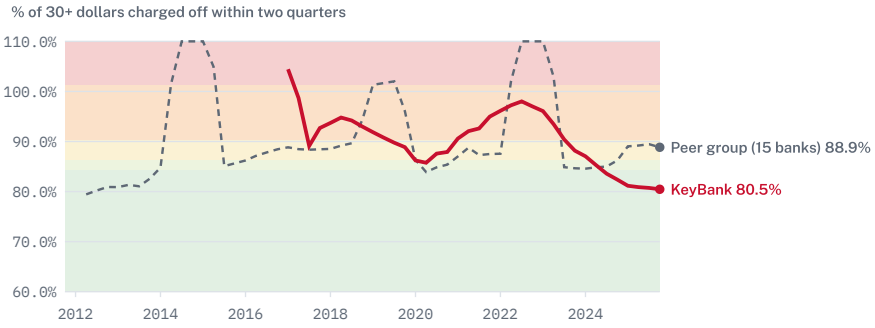
Rated: KeyBank · Lines set from: Peer group (15 banks)

RATING NOW (2025Q4)

1 Low

80.5%

Width of Moderate **0.5** · Moderate-Low begins **1** · How rare High is (z) **2.50**



Rating	Begins	Share of quarters
1 Low	below 84.3%	17%
2 Moderate-Low	from 84.3%	8%
3 Moderate	from 86.3%	19%
4 Moderate-High	from 90.4%	53%
5 High	from 101.4%	3%

Gross charge-offs over the next two quarters ÷ (30–89 + 90+ + nonaccrual) at the quarter-end, four-quarter average. KeyBank shown from 2017 (its card book was small before). KeyBank's 30+ includes its nonaccrual card loans; the internal check (nonaccrual card balances by days past due) decides whether that flatters it.

▼ Numbers behind it: KeyBank card, $ as filed (one row per year, the fourth quarter, plus the latest)

Quarter-end	30–89	90+ accruing	Nonaccrual	All 30+ (sum)	Gross charge-offs, next two quarters	Share charged off (this quarter-end)
2017Q4	$11m	$11m	$2m	$25m	$24m	96.5%
2018Q4	$11m	$12m	$2m	$25m	$23m	91.6%
2019Q4	$11m	$12m	$2m	$26m	$23m	88.0%
2020Q4	$8m	$7m	$2m	$17m	$15m	89.3%
2021Q4	$7m	$6m	$3m	$15m	$15m	98.8%
2022Q4	$9m	$8m	$3m	$19m	$18m	94.9%
2023Q4	$11m	$12m	$5m	$28m	$24m	85.9%
2024Q4	$10m	$12m	$7m	$29m	$24m	81.2%
2025Q4	$9m	$10m	$7m	$26m	$21m	80.2%

▼ Numbers behind it: Peer group card, $ as filed (15 banks summed) (one row per year, the fourth quarter, plus the latest)

Quarter-end	30–89	90+ accruing	Nonaccrual	All 30+ (sum)	Gross charge-offs, next two quarters	Share charged off (this quarter-end)
2017Q4	$4.3bn	$4.2bn	$93m	$8.6bn	$7.5bn	86.8%
2018Q4	$4.6bn	$4.6bn	$98m	$9.3bn	$9.8bn	105.5%
2019Q4	$6.0bn	$6.2bn	$111m	$12.3bn	$10.0bn	81.7%
2020Q4	$4.3bn	$4.6bn	$131m	$9.0bn	$7.6bn	83.9%
2021Q4	$3.2bn	$3.0bn	$111m	$6.3bn	$5.3bn	84.7%
2022Q4	$7.2bn	$6.6bn	$57m	$13.9bn	$11.4bn	82.6%
2023Q4	$10.6bn	$10.8bn	$74m	$21.4bn	$17.5bn	81.8%
2024Q4	$10.7bn	$12.0bn	$72m	$22.8bn	$19.7bn	86.5%
2025Q4	$12.9bn	$13.5bn	$65m	$26.5bn	$22.2bn	84.1%

Other consumer · 30+ to charge-off

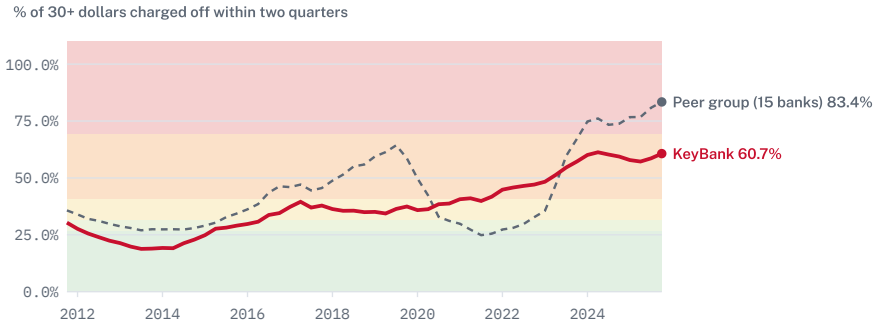
Rated: KeyBank · Lines set from: Peer group (15 banks)

RATING NOW (2025Q4)

4 Moderate-High

60.7%

Width of Moderate **0.5** · Moderate-Low begins **1** · How rare High is (z) **2.50**



Rating	Begins	Share of quarters
1 Low	below 26.7%	21%
2 Moderate-Low	from 26.7%	12%
3 Moderate	from 31.5%	35%
4 Moderate-High	from 41.0%	32%
5 High	from 69.2%	0%

Every source stepped up from about 35% to 60–85% from 2023, which looks like reporting and product mix; lines across the whole history flag everything since. Leaving out the quarters before 2023 is the alternative, with a short history.

▼ Numbers behind it: KeyBank other consumer, $ as filed (one row per year, the fourth quarter, plus the latest)

Quarter-end	30–89	90+ accruing	Nonaccrual	All 30+ (sum)	Gross charge-offs, next two quarters	Share charged off (this quarter-end)
2012Q4	$201m	$99m	$37m	$336m	$63m	18.7%
2013Q4	$152m	$78m	$50m	$280m	$54m	19.3%
2014Q4	$86m	$42m	$23m	$150m	$38m	25.6%
2015Q4	$67m	$33m	$13m	$113m	$33m	29.3%
2016Q4	$59m	$29m	$11m	$99m	$32m	32.6%
2017Q4	$44m	$17m	$14m	$74m	$27m	36.1%
2018Q4	$44m	$28m	$13m	$85m	$29m	33.4%
2019Q4	$37m	$23m	$13m	$73m	$27m	37.7%
2020Q4	$28m	$12m	$11m	$52m	$20m	39.0%
2021Q4	$24m	$13m	$9m	$46m	$21m	46.5%
2022Q4	$31m	$15m	$7m	$53m	$26m	48.7%
2023Q4	$34m	$17m	$6m	$57m	$34m	59.0%
2024Q4	$31m	$15m	$6m	$52m	$29m	55.5%
2025Q4	$30m	$13m	$5m	$48m	$31m	63.8%

▼ Numbers behind it: Peer group other consumer, $ as filed (one row per year, the fourth quarter, plus the latest)

Quarter-end	30–89	90+ accruing	Nonaccrual	All 30+ (sum)	Gross charge-offs, next two quarters	Share charged off (this quarter-end)
2012Q4	$3.5bn	$2.6bn	$646m	$6.8bn	$1.7bn	24.5%
2013Q4	$3.0bn	$2.4bn	$424m	$5.8bn	$1.5bn	26.3%
2014Q4	$2.3bn	$1.4bn	$587m	$4.3bn	$1.2bn	28.9%
2015Q4	$1.8bn	$948m	$514m	$3.3bn	$1.2bn	35.2%
2016Q4	$1.8bn	$824m	$464m	$3.1bn	$1.4bn	46.4%
2017Q4	$1.3bn	$465m	$260m	$2.0bn	$1.0bn	50.5%
2018Q4	$1.2bn	$386m	$251m	$1.8bn	$998m	54.0%
2019Q4	$1.8bn	$1.4bn	$222m	$3.5bn	$1.1bn	31.1%
2020Q4	$2.1bn	$1.4bn	$302m	$3.8bn	$912m	24.1%
2021Q4	$1.5bn	$1.1bn	$321m	$2.9bn	$770m	26.5%
2022Q4	$1.5bn	$892m	$306m	$2.6bn	$1.0bn	38.7%
2023Q4	$1.3bn	$215m	$401m	$1.9bn	$1.3bn	68.5%
2024Q4	$1.2bn	$176m	$562m	$1.9bn	$1.3bn	70.4%
2025Q4	$1.2bn	$216m	$369m	$1.8bn	$1.5bn	80.5%

RELATIVE PERFORMANCE: KEYBANK AGAINST PEERS AND INDUSTRY

Credit card · KeyBank minus industry, 30+ to charge-off

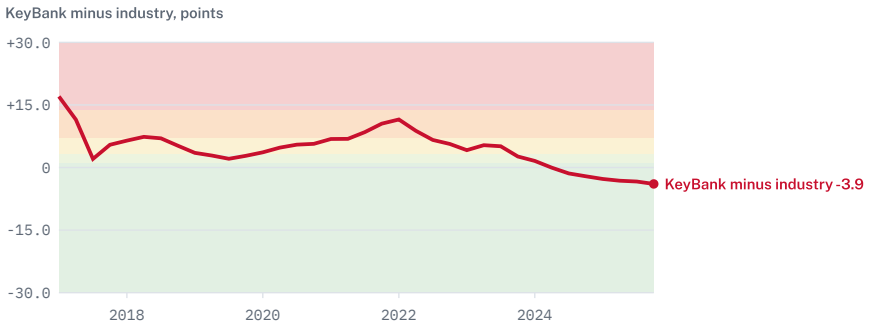
Rated: KeyBank minus industry · Lines set from: its own history

RATING NOW (2025Q4)

1 Low

-3.9 pts

Width of Moderate **0.5** · Moderate-Low begins **1** · How rare High is (z) **2.50** · floor at zero **off**



Rating	Begins	Share of quarters
1 Low	below +1.2	19%
2 Moderate-Low	from +1.2	17%
3 Moderate	from +3.2	44%
4 Moderate-High	from +7.1	17%
5 High	from +13.8	3%

Other consumer · KeyBank minus peers, 30+ to charge-off

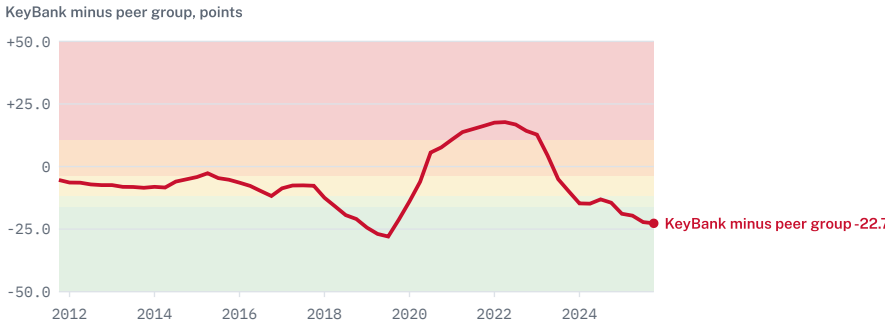
Rated: KeyBank minus peer group · Lines set from: its own history

RATING NOW (2025Q4)

1 Low

-22.7 pts

Width of Moderate **0.5** · Moderate-Low begins **1** · How rare High is (z) **2.50** · floor at zero **off**



Rating	Begins	Share of quarters
1 Low	below -15.9	19%
2 Moderate-Low	from -15.9	12%
3 Moderate	from -11.7	46%
4 Moderate-High	from -3.4	7%
5 High	from +10.6	16%

Measured in the same quarter, so the industry-wide step in 2023 largely cancels out.

Other consumer · KeyBank minus industry, 30+ to charge-off

Rated: KeyBank minus industry · Lines set from: its own history

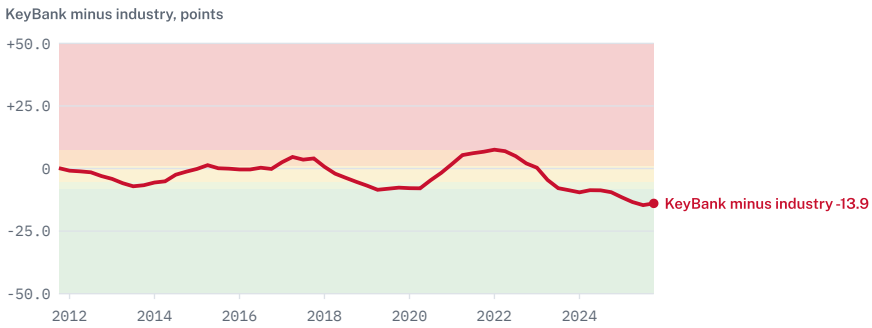
RATING NOW (2025Q4)

1 Low

-13.9 pts

Width of Moderate **0.5** · Moderate-Low begins **1** · How rare High is (z) **1.58** · floor at zero **off**

At z 2.5 the Check says: High would begin beyond the worst quarter kept: nothing kept could rate High. Shown at z 1.58, the highest that fits.



Rating	Begins	Share of quarters
1 Low	below -8.1	18%
2 Moderate-Low	from -8.1	21%
3 Moderate	from -5.1	39%
4 Moderate-High	from +1.0	21%
5 High	from +7.5	2%

BORROWER STRAIN

Credit card · Share of accounts paying only the minimum

Rated: Large banks (Philadelphia Fed) · Lines set from: Large banks (Philadelphia Fed)

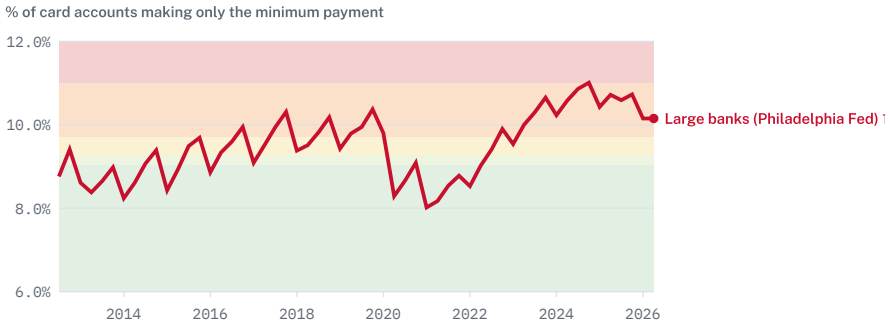
RATING NOW (2026Q2)

4 Moderate-High

10.2%

Width of Moderate **0.5** · Moderate-Low begins **1** · How rare High is (z) **1.56**

At z 2.5 the Check says: High would begin beyond the worst quarter kept: nothing kept could rate High. Shown at z 1.56, the highest that fits.



Rating	Begins	Share of quarters
1 Low	below 9.0%	32%
2 Moderate-Low	from 9.0%	5%
3 Moderate	from 9.3%	21%
4 Moderate-High	from 9.7%	39%
5 High	from 11.0%	2%

Large banks' card accounts, quarterly from 2012. Moves with 30–89 days past due and about a year ahead of charge-offs; tracks the most stretched tenth's utilization (0.92). An industry reading: there is no public KeyBank figure.

▼ Numbers behind it: large banks' card accounts (one row per year, the fourth quarter, plus the latest)

Quarter	Accounts (millions)	Share paying only the minimum	Accounts paying only the minimum (millions, implied)
2012Q4	515.6	9.41%	48.5
2013Q4	518.2	8.98%	46.5
2014Q4	521.5	9.39%	49.0
2015Q4	527.1	9.69%	51.1
2016Q4	539.8	9.95%	53.7
2017Q4	551.7	10.31%	56.9
2018Q4	553.3	10.18%	56.3
2019Q4	557.0	10.37%	57.8
2020Q4	538.0	9.09%	48.9
2021Q4	553.6	8.78%	48.6
2022Q4	589.3	9.90%	58.3
2023Q4	605.0	10.65%	64.4
2024Q4	597.6	11.01%	65.8
2025Q4	600.0	10.73%	64.4
2026Q2	598.8	10.15%	60.8

MORTGAGE

Mortgage · Nonaccrual

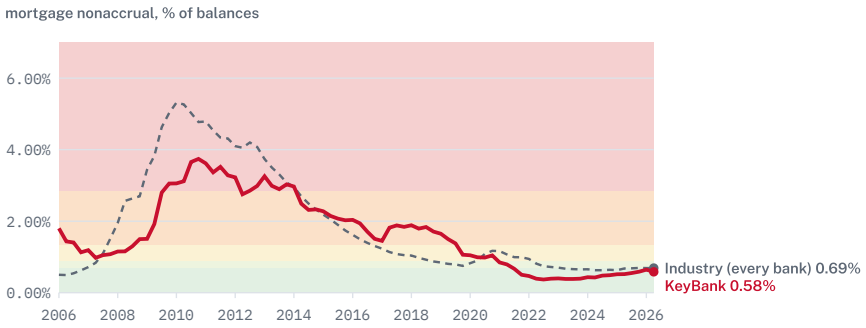
Rated: KeyBank · Lines set from: Industry (every bank)

RATING NOW (2026Q2)

1 Low

0.58%

Width of Moderate **0.5** · Moderate-Low begins **1** · How rare High is (z) **2.50**



Rating	Begins	Share of quarters
1 Low	below 0.69%	24%
2 Moderate-Low	from 0.69%	2%
3 Moderate	from 0.90%	16%
4 Moderate-High	from 1.33%	38%
5 High	from 2.86%	20%

Mortgage rated on delinquency rather than charge-offs (near zero since 2014). Nonaccrual leaves out government-guaranteed 90+ loans that carry no loss. Lines from the whole history are set by 2008; lines from 2016 on are the other choice.

▼ Numbers behind it: KeyBank mortgage (closed-end first liens), $ as filed (one row per year, the fourth quarter, plus the latest)

Quarter-end	Nonaccrual	Balances	Nonaccrual rate	Change from a year earlier
2007Q4	$37m	$3.5bn	1.08%	-0.05 pts
2008Q4	$52m	$3.5bn	1.50%	+0.42 pts
2009Q4	$95m	$3.1bn	3.05%	+1.56 pts
2010Q4	$111m	$3.0bn	3.74%	+0.69 pts
2011Q4	$102m	$3.1bn	3.28%	-0.46 pts
2012Q4	$121m	$4.0bn	2.98%	-0.30 pts
2013Q4	$146m	$4.8bn	3.03%	+0.04 pts
2014Q4	$120m	$5.2bn	2.33%	-0.70 pts
2015Q4	$107m	$5.3bn	2.03%	-0.30 pts
2016Q4	$132m	$8.8bn	1.50%	-0.52 pts
2017Q4	$161m	$8.7bn	1.84%	+0.34 pts
2018Q4	$147m	$8.6bn	1.71%	-0.13 pts
2019Q4	$110m	$10.3bn	1.06%	-0.65 pts
2020Q4	$137m	$13.2bn	1.04%	-0.02 pts
2021Q4	$99m	$19.9bn	0.50%	-0.54 pts
2022Q4	$98m	$24.9bn	0.39%	-0.11 pts
2023Q4	$94m	$24.1bn	0.39%	-0.00 pts
2024Q4	$111m	$22.7bn	0.49%	+0.10 pts
2025Q4	$125m	$21.2bn	0.59%	+0.10 pts
2026Q2	$120m	$20.6bn	0.58%	+0.06 pts

▼ Numbers behind it: Industry mortgage, $ as filed (one row per year, the fourth quarter, plus the latest)

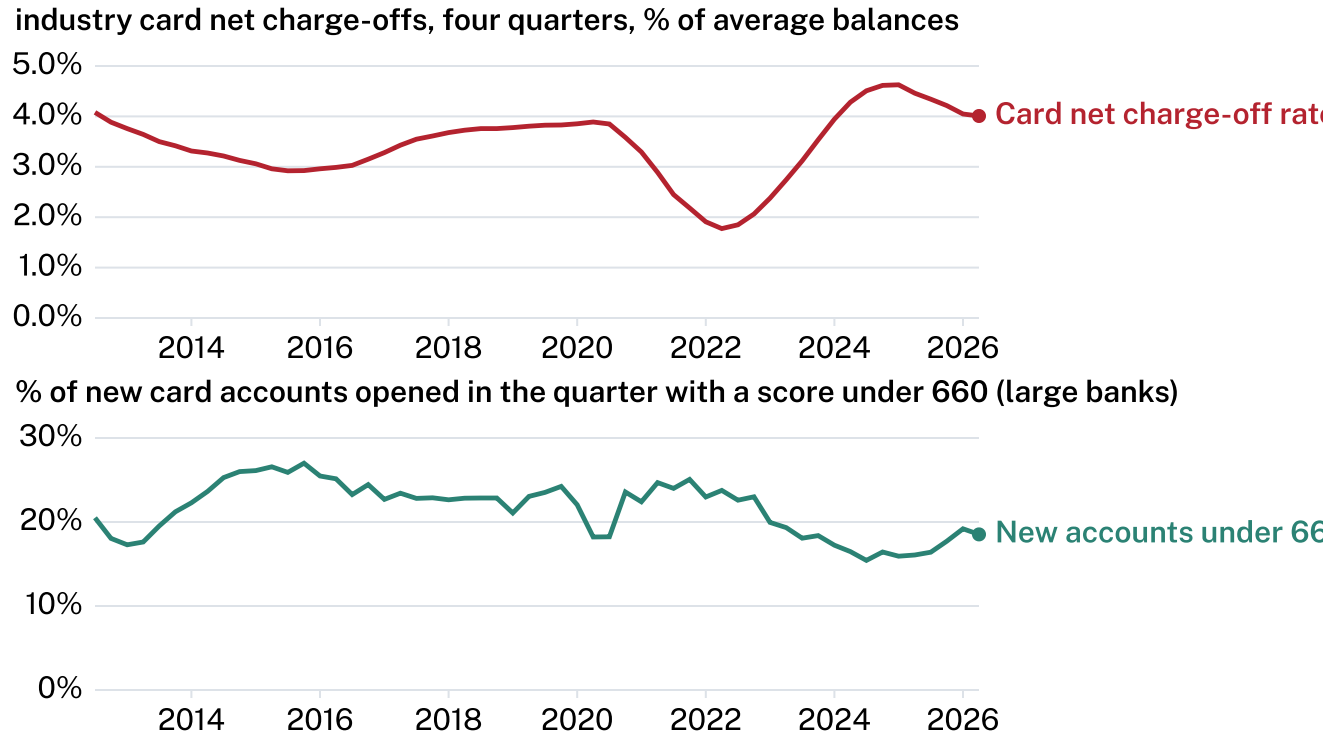
Quarter-end	Nonaccrual	Balances	Nonaccrual rate	Change from a year earlier
2007Q4	$29.9bn	$1978.9bn	1.51%	+0.89 pts
2008Q4	$49.0bn	$1819.1bn	2.69%	+1.18 pts
2009Q4	$87.1bn	$1735.0bn	5.02%	+2.33 pts
2010Q4	$83.6bn	$1752.8bn	4.77%	-0.25 pts
2011Q4	$76.0bn	$1762.7bn	4.31%	-0.46 pts
2012Q4	$73.1bn	$1793.1bn	4.08%	-0.24 pts
2013Q4	$53.7bn	$1743.6bn	3.08%	-1.00 pts
2014Q4	$41.1bn	$1766.4bn	2.33%	-0.75 pts
2015Q4	$31.9bn	$1835.9bn	1.74%	-0.59 pts
2016Q4	$25.2bn	$1934.2bn	1.30%	-0.43 pts
2017Q4	$21.1bn	$2014.4bn	1.05%	-0.26 pts
2018Q4	$18.1bn	$2074.6bn	0.87%	-0.18 pts
2019Q4	$16.2bn	$2160.0bn	0.75%	-0.12 pts
2020Q4	$25.4bn	$2176.5bn	1.17%	+0.42 pts
2021Q4	$22.1bn	$2231.1bn	0.99%	-0.18 pts
2022Q4	$17.6bn	$2447.0bn	0.72%	-0.27 pts
2023Q4	$16.5bn	$2525.0bn	0.65%	-0.07 pts
2024Q4	$16.4bn	$2555.3bn	0.64%	-0.01 pts
2025Q4	$17.9bn	$2587.7bn	0.69%	+0.05 pts
2026Q2	$17.9bn	$2602.4bn	0.69%	+0.01 pts

CONTEXT, SHOWN BESIDE THE RATINGS

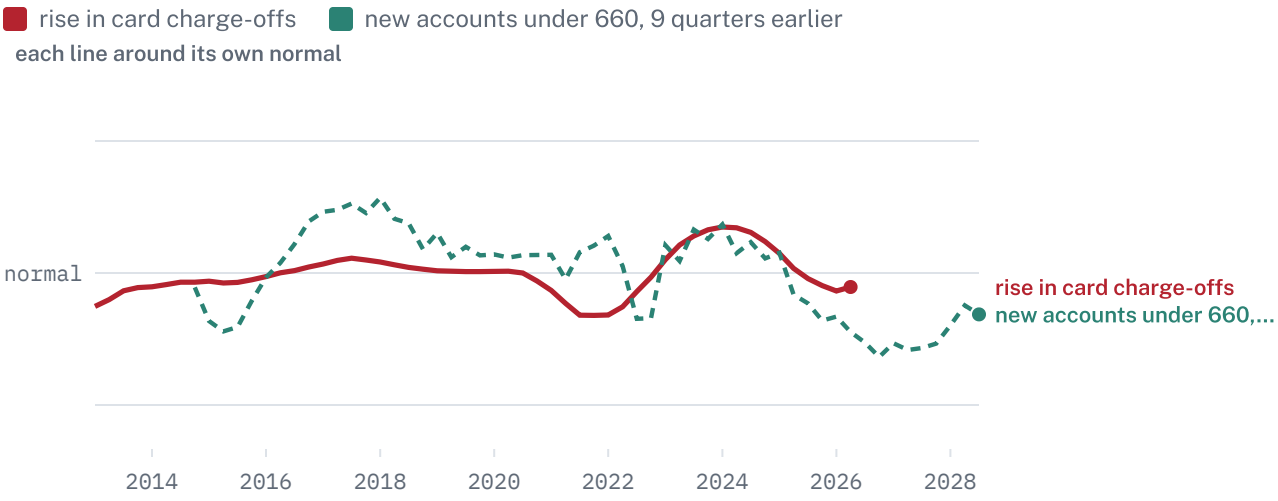
Credit card · New accounts going to scores under 660 (leads charge-offs about 2 years)

Shown beside the ratings, not rated

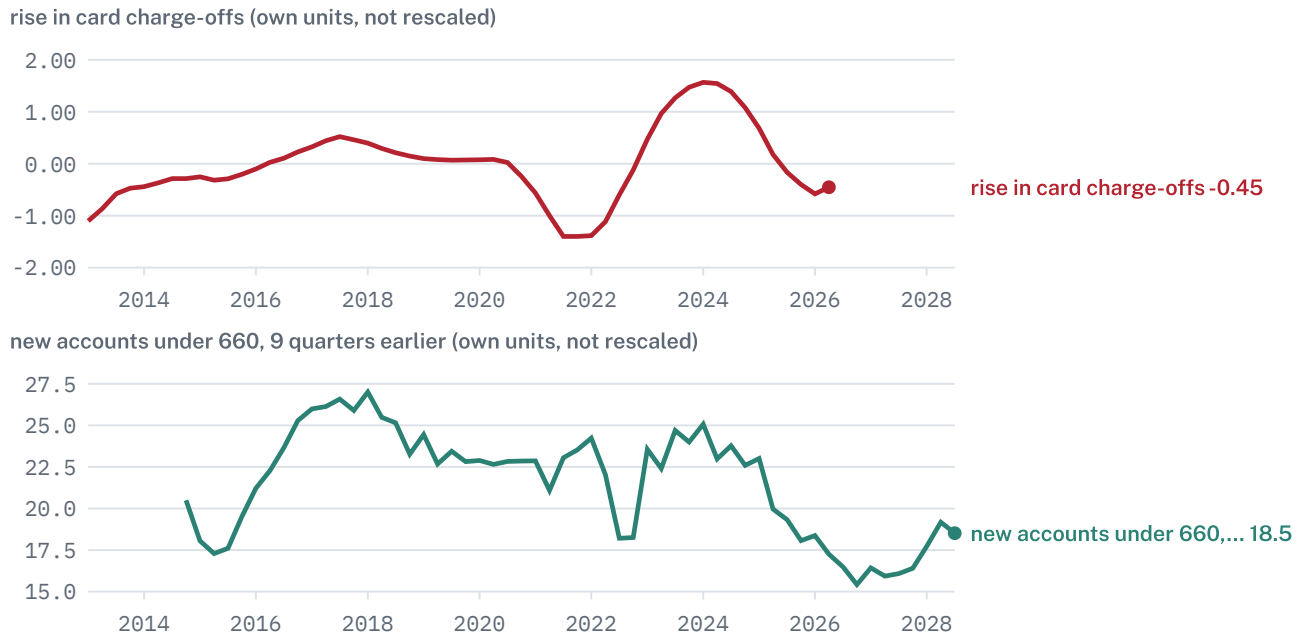
As published: the two series, nothing calculated



Same quarters on both. Below: the same two drawn for timing (rescaled and slid), then the numbers.



▼ The same two series in their own units



"New accounts under 660" is the percent of new card accounts opened in the quarter at the large banks whose borrower scored under 660 (a count of accounts, not dollars). "Rise in card charge-offs" is the change in the industry's net charge-off rate (net charge-offs over four quarters ÷ average balances) from a year earlier, in percentage points. Drawn 9 quarters later; the stretch past today is what it already points to.

▼ Numbers behind it: what "new accounts under 660" and "rise in card charge-offs" are, in counts and dollars (one row per year, the fourth quarter, plus the latest)

Quarter	New card accounts opened (millions)	Share with a score under 660	New accounts under 660 (millions, implied)	New credit lines opened ($bn)	Share of those lines ($) under 660
2012Q4	18.4	18.1%	3.33	66.9	5.2%
2013Q4	20.6	21.2%	4.38	77.2	6.3%
2014Q4	21.1	26.0%	5.49	75.2	8.5%
2015Q4	22.6	27.0%	6.10	84.0	8.6%
2016Q4	21.0	24.4%	5.14	87.4	7.0%
2017Q4	19.9	22.9%	4.55	86.6	6.0%
2018Q4	20.2	22.9%	4.62	90.6	5.9%
2019Q4	21.2	24.2%	5.14	94.4	6.1%
2020Q4	15.8	23.6%	3.73	55.9	5.7%
2021Q4	20.4	25.1%	5.12	93.0	6.9%
2022Q4	21.2	23.0%	4.88	109.3	5.8%
2023Q4	20.4	18.4%	3.75	110.3	4.5%
2024Q4	18.1	16.4%	2.98	103.7	3.9%
2025Q4	19.2	17.7%	3.41	111.3	4.2%
2026Q2	18.5	18.5%	3.43	112.4	4.2%

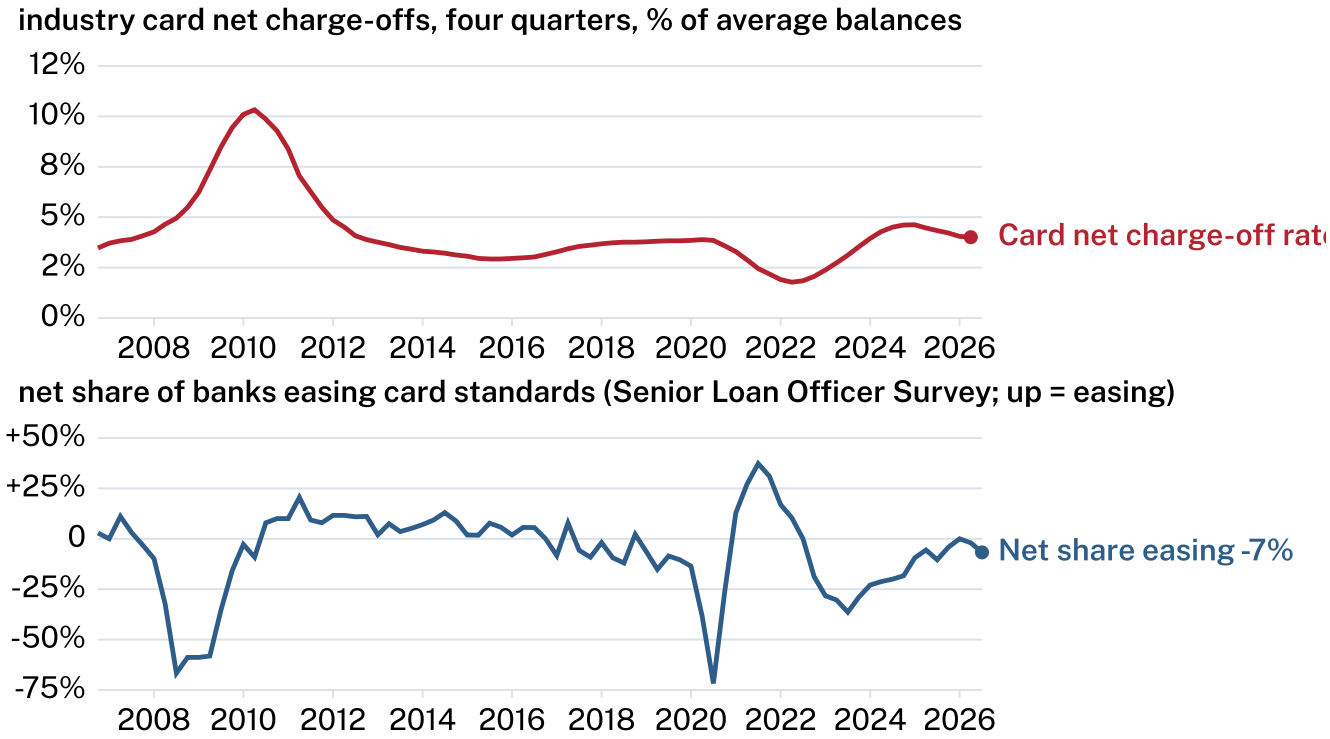
▼ Numbers behind it: industry card net charge-offs (every bank, Call Report) (one row per year, the fourth quarter, plus the latest)

Quarter	Net charge-offs, four quarters	Average balances	Net charge-off rate	Change from a year earlier (the “rise”)
2012Q4	$26.0bn	$669.8bn	3.88%	-2.25 pts
2013Q4	$23.0bn	$674.7bn	3.42%	-0.47 pts
2014Q4	$23.6bn	$684.5bn	3.45%	+0.04 pts
2015Q4	$20.9bn	$713.1bn	2.92%	-0.53 pts
2016Q4	$23.9bn	$757.8bn	3.15%	+0.23 pts
2017Q4	$28.8bn	$799.1bn	3.61%	+0.46 pts
2018Q4	$32.2bn	$854.4bn	3.76%	+0.16 pts
2019Q4	$35.0bn	$893.9bn	3.91%	+0.15 pts
2020Q4	$29.6bn	$824.9bn	3.58%	-0.33 pts
2021Q4	$17.6bn	$807.5bn	2.18%	-1.40 pts
2022Q4	$20.8bn	$924.9bn	2.25%	+0.07 pts
2023Q4	$37.0bn	$1045.3bn	3.54%	+1.28 pts
2024Q4	$51.6bn	$1117.3bn	4.62%	+1.08 pts
2025Q4	$48.6bn	$1153.3bn	4.22%	-0.40 pts
2026Q2	$47.2bn	$1176.6bn	4.01%	-0.45 pts

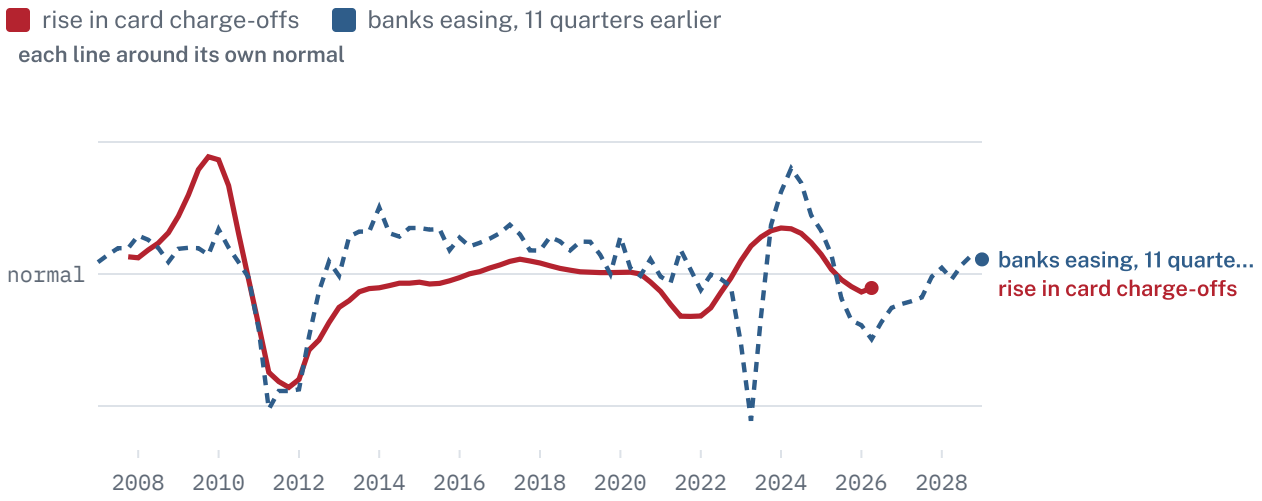
Credit card · Banks easing card standards (leads charge-offs about 3 years)

Shown beside the ratings, not rated

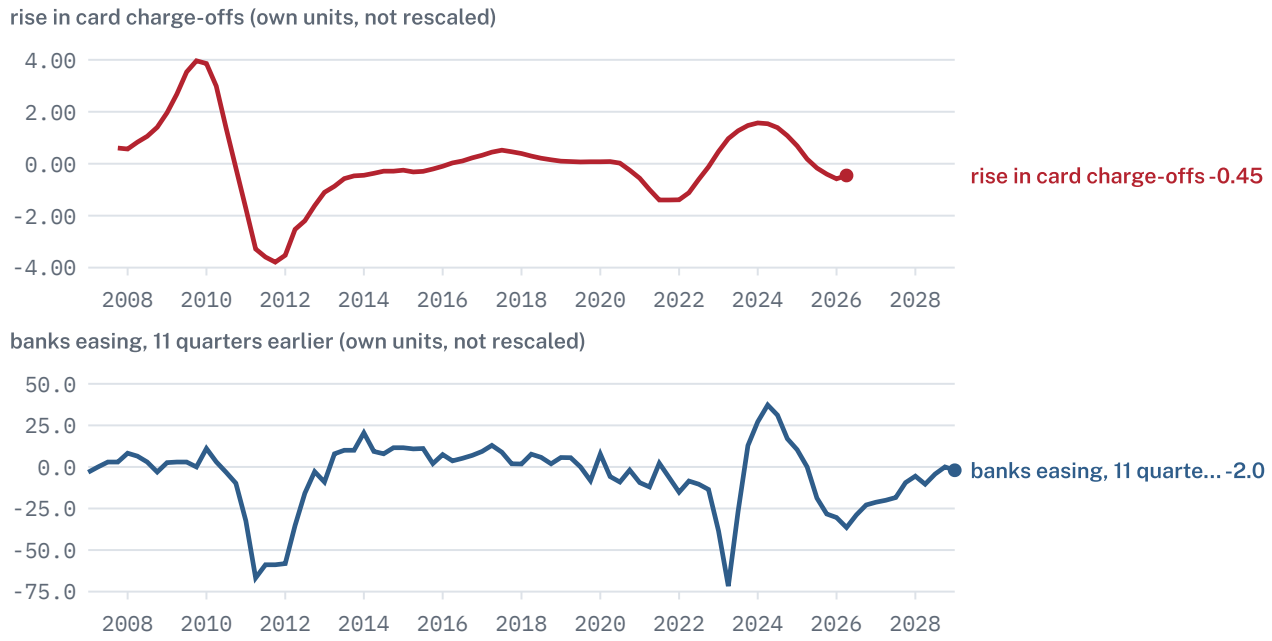
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Same quarters on both. Below: the same two drawn for timing (rescaled and slid), then the numbers.



▼ The same two series in their own units



Senior Loan Officer Survey, 1996 on. Banks tighten while losses rise (a reaction); easing is the early signal. Closer to an economic indicator, so it may sit with Book 2. The survey number is the share of banks reporting tighter card standards minus the share reporting easier, sign flipped so easing is up: a count of banks, not dollars.

▼ Numbers behind it: what the survey number is (one row per year, the fourth quarter, plus the latest)

Quarter	Net share of banks easing card standards
2012Q4	+11.1%
2013Q4	+5.2%
2014Q4	+8.8%
2015Q4	+5.8%
2016Q4	-0.0%
2017Q4	-9.1%
2018Q4	+2.2%
2019Q4	-10.4%
2020Q4	-26.7%
2021Q4	+31.1%
2022Q4	-18.8%
2023Q4	-28.9%
2024Q4	-18.4%
2025Q4	-4.2%
2026Q2	-2.0%
2026Q3	-6.7%

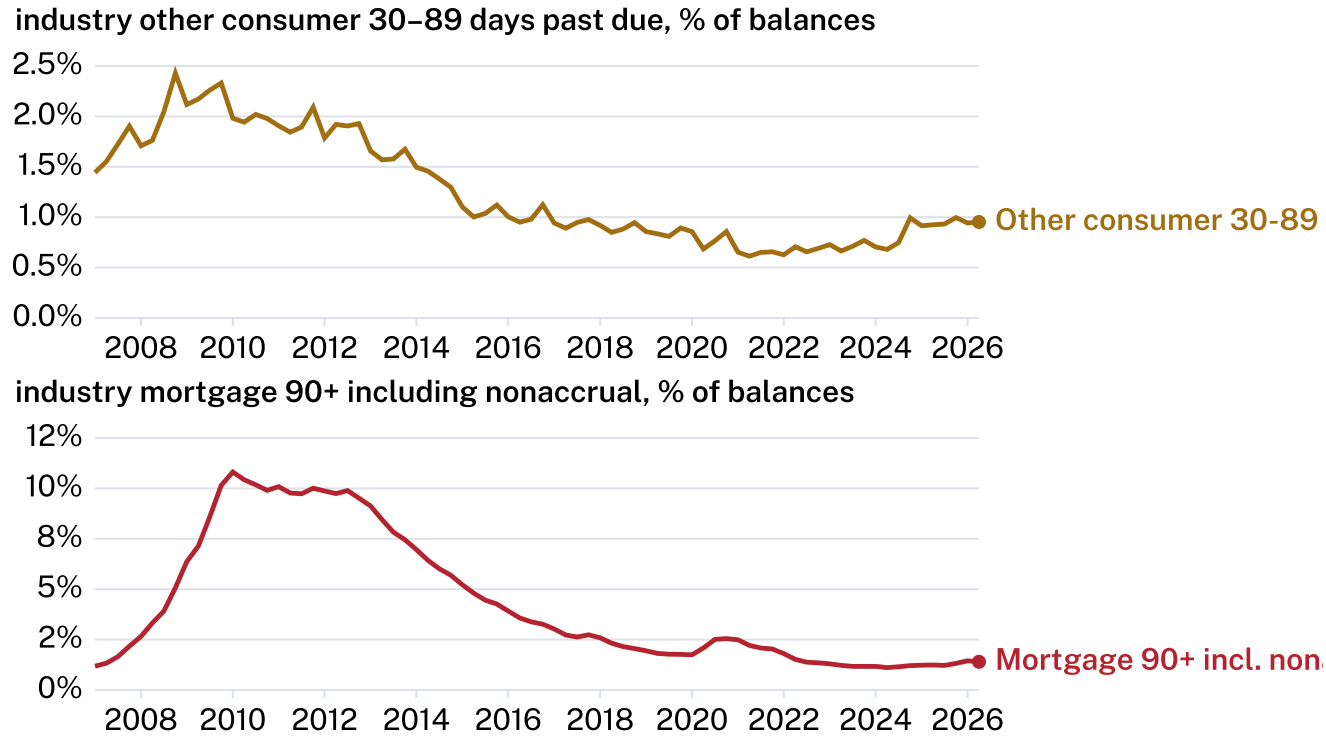
▼ Numbers behind it: industry card net charge-offs (every bank, Call Report) (one row per year, the fourth quarter, plus the latest)

Quarter	Net charge-offs, four quarters	Average balances	Net charge-off rate	Change from a year earlier (the “rise”)
2012Q4	$26.0bn	$669.8bn	3.88%	-2.25 pts
2013Q4	$23.0bn	$674.7bn	3.42%	-0.47 pts
2014Q4	$23.6bn	$684.5bn	3.45%	+0.04 pts
2015Q4	$20.9bn	$713.1bn	2.92%	-0.53 pts
2016Q4	$23.9bn	$757.8bn	3.15%	+0.23 pts
2017Q4	$28.8bn	$799.1bn	3.61%	+0.46 pts
2018Q4	$32.2bn	$854.4bn	3.76%	+0.16 pts
2019Q4	$35.0bn	$893.9bn	3.91%	+0.15 pts
2020Q4	$29.6bn	$824.9bn	3.58%	-0.33 pts
2021Q4	$17.6bn	$807.5bn	2.18%	-1.40 pts
2022Q4	$20.8bn	$924.9bn	2.25%	+0.07 pts
2023Q4	$37.0bn	$1045.3bn	3.54%	+1.28 pts
2024Q4	$51.6bn	$1117.3bn	4.62%	+1.08 pts
2025Q4	$48.6bn	$1153.3bn	4.22%	-0.40 pts
2026Q2	$47.2bn	$1176.6bn	4.01%	-0.45 pts

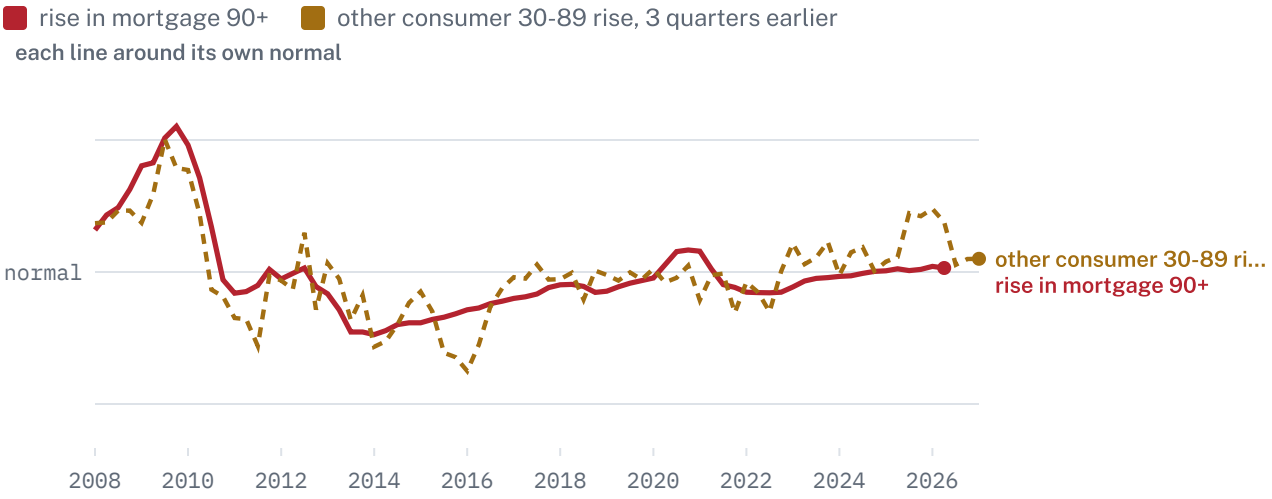
Mortgage · Other consumer 30–89 as a confirming signal for mortgage 90+

Shown beside the ratings, not rated

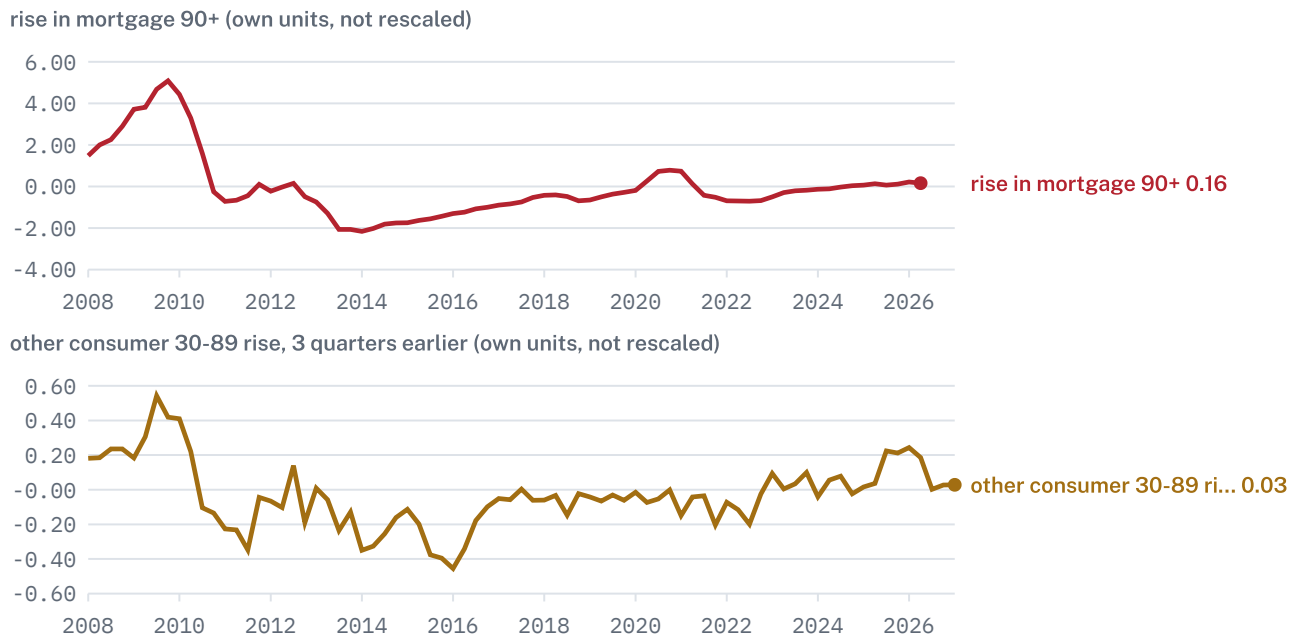
As published: the two series, nothing calculated



Same quarters on both. Below: the same two drawn for timing (rescaled and slid), then the numbers.



▼ The same two series in their own units



Industry, 2007 on. A rise in other consumer 30–89 went with a rise in mortgage 90+ about 3 quarters later (0.80). Mortgage's own 30–89 turns at the same time, so this confirms rather than leads. Both are rates of balances; the rise is the change in the rate from a year earlier, in percentage points.

▼ Numbers behind it: other consumer 30–89 days past due, industry (one row per year, the fourth quarter, plus the latest)

Quarter-end	30–89	Balances	Rate	Change from a year earlier
2007Q4	$12.1bn	$636.6bn	1.90%	+0.24 pts
2008Q4	$15.6bn	$644.2bn	2.43%	+0.53 pts
2009Q4	$14.8bn	$636.3bn	2.33%	-0.10 pts
2010Q4	$12.1bn	$614.3bn	1.98%	-0.35 pts
2011Q4	$6.7bn	$320.0bn	2.09%	+0.11 pts
2012Q4	$5.9bn	$307.7bn	1.93%	-0.16 pts
2013Q4	$5.2bn	$308.2bn	1.67%	-0.25 pts
2014Q4	$4.1bn	$314.6bn	1.30%	-0.38 pts
2015Q4	$3.7bn	$326.6bn	1.12%	-0.18 pts
2016Q4	$3.9bn	$347.8bn	1.12%	+0.00 pts
2017Q4	$3.5bn	$362.9bn	0.98%	-0.15 pts
2018Q4	$3.6bn	$383.8bn	0.95%	-0.03 pts
2019Q4	$3.7bn	$412.3bn	0.89%	-0.05 pts
2020Q4	$3.7bn	$430.4bn	0.86%	-0.04 pts
2021Q4	$3.1bn	$473.1bn	0.66%	-0.20 pts
2022Q4	$3.5bn	$512.8bn	0.69%	+0.03 pts
2023Q4	$3.8bn	$489.1bn	0.77%	+0.08 pts
2024Q4	$3.5bn	$356.2bn	0.99%	+0.22 pts
2025Q4	$3.5bn	$349.3bn	1.00%	+0.00 pts
2026Q2	$3.4bn	$353.5bn	0.95%	+0.03 pts

▼ Numbers behind it: mortgage 90+ including nonaccrual, industry (one row per year, the fourth quarter, plus the latest)

Quarter-end	90+ incl. nonaccrual	Balances	Rate	Change from a year earlier
2007Q4	$43.1bn	$1978.9bn	2.18%	+1.07 pts
2008Q4	$92.2bn	$1819.1bn	5.07%	+2.89 pts
2009Q4	$176.2bn	$1735.0bn	10.15%	+5.08 pts
2010Q4	$173.5bn	$1752.8bn	9.90%	-0.25 pts
2011Q4	$176.4bn	$1762.7bn	10.01%	+0.11 pts
2012Q4	$170.6bn	$1793.1bn	9.52%	-0.49 pts
2013Q4	$129.8bn	$1743.6bn	7.45%	-2.07 pts
2014Q4	$100.6bn	$1766.4bn	5.70%	-1.75 pts
2015Q4	$78.2bn	$1835.9bn	4.26%	-1.43 pts
2016Q4	$63.2bn	$1934.2bn	3.27%	-1.00 pts
2017Q4	$55.2bn	$2014.4bn	2.74%	-0.52 pts
2018Q4	$42.5bn	$2074.6bn	2.05%	-0.69 pts
2019Q4	$38.1bn	$2160.0bn	1.77%	-0.28 pts
2020Q4	$55.4bn	$2176.5bn	2.55%	+0.78 pts
2021Q4	$45.3bn	$2231.1bn	2.03%	-0.52 pts
2022Q4	$32.9bn	$2447.0bn	1.35%	-0.68 pts
2023Q4	$29.6bn	$2525.0bn	1.17%	-0.18 pts
2024Q4	$31.0bn	$2555.3bn	1.21%	+0.04 pts
2025Q4	$34.3bn	$2587.7bn	1.32%	+0.11 pts
2026Q2	$36.4bn	$2602.4bn	1.40%	+0.16 pts

D. Data, scripts and how to rerun

Every chart above is drawn by the scripts in this folder from the data beside them.

Sources

Source	In this record	Used for	Where
FDIC Call Reports, every filer, 2006Q1–2026Q2	the book's cache (`peers pull`); the series files under `docs/walkthrough-book/out/series`	charge-offs, delinquency, balances; 30+ to charge-off; roll steps	banks.data.fdic.gov
Philadelphia Fed, Large Bank Credit Card and Mortgage Data, 2012Q3–2026Q2	`data/philly/26Q2-*.csv`	scores, minimum payments, utilization, originations, mortgage DTI/LTV	www.philadelphiafed.org
FRED copies of the same score series	`data/RCC*.csv`, `data/RCM*.csv`	a second reading; 56 of 56 quarters agree	fred.stlouisfed.org
Federal Reserve Senior Loan Officer Opinion Survey, 1996–2026	`data/sloos/DRTSCLCC.csv`	banks tightening or easing card standards	fred.stlouisfed.org
SEC 10-Q and 10-K XBRL, 11 holding companies, 2019Q4–2026Q2	`data/peerfico/peers.json` (from `peerfico_pull.py`, `bands.py`, `peers.py`); `data/keyhist/`	each bank's refreshed FICO bands by class; KeyCorp back to 2018	www.sec.gov
FHFA National Mortgage Database aggregate statistics, 1998–2026	`data/nmdb/nmdb.json`, `mtm.json`	mortgage terms at origination; current LTV of the outstanding book	www.fhfa.gov

Rerun

From `threshold-engine`, after the book has been built (its series files are read for the Call Report measures):

```
python docs/early-warning-2026-10/record/evidence.py
# the derived data, when the caches are to hand:
python docs/early-warning-2026-10/record/rolls.py CACHE config/peers.json docs/early-warning
python docs/early-warning-2026-10/record/fdic_numbers.py CACHE config/peers.json docs/early-
python docs/early-warning-2026-10/record/peerfico_pull.py docs/early-warning-2026-10/record/
python docs/early-warning-2026-10/record/peers.py docs/early-warning-2026-10/record/data/pee
```

The SEC facts file (14 MB) is not in the repository; `peers.json`, read from it, is. The FHFA zips (21 MB) are not; `nmdb.json`, the national series read from them, is.

What was not checked

- Competitors' FICO figures come from their filings' tagged data; only KeyCorp's 2026Q2 amounts were tied to a printed table.
- Whether every KeyBank nonaccrual card loan is past due: the internal check that decides whether KeyBank's 30+ share is flattered.
- Which FICO version each bank uses, and whether the Philadelphia Fed counts every open account or only active ones for the score percentiles.
- The lines in sections A and B are set from the same history they rate; the book's Backtest, which sets each quarter's lines from earlier quarters only, is the strict test and runs in the workbook.
- With 2020–2023 left out the card matches rest on 20 to 37 quarters; the mortgage matches rest mostly on 2005–2010.